Built 2026-11-03 from github.com/SdWa5/docs. The Markdown pages there are the current version.

SdWa5 — Sound reinforcement

A handbook on live sound for the people who set up, run and plan SdWa5's events, and for anyone else who wants one. It covers the physics of sound, the equipment from the microphone to the loudspeaker, hearing and the limits that protect it, and the design and tuning of a system for a real event. It is written for today's practice, in which consoles, processors and amplifiers are digital, loudspeakers are designed and aligned with measurement and prediction software, signals travel over networks, and hearing protection is a matter of regulation. Its chapters, their order and their figures are its own, and each chapter names the literature it draws on in its *Sources*.

Status. This is pass 1, a full draft written quickly so that the whole book can be read. Every chapter carries a draft banner and ends with a list *To verify in pass 2*, which names each claim that has not yet been checked against its primary source. Pass 2 works through those lists chapter by chapter, adds generated figures and removes the banners. Until then, treat named standards, editions and limit values as pointers to check, not as quotations.

The whole book is also available as one PDF for reading. It is rebuilt by hand, so the Markdown pages here are the current version when the two differ.

Conventions

- **Units** are SI. Distances are in metres, pressures in pascals, and electrical supply is 230/400 V at 50 Hz. The speed of sound is 343 m/s (20 °C) unless a section states a temperature.
- **Levels** always carry their reference, as in dBu, dBV, dBFS or dB SPL. A level written as plain dB is a difference or a gain. Weighted sound levels are written as dB(A) and dB(C). Chapter 3 defines all of them.
- **Gain** means voltage gain unless a section says otherwise.
- **Standards** are cited by number. International (IEC, ISO) and AES standards come first, then European directives and EN standards, then Austrian (ÖVE, ÖNORM) and German (DIN) ones where they are the regional reference. SdWa5 is based in Austria, so national rules are Austrian unless a section says otherwise.
- **Numbers** in tables and worked examples are computed, not copied, and rounded as the text states.
- **Diagrams** are Mermaid flowcharts rendered by GitHub from the page source. Signal flows from left to right, a box is a device or a function, a solid arrow carries audio and a dotted arrow carries control or clock. Chapter 1 explains the conventions.
- **Formulas** use GitHub's math rendering, so they read as formulas in the browser and as plain LaTeX in the source.
- **The SdWa5 rig** appears as a worked example where its own documentation exists. Those links point into the sibling repositories `sdwa5-3d` and `sdwa5-dsp` and resolve in a local checkout with the siblings next to this repository.

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3	Decibels, levels and loudness	dB units, analog and digital levels, SPL, weighting, loudness, LUFS
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20	System design and tuning	Brief, prediction, coverage, subwoofer design, alignment, tuning, a worked event

Acoustics is a companion page with reflection, absorption and transmission data for common materials. Chapter 7 uses it as its reference table.

Sources

- Each chapter lists the standards, papers, books and SdWa5 documents it relies on in its own *Sources* section.
- General references are McCarthy, B.: *Sound Systems: Design and Optimization*, Focal Press, 3rd ed. 2016, and Davis, G., Jones, R.: *The Sound Reinforcement Handbook*, 2nd ed., Yamaha / Hal Leonard 1989, ISBN 0-88188-900-8.

1. Signals and systems

> **Draft, pass 1.** This chapter was written from general knowledge and checked against reference literature for coverage. Standards, editions and figures are named but have not yet been verified against their sources. The list *To verify in pass 2* at the end names every claim that still needs one.

This chapter answers what travels through a sound system and what the system is made of. It starts with sound as a pressure wave, explains phase and why it decides how two signals add, follows a modern signal chain from the microphone to the listener, and ends with how to read and draw the block diagrams the rest of the book uses.

Sound as a pressure wave

Sound is a small, fast variation of air pressure around the static atmospheric pressure. A loudspeaker cone that moves forward pushes the air in front of it together, and the region of slightly higher pressure moves away from the cone. When the cone moves back, a region of slightly lower pressure follows. The air itself barely travels. What travels is the pattern of compression and rarefaction, passed on from one layer of air to the next.

Three quantities describe a steady tone.

- **Frequency** f is the number of complete cycles per second, measured in hertz (Hz). Human hearing spans roughly 20 Hz to 20 kHz, with the upper limit falling with age (chapter 5).
- **Period** $T = 1/f$ is the duration of one cycle. A 1 kHz tone has a period of 1 ms, a 50 Hz tone one of 20 ms.
- **Wavelength** $\lambda = c/f$ is the distance one cycle occupies in the air, where c is the speed of sound.

The speed of sound in air depends on temperature and hardly at all on air pressure or frequency. A linear approximation that is good enough for sound system work is

$$c \approx 331.3 + 0.606\,T \quad \mathrm{m/s}, \text{ with } T \text{ in } ^\circ\mathrm{C}.$$

Air temperature	0 °C	10 °C	15 °C	20 °C	25 °C	30 °C	35 °C
Speed of sound (m/s)	331.3	337.4	340.4	343.4	346.4	349.5	352.5

The difference matters wherever distances are converted into time. A delay tower 30 m behind the main system needs 88.9 ms at 10 °C and 85.8 ms at 30 °C, so an alignment set in the cool afternoon is 3 ms off after a hot evening. This book uses 343 m/s (20 °C) unless a section says otherwise.

Wavelength is the quantity that explains most of what loudspeakers, rooms and microphones do, because an object only influences sound much when it is comparable in size to the wavelength or larger. The audible range covers three decades of frequency and therefore three decades of wavelength.

Frequency	20 Hz	50 Hz	100 Hz	250 Hz	500 Hz	1 kHz	2 kHz	4 kHz	8 kHz	16 kHz	20 kHz
Wavelength at 20 °C	17.2 m	6.87 m	3.43 m	1.37 m	0.69 m	0.34 m	17 cm	8.6 cm	4.3 cm	2.1 cm	1.7 cm

A 20 Hz wave is longer than most stages are deep, so no cabinet of portable size can steer it by its shape alone, and subwoofer directivity has to come from several sources working together (chapter 14). A 16 kHz wave is shorter than a thumb, so the grille of a microphone or a person's head is already a large obstacle to it.

The electrical signal in a cable is an analogue of the pressure wave. Within the working range of the equipment, the voltage follows the pressure in time and is proportional to it. The size of the signal is its level, which is always given in decibels relative to a stated reference. Chapter 3 is devoted to that.

Phase, polarity and how signals add

Phase describes where in its cycle a periodic signal is at a given moment, measured against a reference. One full cycle is 360°. A reference can be a fixed instant or another signal of the same frequency. When the output of a device is compared with its input, a phase shift of 90° means the output crosses zero a quarter period later than the input. A phase shift is always a statement about one frequency. A real device usually shifts different frequencies by different amounts, which is why chapter 2 treats phase as a curve over frequency.

A **time delay** produces a phase shift that grows with frequency. One millisecond is 36° at 100 Hz, 360° at 1 kHz and 3600° at 10 kHz. Delay and phase are therefore related but not the same thing. A delay shifts all frequencies by the same time. A filter shifts different frequencies by different times.

Polarity is not phase. Swapping the two wires of a balanced connection, or pressing a "Ø" button, turns every positive pressure into a negative one at all frequencies at once and without any delay. It looks like a 180° phase shift on a sine wave, but on a kick drum it turns the first push of air into a pull, which no amount of delay can reproduce. Equipment buttons labelled "phase" invert polarity, and this book calls them polarity switches.

When two signals meet, in a mixing bus or in the air, their instantaneous values add. For two signals of the same level and frequency, the result depends on the phase between them.

Phase between the two	Resulting amplitude	Level change against one signal
0° (in phase)	2×	+6.0 dB
90°	1.41×	+3.0 dB
120°	1×	0 dB
180° (opposite)	0	complete cancellation

Two cases matter in practice and should not be confused.

- **Coherent** sources carry the same signal, such as two loudspeakers fed from the same output or a direct sound and its reflection. Their sum depends on the phase at

each frequency, so it can be anything from +6 dB to a deep notch. Chapter 2 shows how a fixed time offset turns this into a comb of peaks and notches.
- **Incoherent** sources carry unrelated signals, such as two singers, two noise generators or the many reflections in a reverberant room. Their phases wander randomly against each other, and on average their powers add. Two equal incoherent sources are 3 dB louder than one.

This is why doubling the number of loudspeakers playing the same signal gives up to 6 dB at low frequencies, where the boxes stay in phase across the audience, and closer to 3 dB at high frequencies, where small path differences already scramble the phase.

What a sound system is for

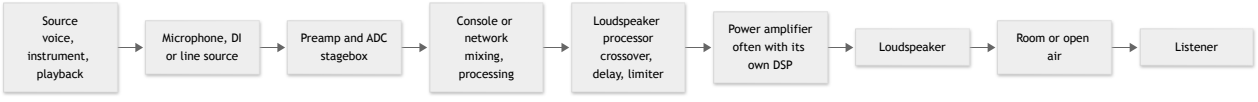
A sound reinforcement system makes sound louder, clearer or available somewhere else. The reasons fall into a few groups.

1. **Intelligibility.** A speaker on a stage should be understood in the last row as well as in the first. The goal is clarity at a natural level, not loudness.
2. **Impact.** A band or a DJ set should feel larger than the unamplified source could. Here the level and the low-frequency energy are part of the artistic intent.
3. **Reach.** Overflow rooms, delay towers at a festival, a livestream or a hearing loop take the programme to people who are not in front of the source.
4. **Playback.** Recorded or generated material, from a DJ controller, a playback computer or a streaming device, is reproduced for an audience. Feedback is then no concern, but level, coverage and noise limits are.

The same physical principles apply to all four, and most real events mix them. A club night with a short speech is a playback system for four hours and a speech reinforcement system for five minutes, and it has to be set up for both.

The signal chain

Every sound system converts sound into an electrical signal, processes and amplifies it, and turns it back into sound in a room where people listen. In its simplest form this is three blocks, an input transducer, processing and an output transducer. A modern system has that skeleton, but most of the processing happens on numbers rather than voltages, and the room and the listener are part of the chain rather than a footnote.



Input transducers turn sound or another physical signal into an electrical one. Microphones turn air pressure into voltage (chapter 9). Magnetic and piezo pickups turn string or body vibration into voltage. A DI box adapts the high-impedance, unbalanced output of an instrument to a balanced microphone input. Playback devices skip the acoustic step entirely, and many deliver a digital signal directly over USB, AES3 or a network.

Conversion and transport. A microphone delivers microvolts to millivolts. A preamplifier raises this to line level, and an analog-to-digital converter (ADC) turns it into a stream of numbers. In a modern rig both sit in a stagebox on stage, and the numbers travel over a network or a single digital multicore to the console (chapter 15).

Mixing and processing. The console sets the gain of each source, equalises, compresses, routes and mixes them into the outputs the event needs (chapters 11 and 12). Today this is software running on dedicated hardware, controlled from a surface, a laptop or a tablet.

Loudspeaker processing. A loudspeaker processor splits the mix into frequency bands for the different drivers, delays the outputs so that boxes at different distances arrive together, equalises the boxes and limits each band to what its driver survives (chapters 12, 14 and 20). In many current systems this happens inside the power amplifier.

Amplification. The power amplifier turns a line-level signal of about a volt into tens or hundreds of volts at the current the loudspeaker demands (chapter 13).

Output transducers turn the amplified signal back into sound. Woofers, midrange drivers, compression drivers and their enclosures each cover part of the spectrum (chapter 14). Monitors and in-ear systems send a separate mix back to the performers (chapter 18).

Room and listener. The loudspeaker's sound reflects, is absorbed, bends around obstacles and loses its highs over distance (chapters 6 and 7). What reaches a person is that sound filtered through human hearing (chapter 5). A system that measures well at one microphone position can still fail in the room, and a room can ruin a good system. Design starts from the audience area and works backwards to the boxes.

The same chain has a **control layer** alongside the audio. Scene recall, remote control of amplifiers and processors, clock synchronisation and timecode run beside the audio and sometimes over the same network (chapter 19).

Digital audio in brief

Nearly every modern system converts to digital audio at the stagebox and back to analog at or inside the amplifier. Four properties of that conversion appear throughout the book.

Sample rate. The ADC measures the voltage at fixed intervals, 48 000 times per second in most live systems (48 kHz), sometimes 96 kHz. A sampled signal can represent frequencies up to half the sample rate, the Nyquist frequency, which is 24 kHz at 48 kHz. Anything above it must be filtered out before sampling, or it folds back as an audible alias. Modern converters oversample internally and filter digitally, so this filter no longer shapes the audible band the way the steep analog filters of early converters did.

Bit depth. Each sample is stored as a number with a fixed number of bits. Every bit doubles the number of possible values and adds about 6.02 dB to the theoretical dynamic range of the format.

Format	Theoretical range	Typical use
16 bit	96.3 dB (98.1 dB for a full-scale sine)	CD, many playback files

Format	Theoretical range	Typical use
24 bit	144.5 dB (146.2 dB for a full-scale sine)	converters, network audio, recording
32 bit float	far beyond any converter	processing inside consoles and DSPs

The theoretical figure is a property of the number format. Real 24-bit converters reach about 110 to 120 dB, because analog noise in the converter sets the floor long before the last bits matter. Chapter 4 shows why this makes the preamp gain the one decision that sets the noise floor.

Full scale. A digital signal has a hard ceiling, the largest number the format can hold, called 0 dBFS. Above it the signal does not distort gradually as an overdriven analog stage might. It clips flat. Chapter 3 explains how 0 dBFS relates to analog levels.

Latency. Every conversion and every processing block takes time. An ADC and a DAC each add some tenths of a millisecond to about a millisecond, a console path adds about one to a few milliseconds, a network hop adds a configurable buffer, and a linear-phase FIR filter can add several more. Latency is a property of the whole system, not of one device, and it adds up.

Latency	0.5 ms	1 ms	2 ms	5 ms	10 ms
Equivalent distance at 20 °C	0.17 m	0.34 m	0.69 m	1.72 m	3.43 m

The conversion to distance is the useful way to think about it. A total of 2 ms between a singer's mouth and the front-of-house loudspeaker is the same as standing 0.7 m further away, which nobody notices. The same 2 ms in a singer's in-ear monitor competes with the sound conducted through their own skull, which some performers notice as a phasey or hollow voice (chapter 18). Latency differences between paths that are mixed together are worse than latency itself, because they create comb filters (chapter 2). Consoles compensate internal path differences automatically, but a signal routed out to an external processor and back may not be compensated.

Reading and drawing block diagrams

A block diagram shows how a signal flows through a device or a system. It treats each function as a box and leaves out the components that implement it. A schematic is the opposite. It shows every resistor, transistor and connection, including power and ground, and is drawn for the person who repairs the device. The user needs the block diagram, because it answers how the device works and where its controls act.

Manufacturers still publish block diagrams for consoles, processors and amplifiers, although the symbol sets have drifted apart over the years. Older references collect symbol sheets for amplifiers, switches, transformers, grounds, meters, connectors and filters. Such sheets are less useful today, because a digital device is mostly software and its diagrams are mostly labelled boxes. The rules for reading a diagram have outlived the symbols.

1. **Signal flows from left to right**, and top to bottom where a diagram wraps. An arrow is drawn only where a path runs against that direction, such as a feedback path or a side chain.

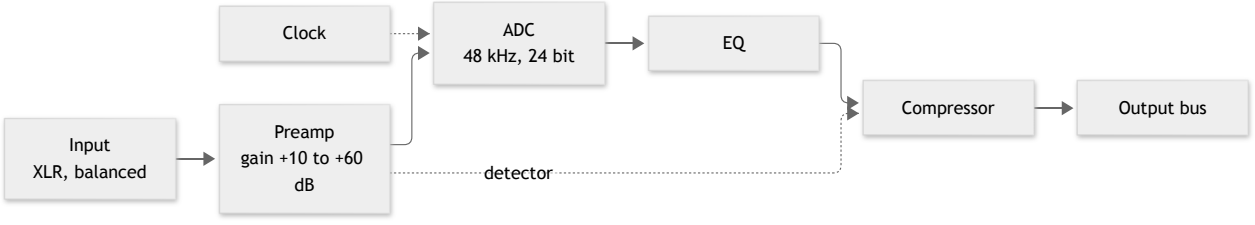
2. **Follow one path at a time.** Start at an input, follow it to an output, and only then return to the branches.
3. **Find where each control acts.** A gain control before a processing stage changes how hard that stage is driven. The same control after it only changes the output level. This difference decides noise and headroom (chapter 4).
4. **Separate audio from control.** Side-chain detectors, clocks and remote control paths carry no audio, even when they are drawn with the same lines.
5. **Do not assume what the diagram does not say.** Polarity is the classic example. A diagram may show a balanced input and output without saying whether the device inverts the signal. The data sheet or a measurement settles it (chapter 16). Since the AES and IEC standardised pin 2 as the positive pin of an XLR connector, inverting devices have become rare, but they still exist.
6. **When in doubt, draw a box and label it.** A well-labelled rectangle is better than a symbol the reader has to guess.

The same technique applies to whole systems. A system diagram that shows every source, every bus, every processor output and every amplifier channel is the best tool for finding a missing feed or a doubled path before the event, and the fastest map for troubleshooting during it.

Diagram conventions in this book

The diagrams in this book are Mermaid flowcharts that GitHub renders from the text of the page. They follow the reading rules above.

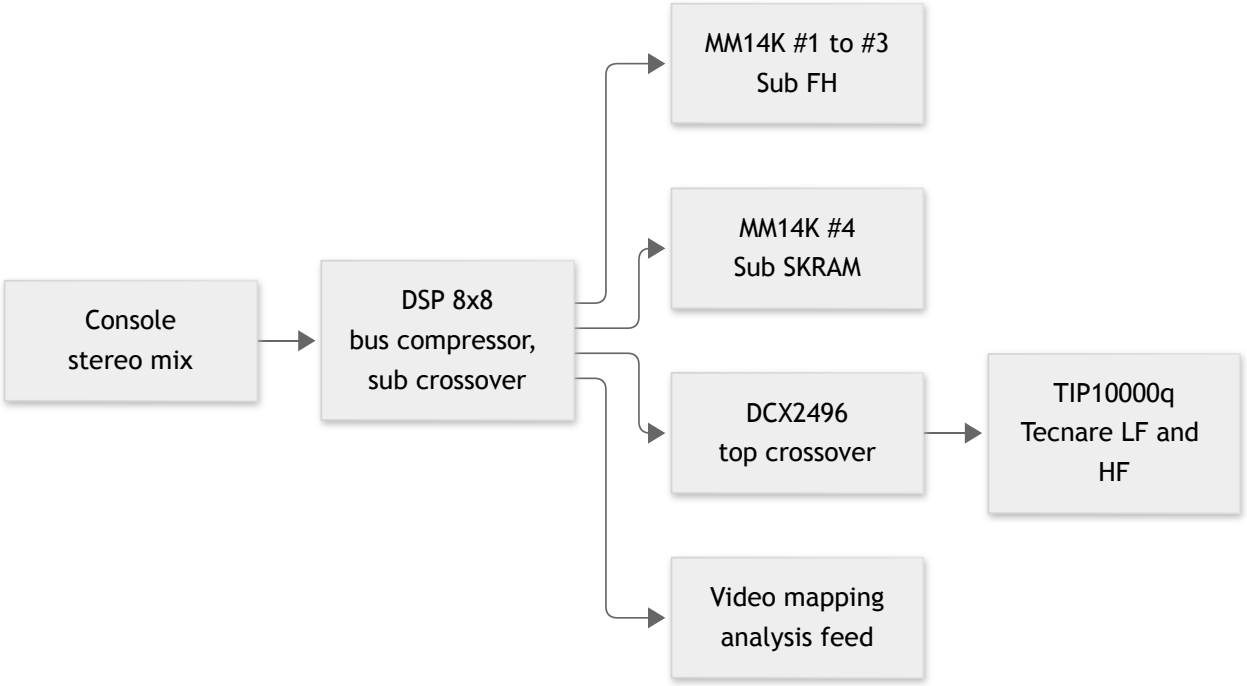
- Flow runs left to right (`flowchart LR`).
- A box is a device or a function, and its label names it in plain words, with the second line giving its role.
- An arrow is a signal path. A dotted arrow (`-.->`) is a control or clock path that carries no audio.
- Acoustic paths, such as loudspeaker to room, are drawn like electrical ones, because for the system they are just another stage in the chain.



The second example reads as follows. A balanced input feeds a preamp with a variable gain, the signal is converted at 48 kHz and 24 bits and then equalised and compressed on its way to an output bus. The compressor's detector is fed from before the converter, which is unusual and worth noticing, and the converter runs from a clock that does not carry audio. Each of these facts could matter during an event, and the diagram makes them visible in seconds.

In the SdWa5 rig

The SdWa5 event rig follows the chain above with two loudspeaker processors in series with the console's output. A DSP 8x8 feeds the subwoofer amplifiers directly and passes the tops to a Behringer DCX2496, which splits them into low and high bands for a four-channel amplifier. The rig's patch, the reason the second processor was needed and the complete limiter table are documented in Signal chain.



The rig shows two points of this chapter. The audio leaves the 8x8 on two outputs and returns on two inputs to pass through the bus compressor, so a block diagram without that loop would misstate where the compressor acts. The video mapping machine takes a copy of the mix as a control signal, which is the kind of path rule 4 above separates from audio.

To verify in pass 2

- Speed of sound approximation $331.3 + 0.606\,T$ and its error against the exact formula across −10 to 40 °C; check against ISO 9613-1 or a physics reference.
- Typical ADC, DAC and console latencies quoted as tenths of a millisecond to a few milliseconds; check against current converter and console data sheets.
- Real 24-bit converter dynamic range of about 110 to 120 dB; check against current converter data sheets and AES17 measurement conditions.
- Pin 2 as the positive XLR pin, attributed to the AES and IEC; check AES26 and IEC 60268-12.
- Upper hearing limit falling with age; check ISO 7029 (handled in chapter 5).

Sources

- ISO 9613-1, attenuation of sound during propagation outdoors, part 1, for the speed of sound and air absorption.

- AES17, measurement of digital audio equipment, for full scale and dynamic range.
- AES26 and IEC 60268-12, for connector pin assignment.
- SdWa5: Signal chain, the rig documented in this chapter.
- Davis, G., Jones, R.: *The Sound Reinforcement Handbook*, 2nd ed., Yamaha / Hal Leonard 1989, §1 (pp. 1–8) and §7 (pp. 61–70).

2. Frequency, phase and time

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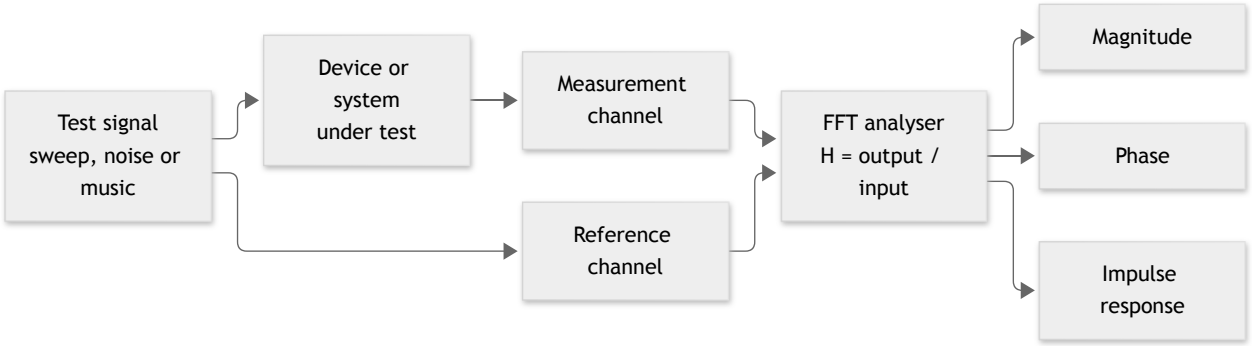
This chapter answers how a device or a system changes a signal depending on its frequency, and why time and phase belong in the same picture as level. It introduces the transfer function, the frequency bands used for measurement and equalisation, filters and their slopes, the frequency content of voices and instruments, and the comb filter that appears whenever a signal meets a delayed copy of itself.

The transfer function

The classic definition of frequency response uses a sine generator, a black box and a level meter. The generator is swept across the audio band at constant level, and the output level is plotted against frequency. The plot shows which frequencies pass and how much each one is raised or lowered.

That plot is half of the description. Any linear device changes two things at each frequency, the **magnitude** (how much larger or smaller the output is) and the **phase** (how far the output is shifted in time against the input). Together they form the **transfer function** $H(f)$. Two devices with identical magnitude curves can sound different and combine differently with other devices if their phase curves differ.

The transfer function has a twin in the time domain, the **impulse response**. It is the output of the device when the input is a single, infinitely short click. The impulse response and the transfer function contain exactly the same information, and the Fourier transform (computed as an FFT) converts one into the other. Modern measurement software never sweeps a sine slowly past a meter. It plays a known signal, such as a sweep, pink noise or the programme itself, records the output, and computes the transfer function by comparing output and input (chapter 16).



Reading a magnitude plot needs three habits.

- **Look for the tolerance.** "30 Hz to 18 kHz" says nothing until a tolerance such as ±3 dB is added, and nothing reliable until the measurement conditions are stated. Chapter 8 deals with reading specifications.
- **Look for the smoothing.** A curve averaged over a third of an octave hides narrow peaks and dips that a 1/24 octave curve shows. Both are honest, but they answer different questions.
- **Look at the axes.** A vertical scale of 50 dB makes a ±3 dB ripple disappear. A scale of 10 dB makes it look dramatic.

Electronic devices such as preamps, mixing busses and power amplifiers are flat across the audio band to within a fraction of a decibel, unless they are meant to shape it. Microphones and loudspeakers are not flat, and the room is the least flat element of all. The response of the whole chain is the product of the individual responses, which in decibels means the sum. A loudspeaker 3 dB down at 60 Hz in a room that adds 8 dB at 60 Hz produces a 5 dB peak at the listener.

Octaves, decades and standard bands

Hearing judges frequency by ratio, not by difference. The step from 100 Hz to 200 Hz sounds like the same musical interval as the step from 1 kHz to 2 kHz, although the second spans ten times as many hertz. A ratio of 2:1 is an **octave**, a ratio of 10:1 a **decade**, and one decade holds about 3.32 octaves. Frequency axes in audio are therefore logarithmic, with each decade taking the same width.

For measurement and for graphic equalisers, the audio band is divided into standard bands of one octave or one third of an octave. Their centre frequencies are defined in ISO 266 and the filters in IEC 61260-1. The exact centres follow the base-ten series $f_m = 1000 \cdot 10^{n/10}$ Hz for third-octave bands, which gives 1000, 1259, 1585, 1995, 2512 Hz and so on. The base-two series $1000 \cdot 2^{n/3}$ gives almost the same values, because $10^{0.1}$ (1.2589) and $2^{1/3}$ (1.2599) differ by less than 0.1 %. Both are rounded to the familiar nominal labels.

Nominal third-octave centres (Hz)									
20	25	31.5	40	50	63	80	100	125	160
200	250	315	400	500	630	800	1 k	1.25 k	1.6 k
2 k	2.5 k	3.15 k	4 k	5 k	6.3 k	8 k	10 k	12.5 k	16 k

The octave centres are every third entry, from 31.5 Hz to 16 kHz. An equaliser whose bands sit on these centres matches a real-time analyser that uses the same standard, which is the point of the standard.

Before dual-channel analysers, band measurements with **pink noise** were the main tool. Pink noise has equal energy in every octave, so it shows a flat line on a band analyser when the system is flat. Its weakness is the same as its strength. A third-octave band averages everything inside it, so a deep notch 1/12 octave wide disappears, and in a reverberant room the result says more about the room than about the loudspeaker. Chapter 16 explains why dual-channel FFT measurements have largely replaced it.

Phase, group delay and minimum phase

A **phase response** shows the phase shift at each frequency. On its own it is hard to read, because a pure time delay produces a phase that falls steadily with frequency and wraps around every 360°. Two derived views are more useful.

Group delay is the time by which each part of the spectrum is delayed. For a pure delay it is flat, the same for all frequencies. For a filter it peaks around the filter's corner frequency. A loudspeaker typically delays its low frequencies by several milliseconds more than its highs, which is one reason a kick drum through a large system can sound late against the hi-hat even when every box is aligned.

Minimum phase describes systems whose phase follows directly from their magnitude. Most analog filters, single loudspeaker drivers, and most acoustic effects that are not reflections behave this way. For such systems, an equaliser that corrects the magnitude also corrects the phase. A reflection, or the overlap of two drivers at different distances, is not minimum phase. Equalisation cannot fix it, because its cause is a time difference, and the cure is to change the time difference or the geometry (chapter 20).

Linear phase means the same delay at every frequency, which keeps the shape of a waveform intact. FIR filters (see below) can be designed to have linear phase. Analog and IIR filters cannot.

Filters and slopes

Filters pass some frequencies and attenuate others. The basic types are the **high-pass** (low cut), the **low-pass** (high cut), the **band-pass** and the **notch** (band stop). Equalisers, crossovers and limiter side chains are all built from them.

A filter is described by its corner frequency, its slope and its shape.

- The **corner** or turnover frequency of a high-pass or low-pass filter is usually the point where the level has dropped 3 dB from the pass band. Linkwitz-Riley crossovers are the exception, with the corner at −6 dB (see below).
- The **slope** is how fast the level falls beyond the corner, in dB per octave. Each order, called a pole in analog design, adds about 6 dB per octave. A second-order filter falls at 12 dB per octave, a fourth-order one at 24 dB per octave. Digital filters reach 48 dB per octave and more without effort.
- The **shape** near the corner is set by the filter alignment.

Butterworth high-pass order	1	2	3	4	8
Asymptotic slope (dB per octave)	6	12	18	24	48
Attenuation one octave below the corner	7.0 dB	12.3 dB	18.1 dB	24.1 dB	48.2 dB
Attenuation two octaves below the corner	12.3 dB	24.1 dB	36.1 dB	48.2 dB	96.3 dB

Near the corner the attenuation is a little more than the slope suggests, and from one octave on it settles to the slope. An 80 Hz third-order high-pass is 18 dB down at 40 Hz and 36 dB down at 20 Hz, which removes stage rumble, handling noise and wind without touching a voice. The exact numbers depend on the alignment, and other alignments than Butterworth reach several decibels more or less one octave out.

Three alignments cover nearly all practical use.

- **Butterworth** is as flat as possible in the pass band and −3 dB at the corner. It is the common choice for high-pass filters on input channels and for protecting subwoofers.
- **Linkwitz-Riley** is two Butterworth filters in series and is −6 dB at its corner. Its purpose is crossovers. A Linkwitz-Riley low-pass and high-pass at the same frequency add to a flat magnitude, because both are −6 dB at the crossover and in phase with each other there. The fourth-order version (LR24) is the standard crossover in live sound.
- **Bessel** has the gentlest corner and the most constant group delay of the three. It keeps transients cleanest and is used where that matters more than a sharp corner.

For band-pass and peaking filters, the width is given as a **Q**, the centre frequency divided by the bandwidth between the −3 dB points, or as a bandwidth in octaves. A Q of 1.41 is about one octave wide, a Q of 4.3 about a third of an octave. Equalisers use the same idea for the width of a boost or cut (chapter 12).

IIR and FIR. Digital filters come in two families. IIR (infinite impulse response) filters copy the behaviour of analog filters, with the same slopes, the same phase shifts and almost no latency. FIR (finite impulse response) filters compute each output sample as a weighted sum of many past input samples, the taps. They can shape magnitude and phase independently, so a crossover can be steep and linear-phase at the same time. The price is latency and, for low frequencies, a large number of taps.

FIR length at 48 kHz	512 taps	1024 taps	2048 taps	4096 taps
Latency of a linear-phase design	5.3 ms	10.7 ms	21.3 ms	42.7 ms
Frequency resolution, roughly	94 Hz	47 Hz	23 Hz	12 Hz

This is why FIR filters are common for the mid and high bands of loudspeaker processing, where a few hundred taps suffice, and rare in subwoofer crossovers, where a useful filter would delay the whole system by tens of milliseconds.

Voices, instruments and harmonics

The frequency range a system needs depends on what it reproduces.

Speech carries most of its energy below about 1 kHz, but its intelligibility lives in the consonants between about 1 and 8 kHz. Cutting the highs costs far more understanding than cutting the lows. Classic telephony passed only 300 Hz to 3.4 kHz and was understandable but tiring. Current wideband and super-wideband telephony passes much more, and a speech reinforcement system should be reasonably flat from about 100 Hz to 8 kHz. The octave bands from 125 Hz to 8 kHz are the ones in which speech intelligibility is measured (chapter 7). A broad presence lift of a few decibels between 2 and 5 kHz can raise intelligibility, but it also brings the system closer to feedback at the frequencies where most vocal microphones and monitors are already strong.

Music has no fixed range. The fundamentals of most instruments lie between about 30 Hz and 4 kHz, but the character of a sound comes from its overtones, and electronic music can put energy anywhere from below 30 Hz to the top of hearing.

Source	Approximate fundamental range
Kick drum, body	40–100 Hz
Electric bass, four strings	41–400 Hz
Male speaking voice	85–180 Hz
Female speaking voice	165–255 Hz
Piano	27.5 Hz–4.2 kHz
Violin	196 Hz–3.5 kHz
Synthesiser, sub bass	down to 20 Hz and below

A pitched sound consists of a **fundamental** and **harmonics** at whole-number multiples of it. An A at 440 Hz has harmonics at 880, 1320, 1760, 2200 Hz and upward, and the relative strength of these components decides whether the note sounds like a flute, a violin or an oboe. Sounds whose components are not whole-number multiples, such as most drums and cymbals, lose a clear pitch and approach noise. Because timbre lives in the balance of components across many octaves, a system whose response is rough changes the character of every source it reproduces, even when its nominal range is wide.

The comb filter

When a signal is added to a copy of itself delayed by Δt, the two copies are in phase at some frequencies and opposite at others. They cancel where the delay equals half a period, one and a half periods, and so on, which is at odd multiples of $1/(2\Delta t)$. Between the notches they add up to +6 dB. The magnitude plot looks like the teeth of a comb, evenly spaced on a linear frequency axis and crowding together towards the top of a logarithmic one.

$$f_{\text{notch},k} = \frac{2k+1}{2\,\Delta t}, \quad k = 0, 1, 2, \ldots$$

Delay	Path difference at 20 °C	First notches	Notch spacing
0.5 ms	0.17 m	1, 3, 5, 7 kHz	2 kHz
1 ms	0.34 m	500, 1500, 2500, 3500 Hz	1 kHz
2 ms	0.69 m	250, 750, 1250, 1750 Hz	500 Hz
5 ms	1.72 m	100, 300, 500, 700 Hz	200 Hz

Comb filters appear everywhere in sound reinforcement.

- Two microphones picking up the same source at different distances, mixed together.
- A microphone on a lectern picking up the voice directly and via the lectern surface.
- Two loudspeakers covering the same seats from different distances.
- A loudspeaker and its reflection from a nearby wall or floor.
- Two paths through a console or a network with different latency, mixed into the same bus.

The cure depends on the cause. Mixing two microphones on one source is fixed by using one, by moving one so that its level is at least about 10 dB lower than the other, or by delaying one so the copies line up. Loudspeaker overlap is fixed by aiming, by splitting coverage, or by delaying the nearer box so that both arrive together at the seats that

matter. No equaliser can fix a comb filter, because the notches move with the listening position.

Square waves and other legacy tests

Square-wave testing with an oscilloscope was once a standard bench method. A square wave contains a fundamental and all its odd harmonics in a fixed ratio, so a circuit that changes the magnitude or phase of any of them changes the shape of the square. A sloping top showed a low-frequency roll-off, rounded corners showed limited high-frequency response, and ripples after each edge showed ringing.

The method still works and is quick with a USB oscilloscope (chapter 16), but it has lost its central role. An impulse response and transfer function from a dual-channel FFT analyser show the same information as numbers and curves, for the whole system including loudspeakers and room, and they do it with music as the test signal. Its limits still apply. A square wave is not a musical signal, no loudspeaker can reproduce one, and a circuit that rounds a square wave can sound excellent.

To verify in pass 2

- ISO 266 and IEC 61260-1 as the references for preferred frequencies and fractional-octave filters, the current editions, and whether IEC 61260-1 prescribes base-ten or base-two band edges.
- The Q to bandwidth relations (Q 1.41 for one octave, Q 4.3 for a third octave); check against a filter design reference.
- The attenuation of an 80 Hz third-order high-pass at 40 Hz, 18.1 dB for the Butterworth alignment computed here against 21 dB in Davis and Jones; the difference presumably comes from the alignment and changes no advice.
- The fundamental ranges in the voice and instrument table; check against an acoustics of music reference.
- The speech energy and consonant band figures and the 300 Hz to 3.4 kHz telephony band; check against ITU-T recommendations and IEC 60268-16.
- The statement that loudspeakers typically delay their low frequencies by several milliseconds more than their highs; check against published group delay measurements.

Sources

- ISO 266, preferred frequencies for acoustical measurements.
- IEC 61260-1, octave-band and fractional-octave-band filters.
- IEC 60268-16, objective rating of speech intelligibility by speech transmission index.
- Linkwitz, S. H.: *Active crossover networks for noncoincident drivers*, Journal of the AES, 1976.
- Davis, G., Jones, R.: *The Sound Reinforcement Handbook*, 2nd ed., Yamaha / Hal Leonard 1989, §2 (pp. 9–18), §8.9 (pp. 99–100), §8.10 (pp. 100–103).

3. Decibels, levels and loudness

> **Draft, pass 1.** This chapter was written from general knowledge and checked against reference literature for coverage. Standards, editions and figures are named but have not yet been verified against their sources. The list *To verify in pass 2* at the end names every claim that still needs one.

This chapter answers what a decibel is, which reference each common dB unit uses, how analog and digital levels relate, and how measured sound levels connect to perceived loudness. It is the vocabulary for everything that follows.

Logarithms in one table

The decibel is built on the base-ten logarithm. $\lg x$ is the power to which 10 must be raised to give x, so $\lg 100 = 2$ and $\lg 0.001 = -3$. Three rules cover nearly all sound system arithmetic.

- $\lg(a \cdot b) = \lg a + \lg b$. Multiplying ratios means adding decibels.
- $\lg(a/b) = \lg a - \lg b$. Dividing ratios means subtracting decibels.
- $\lg(a^n) = n \cdot \lg a$. Squaring a ratio doubles its decibel value, which is why voltage and pressure use 20 lg.

Ratio	1.12	1.26	1.41	2	3.16	4	10	100	1000
As a power ratio, $10 \lg$	0.5 dB	1.0 dB	1.5 dB	3.0 dB	5.0 dB	6.0 dB	10 dB	20 dB	30 dB
As a voltage or pressure ratio, $20 \lg$	1.0 dB	2.0 dB	3.0 dB	6.0 dB	10 dB	12 dB	20 dB	40 dB	60 dB

The values worth memorising are $\lg 2 \approx 0.301$, $\lg 3 \approx 0.477$ and $\lg 5 \approx 0.699$. From them follow "double the power is +3 dB", "double the voltage is +6 dB" and "ten times the power is +10 dB". Nine watts against two watts is $10 \lg 4.5 = 6.5$ dB, which a quick estimate confirms (2 → 4 → 8 W is two doublings, 6 dB, and the ninth watt adds a little).

What a decibel is

A decibel expresses the ratio of two quantities. For powers it is

$$L = 10 \lg \frac{P_1}{P_0} \text{ dB},$$

and for field quantities whose square is proportional to power, such as voltage, current and sound pressure, it is

$$L = 20 \lg \frac{V_1}{V_0} \text{ dB}.$$

The factor 20 is not a second definition. Power into a fixed resistance is $P = V^2/R$, so a tenfold voltage gives a hundredfold power, and both are 20 dB. The two formulas give the

same number of decibels for the same physical change, as long as the impedance stays the same.

A decibel figure without a reference is a **relative** level, a difference or a gain. "The fader is at −10 dB" or "the channel has 20 dB of headroom" are relative statements and complete as they stand. A decibel figure with a fixed reference is an **absolute** level, and the reference must be named. "The console clips at +20 dB" says nothing. "The console clips at +20 dBu" says it delivers 7.75 V. The suffix after "dB" carries the reference, and confusing the suffixes is the most common way to misread a data sheet.

Electrical levels

Unit	Reference	Measures	Typical use
dBu	0.7746 V	voltage, any impedance	professional line and mic levels
dBV	1 V	voltage, any impedance	consumer line level, microphone sensitivity
dBm	1 mW	power into a stated impedance	legacy, telephone lines, RF
dBW	1 W	power	amplifier output, sometimes
dBFS	digital full scale	digital value	converters, digital consoles and files

dBu is the professional voltage unit. Its odd reference of 0.7746 V is the voltage that dissipates 1 mW in 600 Ω, a legacy of the telephone lines on which early studio and broadcast systems ran. The "u" stood for "unloaded", meaning the voltage is quoted regardless of load. Modern line inputs have impedances of 10 kΩ or more and outputs of a few tens of ohms, so a source delivers the same voltage into any reasonable input, and a dBu figure stays valid whatever is connected.

dBV uses 1 V as reference. It is the consumer line-level unit and the unit for microphone sensitivity (chapter 9). The conversion is fixed.

$$L_{\mathrm{dBu}} = L_{\mathrm{dBV}} + 2.22\ \mathrm{dB}$$

dBm is a power unit and only equals dBu when the load is 600 Ω. It belongs to the era of 600 Ω matched lines and appears today in older documentation, in RF and in some broadcast contexts. A line output "rated +20 dBm into 600 Ω" delivers 7.75 V. The same output into a 10 kΩ input delivers about the same voltage but far less power, so its dBm figure changes with the load while its dBu figure does not. That is why voltage units replaced it.

dBW is a power unit for amplifiers. A 100 W amplifier delivers 20 dBW, a 1 kW one 30 dBW.

dBu	−60	−40	−20	−10	0	+4	+10	+18	+20	+24
Voltage (V rms)	0.00077	0.0077	0.077	0.245	0.775	1.23	2.45	6.15	7.75	12.3
dBV	−62.2	−42.2	−22.2	−12.2	−2.2	+1.8	+7.8	+15.8	+17.8	+21.8

Nominal levels and the three classes

Equipment is designed around a **nominal** operating level, the average level the circuits expect, with headroom above it for peaks (chapter 4).

- **Mic level** is anything from the noise floor to around −20 dBu. A dynamic microphone with 2 mV/Pa delivers about −52 dBu at 94 dB SPL and about −16 dBu at 130 dB SPL. A condenser with 20 mV/Pa reaches +4 dBu at 130 dB SPL, which is why condenser microphones in kick drums and guitar cabinets need pads.
- **Line level** runs from about −20 dBu to the clip point of the equipment, typically +20 to +28 dBu. Two nominal standards coexist. Professional equipment uses **+4 dBu** (1.23 V). Consumer and much DJ equipment uses **−10 dBV** (0.316 V).
- **Speaker level** is the output of power amplifiers, from a few volts to well over 100 V (chapter 13).

The difference between the two line standards is

$$20 \lg \frac{1.228 \text{ V}}{0.316 \text{ V}} = 11.78 \text{ dB}.$$

A consumer device plugged into a +4 dBu input arrives about 12 dB low and needs more gain, which costs noise. A +4 dBu output plugged into a −10 dBV input arrives about 12 dB hot, and since consumer inputs often clip at around +4 to +8 dBu, a professional output driven to its normal peaks will overload them. USB interfaces, DJ mixers and playback devices often have switchable or poorly documented levels, so check the clip point of both sides rather than the label.

Voltage gain and power gain

Gain without a qualifier is ambiguous. A power amplifier fed 1.23 V into its 10 kΩ input and delivering 1.23 V into an 8 Ω loudspeaker has a voltage gain of 0 dB and a power gain of 31 dB, because the input draws 0.15 mW and the loudspeaker 189 mW. In a modern system nearly every gain is a voltage gain, and every stage except the power amplifier is designed to deliver voltage into a light load. This book means voltage gain whenever it says gain.

Three words are often confused, volume, level and gain. "Level" is a quantity relative to a stated reference, "gain" is a change of level through a stage, and "volume" is a colloquial word best avoided in technical writing.

Digital levels

A digital signal is a sequence of numbers with a largest possible value, **full scale**. Levels are given in **dBFS**, decibels relative to full scale, so every real digital level is negative and 0 dBFS is the ceiling. AES17 defines full scale by a sine wave whose peaks just reach the largest digital value. Under that definition a full-scale sine reads 0 dBFS, and a square wave touching the same peaks would read +3 dBFS.

How a digital level corresponds to an analog voltage is a design choice of each converter, called **alignment**. Two conventions are common.

Alignment	0 dBu corresponds to	+4 dBu corresponds to	0 dBFS corresponds to
EBU R68	−18 dBFS	−14 dBFS	+18 dBu (6.15 V)
SMPTE RP 155	−24 dBFS	−20 dBFS	+24 dBu (12.3 V)

Live consoles and stageboxes do not follow either strictly. Many map 0 dBFS to somewhere between +18 and +24 dBu at their analog outputs, and some let the user choose. The practical rule is to look up the maximum output level of every analog output and the maximum input level of the device it feeds, and to match them (chapter 4). The SdWa5 rig's measurements showed that a processor's nominal ceiling and its real one can differ by several decibels.

Peak versus true peak. A digital meter that reads sample values can miss peaks that occur between samples and appear after reconstruction in the DAC. A **true-peak** meter oversamples to estimate them. The difference is usually below 1 dB, but heavily limited material can exceed 0 dBFS true peak while every sample stays below it.

Acoustic levels

Sound pressure level uses 20 µPa as reference, roughly the threshold of hearing at 1 kHz.

$$L_p = 20 \lg \frac{p}{20\ \mu\mathrm{Pa}}\ \mathrm{dB\ SPL}$$

A pressure of 1 Pa is 94 dB SPL, which is why microphone sensitivities and acoustic calibrators use 94 dB.

Situation	Approximate level
Quiet bedroom at night	25–30 dB(A)
Conversation at 1 m	55–65 dB(A)
Busy street at the kerb	75–85 dB(A)
Loud club, dance floor	95–105 dB(A)
Front of a rock concert	105–110 dB(A)
Kick drum, microphone inside the shell	140 dB SPL peak and above

Sound power level describes how much acoustic power a source radiates in all directions, with 1 pW as reference.

$$L_W = 10 \lg \frac{W}{1\ \mathrm{pW}}\ \mathrm{dB}$$

One acoustic watt is 120 dB re 1 pW. Sound power is a property of the source, while sound pressure depends on where it is measured. Noise regulations for machines and the prediction of levels in rooms use sound power (chapters 6 and 7). The acoustic power a loudspeaker radiates is a small fraction of the electrical power it receives, typically a few percent for direct-radiating cones and somewhat more for horn-loaded designs.

Adding levels

Levels in decibels cannot be added directly. Incoherent sources, such as two different instruments or a band and the crowd, add their powers.

$$L_{\mathrm{sum}} = 10 \lg \left(10^{L_1/10} + 10^{L_2/10} + \dots \right)$$

Difference between two levels	0 dB	1 dB	2 dB	3 dB	4 dB	6 dB	10 dB
Add to the louder level	3.0 dB	2.5 dB	2.1 dB	1.8 dB	1.5 dB	1.0 dB	0.4 dB

Two sources of 90 dB give 93 dB. A source 10 dB quieter than another hardly changes the total. Four equal sources give +6 dB. Coherent sources, such as two loudspeakers playing the same signal, can give up to +6 dB for two, or a notch, depending on phase (chapter 1).

RMS, peak and crest factor

The **RMS** (root mean square) value of a signal is the square root of the mean of its squared instantaneous values. It is the value that produces the same heating in a resistor as a DC voltage of the same size, and it tracks the energy of a signal and roughly its loudness. The **peak** value is the largest instantaneous value. Their ratio in decibels is the **crest factor**.

Signal	Crest factor
Square wave	0 dB
Sine wave	3.01 dB
Pink noise, unfiltered	about 12 dB
Heavily limited modern master	about 6–10 dB
Speech, live drums, unprocessed acoustic music	15–20 dB and more

The crest factor of the programme decides how much of a system's peak capability is left for its average level, which is the subject of chapter 4. It also explains why the phrase "RMS power" on amplifiers is a misnomer. What is meant is the average power of a continuous sine, computed from the RMS voltage. Chapter 13 handles amplifier ratings.

Meters differ in what they show. A **peak** meter shows the highest instantaneous value and warns of clipping. An **RMS** or VU-style meter averages over a few hundred milliseconds and follows loudness. A digital console meter usually shows sample peaks in dBFS, and the gain structure of the console is set by it.

Weighting and time constants

A sound level meter does not hear like a person, so it applies frequency weightings and time constants defined in IEC 61672-1.

- **A-weighting** follows the ear's sensitivity at low levels, cutting about 30 dB at 50 Hz and 50 dB at 20 Hz. It is the weighting of most noise regulations and hearing-protection rules. It underestimates bass-heavy music at concert levels.
- **C-weighting** is nearly flat from about 30 Hz to 8 kHz and is used for loud music, for low-frequency complaints and for peak measurements.
- **Z-weighting** is flat across the measurement range and replaces the older "linear" setting.
- B and D weightings are obsolete.

The time constants are **F** (fast, 125 ms) and **S** (slow, 1 s). The quantities that matter for events are averages over time.

- $L_{\mathrm{Aeq},T}$ is the A-weighted energy average over a period T, such as 15 minutes or 60 minutes. Limits for audience exposure and for neighbours are usually given this way (chapters 5 and 6).
- $L_{\mathrm{Ceq},T}$ is the same with C-weighting, increasingly used to limit bass.
- L_{Cpeak} is the highest C-weighted instantaneous peak, used in hearing-protection rules.

Writing a level as "dB(A)" names the weighting. A level written as "dB" alone is unweighted or ambiguous, and an SPL figure in a specification without weighting and time constant is incomplete.

Loudness

Loudness is what a listener perceives, and it is not the same as sound pressure level. The ear is most sensitive between about 2 and 5 kHz, largely because of the resonance of the ear canal, and much less sensitive at low and very high frequencies, especially at low levels. **Equal-loudness contours** show which levels at each frequency sound as loud as a 1 kHz tone. The original curves by Fletcher and Munson (1933) and Robinson and Dadson (1956) have been replaced by ISO 226, last revised in 2023.

The **phon** names a contour by the level of the 1 kHz tone on it. A 60 phon sound is as loud as 60 dB SPL at 1 kHz, whatever its frequency. Above about 40 phon, a rise of 10 phon is heard as roughly twice as loud. The **sone** scale expresses that directly, with 1 sone at 40 phon.

Loudness level (phon)	40	50	60	70	80	90	100	110
Loudness (sone)	1	2	4	8	16	32	64	128

This gives the rules of thumb that every engineer uses. A change of 1 dB is barely noticeable on music, 3 dB is easily audible, and 10 dB sounds about twice as loud. Doubling the amplifier power gives 3 dB, so twice the perceived loudness needs ten times the power.

The contours flatten at higher levels. At concert levels the ear hears bass and treble nearly as strongly as the midrange, while at background levels the same mix sounds thin. A system set up for a quiet dinner may need more low and high end than the same system at a party, which is what the loudness button of hi-fi amplifiers was for. Current DSPs can apply level-dependent loudness compensation based on ISO 226 automatically.

Programme loudness is measured in LUFS (loudness units relative to full scale) according to ITU-R BS.1770, with EBU R 128 setting −23 LUFS as the broadcast target in Europe and streaming services normalising to targets around −14 to −16 LUFS. LUFS matter in live sound wherever programme is exchanged with broadcast or streaming. A livestream feed, a playback file for an event or a pre-show playlist should be loudness-normalised so that the first track does not arrive 8 dB louder than the walk-in music. LUFS do not replace dB SPL in the room, because they measure the signal, not the sound.

To verify in pass 2

- AES17 definition of full scale and the resulting +3 dBFS for a full-scale square wave.

- EBU R68 alignment of −18 dBFS for 0 dBu, and SMPTE RP 155 alignment of −20 dBFS for +4 dBu, against the current editions.
- Typical clip levels of consumer line inputs (+4 to +8 dBu).
- The SPL examples in the table of situations; check against a noise reference and the WHO and DIN 15905-5 material used in chapter 5.
- Loudspeaker efficiency of a few percent for direct radiators; check against a loudspeaker reference (chapter 14).
- Crest factors of pink noise, limited masters and live programme; check against AES2-2012 and published programme analyses.
- IEC 61672-1 weightings (A-weighting −30.2 dB at 50 Hz and −50.5 dB at 20 Hz), time constants and the status of B and D weightings.
- ISO 226:2023 as the current edition of the equal-loudness contours.
- ITU-R BS.1770 and EBU R 128 targets, and the streaming normalisation range of −14 to −16 LUFS.

Sources

- AES17, measurement of digital audio equipment.
- EBU R68, alignment level in digital audio production equipment.
- SMPTE RP 155, reference levels for analog and digital audio systems.
- IEC 61672-1, sound level meters.
- ISO 226, normal equal-loudness-level contours.
- ITU-R BS.1770, algorithms to measure audio programme loudness and true-peak audio level.
- EBU R 128, loudness normalisation and permitted maximum level of audio signals.
- Davis, G., Jones, R.: *The Sound Reinforcement Handbook*, 2nd ed., Yamaha / Hal Leonard 1989, §3 (pp. 19–32), §8.7 (pp. 92–96), Appendix A (pp. 409–412).

4. Dynamic range, headroom and gain structure

> **Draft, pass 1.** This chapter was written from general knowledge and checked against reference literature for coverage. Standards, editions and figures are named but have not yet been verified against their sources. The list *To verify in pass 2* at the end names every claim that still needs one.

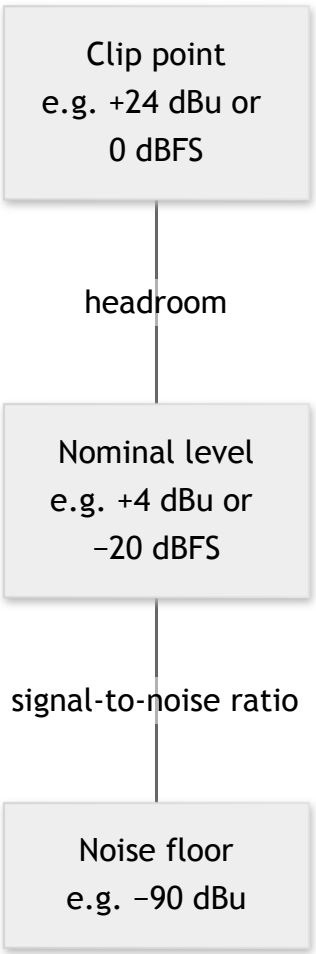
This chapter answers how loud and how quiet a system can be at the same time, where its limits sit, and how to set the gain of every stage so the programme stays between the noise floor and the clip point. It defines dynamic range, headroom and signal-to-noise ratio, works through a gain chain from microphone to converter, shows how the outputs of a console, a processor and an amplifier are matched, and explains how compression fits a wide programme into a narrower system.

Three ranges

Dynamic range is the distance in decibels between the loudest and the quietest part of something. For a programme it is the distance from its highest peak to its quietest passage, or to the noise around it if that noise masks the quietest passage. For a system it is the distance from its maximum output, where it clips, to its noise floor.

Headroom is the distance from the nominal level, where the programme sits most of the time, to the clip point. It is the reserve for peaks.

Signal-to-noise ratio (S/N) is the distance from the nominal level down to the noise floor.



All three are differences, so they are given in plain dB, never in dBu or dB SPL. A system with a nominal level of +4 dBu, a clip point of +24 dBu and a noise floor of −90 dBu has 20 dB of headroom, 94 dB of S/N and a dynamic range of 114 dB.

Headroom plus S/N equals the dynamic range as defined here. One subtlety remains. A listener can follow structured programme such as a voice or a tone several decibels below broadband noise, so the usable range of a system is slightly larger than its measured one. Measuring from the noise floor is the safer and repeatable convention.

The programme and the system

A loud concert can span 85 dB at the microphone, from 40 dB SPL during a pause to 125 dB SPL on a shouted peak. In a linear chain, with no compression or limiting, every stage reproduces those 85 dB. The console output swings 85 dB in voltage, the amplifier 85 dB in power, and the loudspeakers 85 dB in sound pressure. A decibel of change at the microphone is a decibel of change everywhere, whether the stage is described in volts, watts or pascals.

A chain is only as wide as its narrowest stage. If one stage offers 74 dB between its noise and its clip point, an 85 dB programme loses 11 dB somewhere. The loss shows up as clipped peaks, as quiet passages buried in hiss, or as both. Since the 1980s the narrow stages have moved. Analog consoles, tape and the summing amplifiers that limited the systems of that time have given way to converters with well over 100 dB of range and

floating-point mixing that does not clip internally. In a modern rig the narrowest stages are usually

- the **room**, where the audience noise sets a floor of 60 to 80 dB(A) at a busy event,
- the **loudspeakers and amplifiers**, whose maximum output sets the ceiling, and
- the **limits** for audience exposure and for neighbours (chapters 5 and 6), which set a ceiling below the equipment's own.

Every extra 3 dB of maximum level costs twice the amplifiers and loudspeakers, and that is the main reason to manage the programme's range rather than build a system wide enough for everything.

How much headroom

The headroom a system needs is set by the crest factor of its programme, the distance from its average to its peaks (chapter 3).

Programme	Typical crest factor	Headroom to plan for
Paging, announcements, compressed speech	6–10 dB	about 10 dB
Modern mastered pop and electronic music	6–10 dB	about 10–12 dB
Live band, mixed with moderate compression	12–15 dB	about 15 dB
Acoustic ensembles, orchestra, theatre	15–20 dB and more	about 20 dB

These are planning figures, not limits. A system that plays electronic music at a nominal 100 dB(A) needs peak capability around 110 to 112 dB at the same position, while a chamber orchestra at a nominal 85 dB(A) needs peaks of 105 dB. The same equipment can serve both, because the orchestra needs a lower average. More headroom does not always mean more amplifiers.

The headroom is usually the first thing given away when a system is too small. Running out of headroom does not announce itself on an average-reading meter. It shows as limiters working continuously, as a compressed and fatiguing sound, and on the amplifier as clip indicators flashing on every kick drum.

Gain structure in a digital chain

Gain structure means setting each stage so the programme uses its range well, far enough above the noise and far enough below the clip point. The classic rule is to take as much gain as possible at the first stage and to avoid making up level later. In a digital system the rule becomes more precise. **The preamp gain is the one decision that sets both the noise floor and the headroom of a channel.** Everything after the converter is computed with so much range that it neither adds noise nor clips inside the mixing engine, until the signal is converted back or leaves on a fixed-point output.

A worked example

A vocalist sings into a dynamic microphone with a sensitivity of 2 mV/Pa. The loudest peaks reach 125 dB SPL at the capsule, and the quietest phrase is 45 dB SPL. The

stagebox has a preamp with an equivalent input noise of −127 dBu and a converter with a noise floor of −112 dBFS, aligned so that 0 dBFS equals +18 dBu.

Quantity	Value
Microphone output on peaks (125 dB SPL)	−20.8 dBu
Microphone output on the quiet phrase (45 dB SPL)	−100.8 dBu
Target for peaks at the converter	−12 dBFS, which is +6 dBu
Preamp gain needed	26.8 dB
Preamp noise after gain	−100.2 dBu, which is −118.2 dBFS
Total noise with the converter's −112 dBFS	−111.1 dBFS
Quiet phrase at the converter	−92.0 dBFS, 19 dB above the noise

With the gain set 15 dB lower, "to be safe", the peaks reach only −27 dBFS. The preamp's noise falls to −133 dBFS, but the converter's −112 dBFS stays where it is, so the total noise barely moves. The quiet phrase arrives at −107 dBFS, now only 5 dB above the noise. The 15 dB of gain that was given away at the preamp cannot be recovered later, because every fader and every make-up gain afterwards raises noise and signal together.

With the gain set 15 dB higher, the peaks hit +3 dBFS and clip at the converter. No fader afterwards removes that distortion.

The example shows the shape of the decision. Up to the point where the preamp's own noise dominates, more gain is always better. Beyond the point where peaks approach full scale, more gain is destructive. Between the two there is a window of 15 to 20 dB, and the target of peaks around −12 to −18 dBFS keeps a source in it while leaving room for a singer who gets louder in the final chorus.

Setting the gain in practice

1. Put the channel fader and the master at their nominal position, 0 dB on a digital console.
2. Ask for the loudest realistic performance, not a polite soundcheck level. Drummers and singers play louder in the show than in the check.
3. Raise the preamp gain until the channel meter peaks around −12 to −18 dBFS, depending on how predictable the source is.
4. Engage a pad only if the gain at its minimum still lets the source clip. A pad is an attenuator before the preamp, so it reduces signal and preamp noise alike, but the converter noise then sits closer to the signal. With condenser microphones in kick drums or guitar cabinets a pad is often necessary (chapter 3).
5. Mix with the faders. If a channel fader ends up far below 0 dB for the whole show, the gain is set too high, and if it is at the top, the gain is too low.

Split systems and gain sharing

When the front-of-house and monitor consoles share one stagebox, one preamp gain serves both. Whoever changes it changes the other console's level too. Digital systems solve this with **gain compensation** or gain sharing. One console owns the analog gain, and the other receives the signal with a digital trim that automatically offsets every change of the analog gain, so its level stays constant. The same principle applies to a

broadcast or recording split taken from the same preamps. Chapter 11 covers splits in detail.

Floating point inside, fixed point at the edges

Consoles and processors compute internally in 32-bit or 64-bit floating point, which has a range of more than 1500 dB. A bus that is driven 20 dB over 0 dBFS inside the engine does not clip, as long as the level is brought down before it leaves. Analog summing amplifiers clipped when too many channels were pushed together, and much of the older gain-structure advice addressed that. In a digital console the risk has moved to the outputs. A DAC, an AES output, a network output and the input of the next device are fixed-point and clip at their full scale.

Recorders that write 32-bit floating-point files have the same property for recording. A take that goes over full scale can be turned down afterwards without distortion, provided the converters themselves did not clip. Some recorders use two converters at different gains for this purpose. This makes recording safer, but it does not change the gain structure of the live chain, where a fixed-point output still clips.

Matching the stages after the console

After the console the chain is console output, loudspeaker processor, power amplifier and loudspeaker. Each stage has a maximum level, and the goal is that each one reaches its maximum at about the same time, so no stage clips early and none wastes range.

- The **console output** has a maximum analog level, such as +18 or +24 dBu.
- The **processor input** has a maximum level that should be at least as high as the console's output.
- The **processor output** has a maximum level and, in every serious system, a limiter per band.
- The **amplifier** has a voltage gain or an input sensitivity, the input level that drives it to full output.
- The **loudspeaker** has a maximum voltage it survives, derived from its power rating and impedance.

A typical mismatch looks like this. A processor that delivers +18 dBu (6.15 V) feeds an amplifier with 32 dB of gain. At full processor output the amplifier would try to produce 245 V. The amplifier can only deliver 89 V into 4 Ω, its 2000 W rating. It reaches full output with a processor level of +9.2 dBu, so the top 8.8 dB of the processor's range drive the amplifier into clipping. The limiter in the processor must therefore act at or below +9.2 dBu, and lower still if the loudspeaker cannot take the amplifier's full output.

The arithmetic that turns a loudspeaker's rating, an amplifier's gain and a processor's ceiling into a limiter threshold is four lines long. It is written out, with the reasons for its two common mistakes, in the SdWa5 rig documentation under Why 32 dB and 26 dB. Two of its conclusions apply to any rig.

- **The lowest amplifier gain that leaves a safe margin is the right one.** A higher gain moves the limiter threshold down one for one and wastes the processor's range near its noise floor. The amplifier is not louder for it.

- **Paralleling loudspeakers on one amplifier channel does not change the threshold**, because each box still sees the same voltage. The impedance and power in the calculation are always those of one box.

When an amplifier has no input attenuator, the level reduction belongs in the processor's output gain or the amplifier's gain switch, never in a console master pulled far down. A master fader at −30 dB throws away 30 dB of the console output's range and brings its noise closer.

Compression and limiting as range management

When the programme is wider than the system, or wider than the room and the limits allow, some of its range must be removed deliberately. A **compressor** above its threshold changes the output by less than the input. At a ratio of 2:1, every 2 dB of input above the threshold become 1 dB of output. A **limiter** is a compressor with a high ratio, 10:1 to ∞:1, that holds peaks at or near the threshold.

One way is to compress a whole programme. Fitting 90 dB into 74 dB needs a ratio of about 1.22:1 across the full range, a gentle setting that keeps most of the music's movement. A more common approach is to leave everything below a threshold untouched and compress only above it. A limiter with a threshold of +15 dBu and a ratio of 10:1 lets an input of +25 dBu out at +16 dBu.

In live sound three layers of dynamics control are usual.

1. **Channel compressors** on voices and some instruments keep each source in its window.
2. **Bus or master compression** keeps the mix consistent and controls the crest factor the system has to carry.
3. **Loudspeaker limiters** in the processor or the amplifier protect the drivers and the amplifiers. They should almost never act audibly. When they work continuously, the system is too small for the level asked of it.

Chapter 12 covers the processors and their settings. Tape noise reduction worked the same way. It compressed the signal before the tape and expanded it after. Tape has left the live chain. Compander circuits survive in analog wireless systems (chapter 10).

In the SdWa5 rig

The rig's limiter table, the choice of 32 dB on the MM14K amplifiers and 26 dB on the TIP10000q, and the bench measurement that showed the DSP 8x8 clipping at about +12.5 rather than its documented 18 are in Signal chain. That section is the worked example for the matching rules above, measured on real equipment.

To verify in pass 2

- The equivalent input noise of −127 dBu and the converter noise floor of −112 dBFS used in the worked example; check against current stagebox data sheets.
- The gain target of peaks between −12 and −18 dBFS; check against console manufacturers' guidance and AES or EBU recommendations.

- The crest factors and headroom figures in the programme table; check against AES2-2012, IEC 60268-1 programme simulation noise and published loudness analyses.
- The internal floating-point formats of current consoles and processors (32-bit or 64-bit).
- The behaviour of 32-bit floating-point recorders with dual converters; check manufacturer documentation.
- Typical audience noise of 60 to 80 dB(A) at busy events; check against measurement literature.

Sources

- AES17, measurement of digital audio equipment.
- AES2-2012, specification of loudspeaker components used in professional audio and sound reinforcement.
- SdWa5: Signal chain, sections *The table is generated, not copied* and *Why 32 dB and 26 dB*.
- Davis, G., Jones, R.: *The Sound Reinforcement Handbook*, 2nd ed., Yamaha / Hal Leonard 1989, §4 (pp. 33–42) and §11.7 (pp. 172–175).

5. Hearing

> **Draft, pass 1.** This chapter was written from general knowledge and checked against reference literature for coverage. Standards, editions and figures are named but have not yet been verified against their sources. The list *To verify in pass 2* at the end names every claim that still needs one.

This chapter answers what the ear can hear, how it hides one sound behind another, how it decides where a sound comes from, why a measurement and a listener can disagree about the same system, and how loud an event may be before it harms the people in it. The answer to the last question has changed most in recent decades. Older sound reinforcement literature advised caution above about 95 dB(C) and left the danger levels to others, while today an EU directive protects the crew and a WHO standard and national norms address the audience. The numbers behind decibels, weighting and loudness are in chapter 3.

Range and sensitivity

A healthy young ear hears from about 20 Hz to about 20 kHz. The upper limit falls with age, and by fifty it often lies between 12 and 16 kHz. Noise exposure takes away the range above 3 kHz first, with a typical notch near 4 kHz. Below 20 Hz, sound is felt in the body rather than heard as a tone, which matters for subwoofers at high level.

The reference for sound pressure level, 20 µPa, sits close to the threshold of hearing at 1 kHz. The ear is most sensitive between about 2 and 5 kHz, where the threshold drops a few decibels below 0 dB SPL. Two reasons are mechanical. The ear canal is a tube of about 25 mm that is closed at the eardrum, and a quarter-wave resonance of such a tube lies near $343/(4 \times 0.025) \approx 3.4$ kHz. The eardrum and the middle ear are small and light, so they couple poorly to long wavelengths at the bottom of the range and lose motion at the top.

Sensitivity also depends on level. The equal-loudness contours of ISO 226:2023 show that the ear needs much more pressure at 50 Hz than at 1 kHz to hear the same loudness at a quiet level, and that the contours flatten as the level rises. A mix balanced loud sounds thin when it is played quietly, and a mix balanced quietly sounds boomy when it is played loud. Chapter 3 explains the phon, the sone and the A- and C-weightings that follow from this.

Pressure	Level	What it means
20 µPa	0 dB SPL	Reference, close to the threshold at 1 kHz
0.36 Pa	85 dB SPL	Upper action value for an 8-hour working day (A-weighted)
1 Pa	94 dB SPL	Calibrator level
2 Pa	100 dB SPL	WHO limit for an audience over 15 minutes (A-weighted)
20 Pa	120 dB SPL	Discomfort for most listeners
112 Pa	135 dB SPL	Lower action value for peaks (C-weighted)
200 Pa	140 dB SPL	Exposure limit for peaks (C-weighted), threshold of pain

The smallest level change a listener notices in a direct comparison is about 1 dB for broadband sound. Over minutes, without a direct comparison, a few decibels go unnoticed, which is why levels creep upwards during a show.

Masking and critical bands

A loud sound makes a quieter one inaudible when the two are close in frequency or in time. This is masking, and it explains much of what a mix engineer does with an equalizer.

The inner ear analyses sound in overlapping bands, the critical bands. Two components inside one band compete, and the louder one masks the quieter. Components in different bands are heard separately. One common model of the band width is the equivalent rectangular bandwidth of Glasberg and Moore, $\mathrm{ERB} = 24.7\,(4.37 f/1000 + 1)$ Hz.

Centre	ERB	Relative width
100 Hz	35.5 Hz	35.5 %
500 Hz	78.7 Hz	15.7 %
1 kHz	132.6 Hz	13.3 %
4 kHz	456.5 Hz	11.4 %
10 kHz	1104.1 Hz	11.0 %

A third-octave band is 23.2 % wide, so above a few hundred hertz the ear's bands are about half as wide as the bands of a third-octave equalizer. Below a few hundred hertz they are wider in relative terms.

Masking spreads further upwards in frequency than downwards, and the spread grows with level. A loud bass line therefore masks the lower midrange of a vocal more than a loud vocal masks the bass. Masking also works in time. A loud sound hides quieter sounds that follow it for some tens of milliseconds, and it hides sounds just before it for a few milliseconds.

Three examples show where masking fails.

- **Distortion is not masked by other distortion of a different kind.** A loudspeaker with 1 % of mostly second and third harmonics does not hide an amplifier's much smaller share of high odd harmonics. Higher and odd harmonics are more audible than low and even ones, so a total figure such as THD says little about what is heard.
- **Structured noise is heard below random noise.** The quantization noise of an early digital delay, for example, was audible through the hiss of an analog equalizer. Today the remedy is dither, which turns the structured error into a smooth noise, and converters with enough bits that the error lies far below any analog noise.
- **Noise that rises with the signal is hidden only near the signal.** Modulation noise accompanies a note and is masked within about an octave of it, but a strong low note can let higher noise sidebands through.

Perceptual audio codecs such as MP3, AAC and Opus remove what masking hides, and they are the best-known application of masking today. The same principle sets a limit for live work. An instrument that is masked by another in the same band cannot be made

audible by raising its fader alone without raising the whole band. A cut in the masking instrument, or a different arrangement, usually works better.

Localisation and the precedence effect

Two ears give the brain two cues for direction. The interaural time difference is the delay between the ears, which reaches about 0.66 ms for a source at the side of a head with a radius of 87.5 mm. The interaural level difference comes from the shadow of the head and matters above about 1.5 kHz, where the head is large compared with the wavelength. The outer ear filters sound differently for each direction, which gives the brain cues for elevation and for front and back.

When the same sound arrives twice from two directions, the brain does not hear two sources if the second arrival is close enough in time. This is the precedence effect, also called the law of the first wavefront or the Haas effect.

Delay of the second arrival	What a listener hears
Below about 1 ms	One source between the two, shifted towards the earlier one (summing localisation, as in stereo)
About 1 to 30 ms	One source at the first arrival. The second adds loudness and colour but no direction, even if it is somewhat louder
Above about 30 to 50 ms	The second arrival is heard as an echo

The window depends on the signal. For clicks it closes after a few milliseconds, and for speech and music it stays open for tens of milliseconds. A delayed arrival of 10 ms corresponds to 3.43 m of path difference, and 30 ms to 10.3 m.

Sound reinforcement uses the effect to keep the image on the stage. A delay loudspeaker halfway down a hall is delayed so that its sound arrives a little after the sound from the main system, and listeners near it then localise the source at the stage while hearing the level of the delay. Chapter 20 sets the delay times. The same effect also explains why a reflection from a wall does not shift the image, and why one at 60 ms or more is heard as an echo.

Measurement and listening

Every engineer meets the question why a system that measures well can sound wrong and the reverse. There are several reasons, and the tools have answered some of them.

- **A microphone is one point, and a listener is two moving ones.** At one fixed point, reflections add and cancel into a comb of peaks and notches. A shift of a few centimetres changes the comb at high frequencies. Two ears and a head that moves let the brain form an average of the field around the head, and that average is what the listener calls the sound of the room.
- **An amplitude-only analyser misses time.** The real-time analysers of the 1980s showed level per third-octave band from pink noise and nothing else. A correction made from it could match two loudspeakers in level and leave them sounding different, because the equalizers changed phase as well. Today's dual-channel FFT

analysers, such as Smaart, REW and Open Sound Meter, measure the transfer function with magnitude, phase and coherence, which answers much of this objection. Chapter 16 covers them.
- **One test examines one property, the ear all at once.** A sine sweep shows response, a THD test shows harmonics, and the ear judges both together with the time behaviour of the system.
- **Static signals hide dynamic faults.** A circuit can ring on a fast transient, and an amplifier can misbehave into a reactive loudspeaker load that a resistor on the bench does not present. Bursts and music-like test signals reveal what steady tones do not. Chapters 8 and 13 return to this.
- **Analysers once had less range than the ear.** Analysers of that time showed 30 to 60 dB on screen against about 120 dB for the ear. A current FFT analyser behind a 24-bit interface shows over 100 dB, so this reason has mostly gone.

Two practices follow. A measurement for equalization is taken at several positions across the audience and averaged by power, never by adding microphones electrically, because an electrical sum makes its own comb. A spatial average of five to eight positions shows the trend that the audience hears, while a single position shows that seat only. The second practice is a listening test after every correction, with known programme material, because the analyser answers only the question it was asked.

Listening tests that mean something

A listener can be trained, and an experienced engineer does hear faults that others miss. Such a listener deserves the benefit of the doubt, and what sounds good is partly a matter of taste. One more finding has to be added. A listener who knows which device is playing is influenced by that knowledge, and a louder version of the same sound is usually judged better.

A fair comparison therefore has three properties.

1. **Matched levels.** The two versions are matched within about 0.1 dB, measured with a meter, because a difference of a fraction of a decibel is heard as a difference in quality rather than in level.
2. **Blind.** The listener does not know which version plays, and in an ABX test has to identify an unknown X as A or B often enough that chance is unlikely.
3. **Same conditions.** The same seat, the same programme and quick switching, because auditory memory for detail fades within seconds.

ITU-R BS.1116 describes formal tests for small impairments, and ITU-R BS.1534 (MUSHRA) for intermediate ones. A sound check cannot follow either, but the three properties above can be kept with a level meter and a helper.

How hearing is damaged

Sound damages the hair cells of the inner ear. A loud exposure first causes a temporary threshold shift. Sounds are duller and quieter after the event, often with ringing in the ears, and hearing returns over hours to days. Repeated exposure turns the shift into a permanent one, and the hair cells do not grow back. Tinnitus, a sound heard without a source, can stay after a single exposure or develop over years. Research since about 2009

suggests that an exposure whose threshold shift recovers fully can still damage the nerve connections of the inner ear, which impairs hearing in noise. The damage therefore starts before a hearing test shows it.

The risk depends on level and time together. Most limits use the equal-energy rule, under which 3 dB more level halves the allowed time. The rule turns any exposure into a daily dose, so that 15 minutes at 100 dB(A) carry the same energy as 8 hours at 85 dB(A).

Level, A-weighted	Time with the same dose as 85 dB(A) for 8 h
85 dB	8 h
88 dB	4 h
91 dB	2 h
94 dB	1 h
97 dB	30 min
100 dB	15 min
103 dB	7.5 min
106 dB	3.8 min

A two-hour show at 100 dB(A) gives a daily exposure level of 94 dB(A), which is eight times the dose of the upper action value. A crew member who stands near the stage for the sound check, the support act and the headliner has a larger dose than anyone in the audience.

Limits for crew and audience

Crew

The EU noise directive 2003/10/EC protects workers, and it covers everyone employed at an event, including sound engineers, bar staff and security. It sets three values for the daily exposure level $L_{EX,8h}$ and for the C-weighted peak.

Value	$L_{EX,8h}$	Peak L_{Cpeak}	What it requires
Lower action value	80 dB(A)	135 dB(C)	Information, training, hearing protection made available
Upper action value	85 dB(A)	137 dB(C)	Hearing protection worn, a programme to reduce exposure, marked areas
Exposure limit	87 dB(A)	140 dB(C)	Must not be exceeded at the ear, hearing protection included

The directive asked for a code of conduct for the music and entertainment sector, and the EU later published non-binding guidance for it. Austria transposed the directive with the ordinance on noise and vibration (VOLV), and Germany with the LärmVibrationsArbSchV. Volunteers who are not employed may fall outside the directive's letter, but its values are a sound measure for anyone who works near a PA.

Audience

No EU law limits the level for an audience. The references in use are these.

- **WHO Global Standard for Safe Listening Venues and Events (2022).** It sets a limit of 100 dB $L_{Aeq,15min}$, measured continuously, together with live monitoring, free hearing protection, quiet zones, information for the audience and trained staff. It is a recommendation, and its value of 100 dB for 15 minutes equals the daily dose of the workers' upper action value.
- **DIN 15905-5 (Germany).** The norm for event technology rates the level over 30 minutes at the loudest audience position. The 2007 edition sets 99 dB(A) for the rating level and 135 dB(C) for peaks, with information and hearing protection required from lower levels. German courts and authorities use it as the state of the art, which makes it the DACH reference.
- **Austria.** No federal norm for audience levels is known to the author. Event permits under the event laws of the federal states can set conditions, and authorities often refer to the WHO standard or to DIN 15905-5.
- **Switzerland.** The ordinance on sound and laser (V-NISSG) sets binding limits for events, graded by level.

A practical policy for an SdWa5 event combines these. The level is logged at the mix position as $L_{Aeq,15min}$ and L_{Cpeak}, the offset to the loudest audience position is measured once during the setup, the target stays below 100 dB(A) over 15 minutes at that position, and earplugs are free at the bar.

Hearing protection

Foam earplugs attenuate a lot, but more at high than at low frequencies, so music through them sounds muffled and many people take them out. Earplugs with a flatter attenuation, sold as musician's earplugs, keep the balance and lower the level by about 10 to 25 dB depending on the filter. Custom-moulded plugs with interchangeable filters fit best and are the usual choice for crew. A filter of 15 dB brings 105 dB(A) at the mix position down to 90 dB(A) at the ear, and one of 25 dB to 80 dB(A). The rated attenuation is a laboratory value per EN 352-2 and the SNR of ISO 4869-2, and a plug worn badly gives much less.

In-ear monitors can protect or harm. A well-sealed earpiece blocks the stage sound, so the monitor mix can be quieter than a wedge mix for the same clarity. The risk is that a performer raises the mix until it masks the remaining leakage and the crowd, and that a feedback squeal or a dropped wireless link reaches the ear at full level. A limiter on every in-ear mix, a seal checked by the performer, ambient microphones instead of one earpiece taken out, and a level that is measured rather than guessed make in-ear monitoring the quieter option.

To verify in pass 2

- The upper frequency limit by age (12 to 16 kHz at fifty); check ISO 7029.
- The just noticeable level difference of about 1 dB; check psychoacoustics literature (Zwicker and Fastl, Moore).
- The ERB formula and constants; check Glasberg and Moore 1990.
- The temporal masking durations (tens of ms forward, a few ms backward).
- The precedence windows (1 ms, 30 to 50 ms) and the level a second arrival may have without shifting the image; check Blauert, *Spatial Hearing*, and Haas 1951.
- The ITD model with a head radius of 87.5 mm (Woodworth).

- The level-matching tolerance of 0.1 dB for comparisons; check ITU-R BS.1116 and AES literature.
- The editions and scope of ITU-R BS.1116 and BS.1534.
- The synaptopathy findings since 2009; check Kujawa and Liberman 2009 and later human studies.
- The values and duties of Directive 2003/10/EC (80/85/87 dB(A), 135/137/140 dB(C)), and the EU code of conduct for music and entertainment.
- The Austrian VOLV (title, BGBl. number) and the German LärmVibrationsArbSchV as transpositions.
- Whether unpaid volunteers of an association fall under the directive or the Austrian ASchG.
- The WHO Global Standard for Safe Listening Venues and Events (2022), its 100 dB $L_{Aeq,15min}$ and its features.
- DIN 15905-5, the current edition, its 99 dB(A) over 30 minutes and 135 dB(C), and its graded duties.
- The Austrian situation for audience levels, including the Salzburg event law and any ÖNORM.
- The Swiss V-NISSG values.
- The attenuation of musician's earplug filters and the standards EN 352-2 and ISO 4869-2.

Sources

- ISO 226:2023, normal equal-loudness-level contours.
- ISO 7029, statistical distribution of hearing thresholds related to age.
- Glasberg, B. R., Moore, B. C. J.: Derivation of auditory filter shapes from notched-noise data, *Hearing Research* 47, 1990.
- Blauert, J.: *Spatial Hearing*, MIT Press.
- ITU-R BS.1116 and ITU-R BS.1534.
- Directive 2003/10/EC on the minimum health and safety requirements regarding the exposure of workers to noise.
- WHO: *Global Standard for Safe Listening Venues and Events*, 2022.
- DIN 15905-5, event technology, sound exposure of the audience.
- EN 352-2 and ISO 4869-2, hearing protectors.
- Davis, G., Jones, R.: *The Sound Reinforcement Handbook*, 2nd ed., Yamaha / Hal Leonard 1989, §9 (pp. 105–112) and §3.6 (pp. 29–32).

6. Sound outdoors

> **Draft, pass 1.** This chapter was written from general knowledge and checked against reference literature for coverage. Standards, editions and figures are named but have not yet been verified against their sources. The list *To verify in pass 2* at the end names every claim that still needs one.

This chapter answers how sound behaves when there are no walls, which is the simplest case and the base for every other. It covers the loss with distance from point and line sources, a level budget from the loudspeaker to the last row, what air, wind and temperature do over longer distances, how far a festival carries into the neighbourhood, and how much gain a microphone can have before the system feeds back. Rooms follow in chapter 7.

Loss with distance

Outdoors, far from buildings and above a ground that absorbs some sound, a loudspeaker works in what is called a free field. Its sound spreads without returning, and the level at a listener depends on the distance alone.

Point source

A source that is small compared with the distance to the listener radiates as a point. Its power spreads over the surface of a sphere, which grows with the square of the radius. The intensity therefore falls with the square of the distance, and the sound pressure with the distance itself.

$$L(r) = L(r_0) - 20 \lg \frac{r}{r_0}$$

Every doubling of the distance costs 6.02 dB, which is usually written as 6 dB. Ten times the distance costs 20 dB. The source's sound power level does not change with distance, because it describes the whole output, while the sound pressure level describes one place in the field. Chapter 3 explains the difference.

Line source

A source that is long compared with the distance to the listener, such as a tall line array seen from close by, radiates a cylinder rather than a sphere. The surface grows with the radius alone, and the level falls by 3 dB per doubling of the distance. No real line is infinite, so this holds only in the near field of the array. Beyond a transition distance the array behaves like a point source again and loses 6 dB per doubling.

A common estimate of the transition distance is $r_t \approx H^2 f / (2c)$ for an array of height H at frequency f. It grows with frequency, so an array is a line source for the treble and a point source for the bass at the same listener.

Array height	100 Hz	1 kHz	4 kHz	10 kHz
2 m	0.6 m	5.8 m	23 m	58 m

Array height	100 Hz	1 kHz	4 kHz	10 kHz
4 m	2.3 m	23 m	93 m	233 m
8 m	9.3 m	93 m	373 m	933 m

The consequence is a tonal balance that shifts with distance, with relatively more treble near a long array than near a point source of the same output. A line array is curved and processed so that its far field is shaped as well, which chapter 14 covers.

Distance from 1 m	Point source	Line source (near field)
2 m	−6.0 dB	−3.0 dB
4 m	−12.0 dB	−6.0 dB
10 m	−20.0 dB	−10.0 dB
25 m	−28.0 dB	−14.0 dB
50 m	−34.0 dB	−17.0 dB
100 m	−40.0 dB	−20.0 dB

A level budget

A level budget starts from the level wanted at the last listener and works back to the loudspeaker.

Say the audience should hear 100 dB(A) averaged over time at 50 m, and the programme has peaks 12 dB above that average. A point source then has to produce 134 dB at 1 m for the average and 146 dB at 1 m for the peaks. At 25 m the same targets need 128 dB and 140 dB. A single box of the kind used at small events has a maximum level of 125 to 135 dB at 1 m, so the budget shows at once whether one box per side is enough or whether the system needs more boxes, a line array or delay loudspeakers further back.

Several boxes add up as follows. Boxes that cover the same area and add coherently gain up to 6 dB per doubling of their number, while boxes whose sound adds without a fixed phase relation gain 3 dB. Planning with 3 dB per doubling and treating any further gain as headroom is a safe start.

Covering the audience evenly

The front rows sit much closer to a loudspeaker than the back rows. A box whose acoustic centre lies 6 m above the ears of the audience is 7.8 m from a listener at 5 m and 40.4 m from one at 40 m, a difference of 14.3 dB.

Horizontal distance	Distance to the box	Angle below horizontal	Level relative to the back row
5 m	7.8 m	50°	+14.3 dB
10 m	11.7 m	31°	+10.8 dB
20 m	20.9 m	17°	+5.7 dB
30 m	30.6 m	11°	+2.4 dB
40 m	40.4 m	8.5°	0 dB

Aiming the axis at the back row places the closer listeners off axis, and the box's vertical pattern then lowers their level. In a favourable case, the off-axis loss and the distance gain

cancel within ±2 dB. With a conventional box of about 40° vertical coverage, this works for the rear two thirds of such an audience but not for the front rows, which need front fills or a line array whose curvature spreads the energy by distance. Aiming the axis into the middle of the audience makes it worse, because the front then gets both the shorter distance and the axis. Two systems at the sides of a stage are each aimed towards the centre of the rear edge of the audience, so that they overlap down the middle. Prediction software from the loudspeaker manufacturers makes this calculation for real patterns and is covered in chapter 20.

Air, wind and weather

Over the first tens of metres, distance is all that matters. Further out, the air itself changes the sound.

Air absorption

Air turns sound into heat, and much more so at high frequencies. The cause is a relaxation of oxygen and nitrogen molecules, which water vapour shifts in frequency. The humidity effect is sometimes explained by the lower density of moist air, but that explanation does not hold. The loss depends on temperature and humidity together, and not monotonically, so dry air is not always the worst case. ISO 9613-1 gives the formula. The values below are computed from it for standard air pressure.

Air	125 Hz	500 Hz	1 kHz	2 kHz	4 kHz	8 kHz	16 kHz
20 °C, 70 %	0.3	2.8	5.0	9.0	23.1	77.6	281
20 °C, 50 %	0.4	2.7	4.7	9.9	29.7	105	365
20 °C, 30 %	0.6	2.5	5.0	14.1	48.9	168	466
10 °C, 70 %	0.4	1.9	3.7	9.7	33.1	118	370
10 °C, 30 %	0.5	2.3	6.8	23.6	77.2	188	314

The values are in dB per kilometre. At 20 °C and 50 % humidity the air takes 1.5 dB at 4 kHz and 5.3 dB at 8 kHz over 50 m, and 10.5 dB at 8 kHz over 100 m. On a dry day the 8 kHz loss over 50 m grows to about 11 dB. The far audience of a large event therefore needs high-frequency correction that changes with the weather, and modern line-array systems provide an air-absorption filter per box or per zone for this. Thunder heard from far away is a low rumble for the same reason.

Wind and temperature

Sound travels with the air it moves in, and it travels faster in warm air. The speed is about 331 m/s at 0 °C, 343 m/s at 20 °C and 349 m/s at 30 °C.

A steady wind of 5 m/s changes the speed of sound by only 1.5 %, so a crosswind displaces the sound a little but does not blow it away. What does matter is that wind is slower near the ground than above it. Sound that travels against the wind is bent upwards, away from the audience, and sound that travels with the wind is bent downwards. Behind the stage on a windy day, the PA sounds louder than expected, and the audience in front of it hears it weaker and thinner, often with a level that pumps with the gusts. A gusty crosswind also moves the stereo image.

Temperature layers bend sound in the same way. Sound bends towards the slower, colder air.

- **Warm ground, cool air above** is the usual daytime case on a sunny afternoon. Sound bends upwards, and the level far away is lower than the distance law predicts.
- **Cool ground, warm air above** is an inversion, common after sunset on a clear evening and in valleys. Sound bends back down to the ground and can carry far, sometimes several decibels above the free-field prediction at hundreds of metres.

The inversion explains a familiar pattern of complaints. A festival that is inaudible in the village in the afternoon can be heard well there at ten at night with the same settings, because the weather has turned the air into a duct. The level at the mix position has not changed, only the path.

The ground

Above hard ground, such as asphalt or a paved square, the sound reflected from the ground adds to the direct sound and causes a comb filter at the listener, which chapter 2 explains. Grass and soft ground absorb part of the reflection, and an audience absorbs much more. Raising the loudspeakers above the heads keeps the path to the back rows clear of the people in front, who would otherwise absorb and scatter the high frequencies.

Barriers and neighbours

Barriers

A wall, a building or a ridge between the stage and a listener shields them as long as it blocks the line of sight. Sound bends over the top edge, so the screening depends on how much longer the path over the top is than the direct path, measured in wavelengths. A barrier helps at high frequencies and little at low ones. ISO 9613-2 limits the screening of a single barrier to 20 dB and of a double edge to 25 dB, and once the barrier's own sound reduction lies about 10 dB above that, the sound through it no longer matters. The SdWa5 acoustics page explains this in What it means for the sound system. Wind and temperature layers can bend sound over a barrier, so its effect at a distance is less than the calculation on a still day.

What reaches the neighbours

A system that produces 100 dB(A) at 30 m still gives about 75.6 dB at 500 m and about 69.5 dB at 1 km by distance alone. Air absorption takes the treble away on the way, so the neighbours hear mostly kick drum and bass. A level logged in dB(C) or in octave bands describes this better than one in dB(A). Cardioid subwoofer arrays, a stage orientation that points the system away from the nearest houses, and a curfew for the loudest part of the programme do more for the neighbours than a lower overall level.

Limits for neighbours are set nationally or regionally and, in Austria and Germany, mostly in the event permit.

- **Austria.** The noise from events falls under the event laws and the nuisance rules of the federal states, and the authority that issues the permit sets the conditions, often

with a time limit in the evening. Assessment usually refers to the ÖAL guideline No. 3 for the judgement of noise immissions.
- **Germany.** Events are assessed under the leisure noise guideline of the federal states' working group on immission control (Freizeitlärm-Richtlinie), which follows the method of TA Lärm. In a general residential area it allows 55 dB(A) during the day and 40 dB(A) at night outside the most affected window, and it allows higher values, in the order of 70 dB(A) by day and 55 dB(A) at night, on a limited number of rare events per year.

Both countries treat ten o'clock in the evening as the start of the night period in most rules. A permit that names a limit at the nearest house can be met only if that limit is translated into a limit at the mix position during the setup, by measuring both at the same time with the system running.

Feedback and gain before feedback

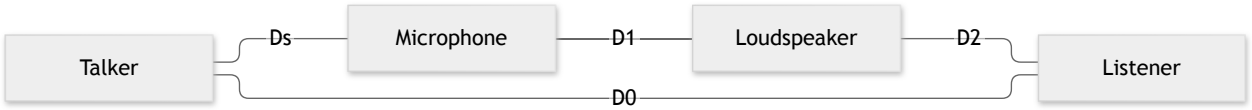
The loop

A microphone in front of a loudspeaker forms a loop. The loudspeaker's sound reaches the microphone, is amplified and comes out of the loudspeaker again. When the loop gain reaches 0 dB at a frequency where the delay around the loop brings the sound back in phase, the system rings continuously. Just below that point, the system still rings briefly after each sound, and it colours everything that passes through it. A system therefore runs at least 6 dB below the onset of feedback.

Potential acoustic gain

The potential acoustic gain (PAG) is how much louder the system can make a talker at the listener than the talker would be without it, at the point of feedback minus a safety margin. With four distances it follows from the inverse square law alone, for an omnidirectional microphone and loudspeaker outdoors.

$$\mathrm{PAG} = 20\lg D_0 - 20\lg D_s + 20\lg D_1 - 20\lg D_2 - 10\lg \mathrm{NOM} - 6\ \mathrm{dB}$$



D_s is the distance from the talker to the microphone, D_0 from the talker to the listener, D_1 from the microphone to the loudspeaker and D_2 from the loudspeaker to the listener. NOM is the number of open microphones, and each doubling of it costs 3 dB. The level of the talker cancels out, so a louder voice does not help.

As an example, a speaker at a lectern talks into a microphone at 0.3 m, the loudspeaker is 5 m from the microphone, and the listener sits 15 m from the talker and 14 m from the loudspeaker. The PAG is 25.0 dB at the point of feedback and 19.0 dB with the margin. With a second open microphone it drops to 16.0 dB.

The needed acoustic gain (NAG) is what the listener needs, and it is set by the equivalent acoustic distance (EAD). That is the distance at which the listener would understand the

talker without a system, typically 1 to 4 m depending on the background noise.

$$\mathrm{NAG} = 20 \lg \frac{D_0}{\mathrm{EAD}}$$

For the listener at 15 m and an EAD of 2 m, the NAG is 17.5 dB. The system with one microphone barely makes it, and with two open microphones it does not. Moving the microphone closer makes up the difference.

Talker to microphone	PAG with 6 dB margin
0.3 m (lectern)	19.0 dB
0.1 m (handheld)	28.6 dB
0.05 m (handheld, close)	34.6 dB
0.02 m (headset)	42.6 dB

What raises it

Three classic rules remain the core, and a modern system adds a few tools.

1. **Microphone close to the source.** Every halving of D_s gives 6 dB, which is why a headset is the strongest single measure for a speaker or a singer who moves.
2. **Loudspeaker far from the microphone.** Every doubling of D_1 gives 6 dB. A flown system and loudspeakers in front of the stage line help, while a loudspeaker behind the microphone is the worst case.
3. **Directional microphones and loudspeakers, aimed.** A microphone that has the loudspeaker in its rejection zone and a loudspeaker that has the microphone off its axis each gain several decibels. About 6 dB together is a realistic expectation in practice, because a loudspeaker becomes omnidirectional at low frequencies and a microphone's pattern changes with frequency. A microphone with smooth off-axis response gains more than one with a sharper pattern and peaks off axis.
4. **Fewer open microphones.** Each doubling of NOM costs 3 dB. Muting unused channels or an automatic mixer that opens only the active microphone, covered in chapter 11, gains this back.
5. **Smooth response.** A peak in the response of the microphone, the loudspeaker or the room is where feedback starts first. A narrow notch at a ringing frequency raises the gain at that frequency, and chapter 12 covers automatic feedback suppressors.
6. **No loudspeaker where it is not needed.** Cardioid subwoofer arrays send less low frequency onto the stage, and in-ear monitors remove the wedge as a feedback path entirely.

A cardioid subwoofer array uses at least two sources, with one delayed and often polarity-inverted so that the two cancel behind the array and add in front. For a gradient pair with the sources a quarter wavelength apart at 100 Hz, the spacing is 0.86 m and the delay 2.5 ms. Such arrays reach well over 10 dB of rear rejection over the subwoofer range, which helps the stage and the neighbours behind it. Chapter 14 covers the variants.

To verify in pass 2

- The transition distance formula $H^2 f / (2c)$ and its variants (Fresnel, Heil and Urban); check the line-array literature.

- Typical maximum levels of 125 to 135 dB at 1 m for small-event boxes; check data sheets.
- The ISO 9613-1 implementation in the scratchpad against the tables of the standard; the 20 °C and 10 °C rows at 70 % match ISO 9613-2 Table 2 as remembered.
- The size of the excess level under inversions at hundreds of metres; check ISO 9613-2 meteorological correction and CONCAWE.
- The barrier limits of 20 and 25 dB in ISO 9613-2.
- Austrian practice for event noise, including the ÖAL guideline No. 3 and the Salzburg event law.
- The German Freizeitlärm-Richtlinie values (55/40 dB(A) in general residential areas, rare events about 70/55 dB(A)), the number of rare events per year and the measuring point.
- The start of the night period at 22:00 in Austria and Germany.
- The rear rejection of cardioid subwoofer arrays (over 10 dB); check manufacturer application notes.
- The EAD range of 1 to 4 m; check Davis and Davis, *Sound System Engineering*.

Sources

- ISO 9613-1, attenuation of sound during propagation outdoors, calculation of the absorption of sound by the atmosphere.
- ISO 9613-2, attenuation of sound during propagation outdoors, general method of calculation.
- ÖAL-Richtlinie Nr. 3, Beurteilung von Schallimmissionen.
- LAI Freizeitlärm-Richtlinie and TA Lärm.
- Davis, D., Davis, C.: *Sound System Engineering*, Focal Press.
- SdWa5: Acoustics, section *What it means for the sound system*.
- Davis, G., Jones, R.: *The Sound Reinforcement Handbook*, 2nd ed., Yamaha / Hal Leonard 1989, §5 (pp. 43–52) and §18.2.3 to §18.3.3 (pp. 351–353).

7. Sound indoors

> **Draft, pass 1.** This chapter was written from general knowledge and checked against reference literature for coverage. Standards, editions and figures are named but have not yet been verified against their sources. The list *To verify in pass 2* at the end names every claim that still needs one.

This chapter answers what a room adds to a sound system. It covers what boundaries do with sound, the resonances of a room at low frequencies, reverberation and how long it lasts, the distance at which the room becomes louder than the loudspeaker, how intelligibility is measured, and how loudspeakers are placed and feedback is controlled when the room takes part. Free-field behaviour, the inverse square law and gain before feedback are in chapter 6.

Boundaries

A sound wave that hits a wall, a floor or a ceiling is split three ways. Part of it is reflected back into the room, part is turned into heat inside the material, and the rest passes through to the other side. The shares depend on the frequency and the angle, but not on the level. An object that is small compared with the wavelength, such as a lectern at 100 Hz, does not cast a shadow at all, because the sound bends around it. That bending is diffraction, which older texts sometimes call refraction.

The absorption coefficient α, introduced by Wallace Sabine, gives the share of energy that does not come back, from 0 for a perfect reflector to 1 for an open window. A surface with $\alpha = 0.15$ reflects 85 % of the energy, which is $10 \lg 0.85 = -0.7$ dB. A reflection is therefore weakened much less than the absorption coefficient suggests, and a room needs many reflections or highly absorbent surfaces before its sound decays.

The SdWa5 acoustics page gives a table of absorption coefficients with the split into reflected, absorbed and transmitted energy, and it adds the sound reduction index that matters for the neighbours. It is the reference for this section, in Reflected, absorbed and through.

Three results from it matter most for a sound system.

- **The audience is the main absorber.** An occupied audience area absorbs nearly all the energy at mid and high frequencies, so a hall sounds very different empty at the sound check and full at the show.
- **Curtains work only at mid and high frequencies.** A curtain hung in front of a wall absorbs most where it hangs a quarter wavelength in front of the wall, where the air moves fastest. A curtain 10 cm from a wall helps from about 850 Hz upwards and does almost nothing at 100 Hz.
- **Bass needs mass or depth.** Low frequencies pass through light walls and are absorbed only by deep porous absorbers or tuned resonators.

Room modes

A wave that reflects back and forth between two parallel walls reinforces itself when a whole number of half wavelengths fits between them. The room then resonates at those frequencies, and the pressure varies through the room with loud places, the antinodes, and quiet ones, the nodes, half a wavelength apart. These resonances are the room modes. Between two walls at distance L they lie at

$$f_n = \frac{n\,c}{2L}, \quad n = 1, 2, 3, \ldots$$

Those between one pair of opposite surfaces are axial modes. Tangential modes involve four surfaces and oblique modes all six, and they are weaker. A rectangular room has all three kinds.

Dimension	1st	2nd	3rd	4th
Hall, length 20 m	8.6 Hz	17.1 Hz	25.7 Hz	34.3 Hz
Hall, width 12 m	14.3 Hz	28.6 Hz	42.9 Hz	57.2 Hz
Hall, height 6 m	28.6 Hz	57.2 Hz	85.8 Hz	114.3 Hz
Small room, length 8 m	21.4 Hz	42.9 Hz	64.3 Hz	85.8 Hz
Small room, width 5 m	34.3 Hz	68.6 Hz	102.9 Hz	137.2 Hz
Small room, height 3 m	57.2 Hz	114.3 Hz	171.5 Hz	228.7 Hz

Modes are spaced widely at the bottom of the range and crowd together further up. Above a certain frequency they overlap so densely that the room no longer has distinct resonances, and the sound field behaves statistically. That limit is the Schroeder frequency, with T the reverberation time in seconds and V the volume in cubic metres.

$$f_S \approx 2000\sqrt{\frac{T}{V}}$$

Room	Volume	Reverberation time	Schroeder frequency
Small room 8 × 5 × 3 m	120 m³	0.5 s	129 Hz
Hall 20 × 12 × 6 m, empty	1440 m³	1.79 s	71 Hz
Hall 20 × 12 × 6 m, occupied	1440 m³	0.93 s	51 Hz
Large hall	10 000 m³	2.5 s	32 Hz

Below the Schroeder frequency, the bass at a seat depends on where the seat lies in the mode pattern, and two seats a few metres apart can differ by 10 dB or more. Above it, the reverberation dominates. The practical consequences of the modes are these.

- **An equalizer cannot fix a mode for the whole room.** A cut at a modal peak helps where the peak is and deepens the dip at a node. Moving the subwoofers, the listening position or the measurement position changes more.
- **Source position matters.** A mode is excited strongly by a source at its antinode and hardly at all by one at its node. Loudspeaker placement does not change the frequencies of the modes, but it does change how strongly they are excited. Subwoofers placed at the middle of a wall do not excite the first width mode, which has its node there.

- **Small rooms are the hard case.** A rehearsal room or a small club has few, widely spaced modes in the range of the kick drum and the bass guitar.

Reverberation

When a source in a room stops, its sound does not stop with it. The reflections go on, losing energy at every boundary and in the air, until they die away. The time it takes for the level to fall by 60 dB is the reverberation time T or RT60.

Calculating it

Sabine found that the reverberation time grows with the volume and falls with the total absorption. In SI units, with V in cubic metres and the equivalent absorption area $A = \sum S_i \alpha_i$ in square metres, his equation reads as follows.

$$T = 0.161 \frac{V}{A}$$

Sabine's equation overestimates the time in rooms with a lot of absorption, because even a room with α = 1 everywhere would still give a finite time. Eyring's equation, with the total surface S and the mean absorption coefficient $\bar{\alpha}$, corrects this.

$$T = 0.161 \frac{V}{-S \ln(1 - \bar{\alpha})}$$

For a hall of 20 × 12 × 6 m, with 1440 m³ and 864 m² of surface, the two give the following.

Mean absorption	Sabine	Eyring
0.10	2.68 s	2.55 s
0.15	1.79 s	1.65 s
0.20	1.34 s	1.20 s
0.30	0.89 s	0.75 s

The same hall with a mean α of 0.15 empty gets an audience on 150 m² of its floor, which raises the absorption there from about 0.1 to about 0.9. The reverberation time at mid frequencies falls from 1.79 s to 0.93 s by Sabine. The audience halves the reverberation time, which is why a sound check in an empty hall misleads. The Fitzroy equation treats rooms where the absorption is concentrated on one pair of surfaces, such as a hall with an absorbent ceiling and hard walls, and it gives longer times than Sabine there.

What the room should have

A target depends on use and volume. The German DIN 18041 gives target times for rooms by use, as functions of the volume, and it is the usual DACH reference for rooms up to about 30 000 m³. For a volume of 1440 m³ its formulas give about 1.5 s for music and about 1.0 s for speech. A room for amplified music can be drier than one for acoustic music, because the system supplies the energy and the room only needs to stay out of the way.

Measuring it

ISO 3382-1 describes the measurement for performance spaces and ISO 3382-2 for ordinary rooms. A loudspeaker excites the room with a sweep or noise, an omnidirectional microphone records the decay at several positions, and the software calculates the time. The full 60 dB of decay is rarely above the background noise, so the standard extrapolates from the slope between −5 and −25 dB (T20) or between −5 and −35 dB (T30). The early decay time (EDT), from the first 10 dB, matches what listeners perceive more closely. REW, which is free, and commercial tools such as Smaart and ODEON's measurement module all report these values from an impulse response.

Direct and reverberant sound

Critical distance

Close to a loudspeaker, the direct sound dominates and falls by 6 dB per doubling of the distance as outdoors. The reverberant sound, on the other hand, is about equally strong throughout the room, as long as the source keeps running. At some distance the two are equally loud. That distance is the critical distance D_c, and beyond it the level stops falling and the room is louder than the loudspeaker.

$$D_c \approx 0.057\sqrt{\frac{Q\,V}{T}}$$

Q is the directivity factor of the loudspeaker on the axis that points at the listener, 1 for an omnidirectional source and 10 for a horn with a directivity index of 10 dB. For the example hall, the critical distance is as follows.

Directivity Q	Directivity index	Empty, 1.79 s	Occupied, 0.93 s
1	0 dB	1.6 m	2.2 m
2	3 dB	2.3 m	3.2 m
5	7 dB	3.6 m	5.0 m
10	10 dB	5.1 m	7.1 m
20	13 dB	7.2 m	10.0 m

The direct-to-reverberant ratio falls by 6 dB per doubling of the distance, from +6 dB at half the critical distance to −6 dB at twice and −12 dB at four times. Even a directional loudspeaker in a hall of this size leaves most of the audience beyond the critical distance.

Why directivity matters indoors

A directional loudspeaker raises the critical distance along its axis, because it sends more energy towards the listener and less into the room. When its axis points at the audience, the energy that misses the listeners' ears lands on the most absorbent surface in the room and does not return. Both effects reduce the reverberant field. Off the axis, the critical distance shrinks, so the coverage pattern must match the audience area. A loudspeaker that covers the side walls as well as the audience feeds the reverberation and gains nothing.

This is the main reason why directivity, and especially directivity that holds down into the lower midrange, is valued so highly in sound reinforcement. A line array or a large horn keeps its pattern to lower frequencies than a small box, and the bass, which no practical box can steer, is what makes a reverberant hall sound boomy.

Intelligibility

Reverberation blurs speech. Each syllable is followed by its own decay, which masks the next, quieter syllable, and consonants suffer most. A loud system cannot help beyond a point, because it raises the reverberation with the direct sound.

Acousticians long related intelligibility to the direct-to-reverberant ratio and left the methods to the specialist. Today, measured indices are routine.

- **Speech Transmission Index (STI), IEC 60268-16.** STI measures how much the modulation of speech survives the path through the system and the room, including noise and reverberation, on a scale from 0 to 1. The standard's categories run from bad below 0.30 to excellent above 0.75. Voice alarm systems usually have to reach about 0.50. STIPA is a simplified test signal for measurement with a hand-held meter.
- **%ALcons (Peutz).** The articulation loss of consonants is an older measure that estimates the share of consonants missed by a listener. Peutz's estimate is $200\,D^2T^2N/(VQ)$, with $N = 1$ for a single source. For the example hall occupied, a horn with $Q = 10$ and a listener at 15 m, it gives about 2.7 %, which counts as good. The same hall empty gives about 10 %, at the edge of poor. The estimate is valid up to about three times the critical distance.
- **Clarity C50 and C80.** These compare the energy arriving in the first 50 ms (for speech) or 80 ms (for music) with the energy after it, in decibels. They come from the same impulse response as the reverberation time and are defined in ISO 3382-1.

Background noise reduces intelligibility as well. As a rule, speech should be at least about 10 to 15 dB above the noise, and much more makes no further difference.

Loudspeakers and feedback in a room

Placement

The outdoor rules from chapter 6 hold indoors, with the room as an extra participant.

- **Aim at the audience, not at the walls.** The audience absorbs, and the walls reflect. A common aim point is about two thirds of the depth of the hall, so that the level falls off towards the back and less sound reaches the rear wall. A slightly lower level at the back is accepted by listeners and often wanted.
- **Treat the rear wall if possible.** A reflective rear wall sends an echo back to the stage and the front rows, which is late enough to be heard as a separate event in a large hall. Heavy curtains help at mid and high frequencies.
- **Several sources rather than one loud one.** Delay loudspeakers further back keep the listeners there inside the critical distance of a closer source, at a lower level overall. Chapter 20 sets them up.

- **Mind the boundaries near the subwoofers.** A subwoofer on the floor gains up to 6 dB at low frequencies over the same box in free space, in a floor-wall edge up to 12 dB and in a corner up to 18 dB, as long as the distances are small compared with the wavelength. At larger distances, the reflection from a nearby wall cancels the direct sound at $f = c/(4d)$, which is 86 Hz for a box 1 m from the wall and 43 Hz for 2 m. A subwoofer against the wall or well away from it avoids a cancellation in its own range.

Feedback indoors

A room makes feedback worse in two ways. The reverberant field adds to the sound at the microphone, and the modes favour some frequencies where the loop gain rises first. A feedback frequency in a room is often a modal peak at the microphone position, and it can shift when the temperature changes the speed of sound, because the room resonates at a fixed wavelength.

Acoustic measures come before equalization.

1. **Move the microphone.** A microphone at a node of the troublesome mode picks it up much less, and a microphone in a damped part of the stage picks up less reverberation.
2. **Use a directional microphone close to the source,** with smooth off-axis response, because a lobe off its axis cancels the advantage of its pattern.
3. **Notch narrowly.** A modal resonance is only a few hertz wide. Its bandwidth is about $2.2/T$, which gives 1.2 Hz for the empty example hall and 2.4 Hz when occupied. A third-octave filter at 250 Hz is 58 Hz wide, a tenth-octave filter 17 Hz and a sixtieth-octave filter 3 Hz. Parametric notches of a tenth octave or narrower, set by ear or by an automatic feedback suppressor, take far less out of the programme than third-octave graphic equalizers. Chapter 12 covers them.
4. **Ring out with care.** The gain is raised slowly until a frequency rings, that frequency is notched, and the step is repeated a few times. After the third or fourth notch, the gain gained per notch is usually small, and the remaining margin is better left unused.

Predicting and measuring a room

Room acoustics software builds a 3D model of the room with the absorption of its surfaces and calculates the reverberation, the coverage of each loudspeaker and the STI before anything is installed. EASE, ODEON and CATT-Acoustic are the established tools for this. The loudspeaker manufacturers' own tools, such as EASE Focus, d&b ArrayCalc, L-Acoustics Soundvision and Meyer MAPP 3D, calculate their own systems quickly, but they mostly treat the room as free field and leave the reverberation out. For an existing room, a measurement with REW, Smaart or a similar tool gives the reverberation time, the modes and the clarity from a few impulse responses in an hour, and that is usually the better start for a temporary system in a venue that has been measured once.

To verify in pass 2

- The quarter-wavelength rule for curtains (850 Hz for 10 cm); check absorber literature for the frequency where a porous layer at that distance becomes effective.
- The Schroeder frequency constant 2000; check Schroeder and Kuttruff, *Room Acoustics*.
- The assumed absorption of an occupied audience area (0.9) against an empty floor (0.1); check published data.
- The Fitzroy equation and when it gives longer times than Sabine.
- DIN 18041:2016, the room groups, the formulas for target reverberation times and their volume limits; and the Austrian ÖNORM B 8115-3 as the corresponding reference.
- ISO 3382-1 and ISO 3382-2, the evaluation ranges for T20 and T30, EDT, C50 and C80.
- The critical distance constant 0.057 and the assumption of a diffuse field.
- IEC 60268-16 edition, STI categories and the 0.50 target for voice alarm systems (EN 54-32, ISO 7240-19).
- Peutz's %ALcons formula, its constant, the meaning of N (some versions use $N+1$, which doubles the result), the categories below 5 % and above 10 %, and its range of validity.
- The 10 to 15 dB speech-to-noise ratio for intelligibility.
- Boundary gains of 6, 12 and 18 dB and the $c/(4d)$ cancellation.
- The mode bandwidth $2.2/T$.

Sources

- Kuttruff, H.: *Room Acoustics*, CRC Press.
- ISO 3382-1 and ISO 3382-2, measurement of room acoustic parameters.
- IEC 60268-16, objective rating of speech intelligibility by speech transmission index.
- DIN 18041, acoustic quality in rooms.
- Peutz, V. M. A.: Articulation loss of consonants as a criterion for speech transmission in a room, *JAES* 19, 1971.
- SdWa5: Acoustics, section *Reflected, absorbed and through*.
- Davis, G., Jones, R.: *The Sound Reinforcement Handbook*, 2nd ed., Yamaha / Hal Leonard 1989, §6 (pp. 53–60) and §18.3.4 to §18.3.5 (p. 354).

8. Specifications

> **Draft, pass 1.** This chapter was written from general knowledge and checked against reference literature for coverage. Standards, editions and figures are named but have not yet been verified against their sources. The list *To verify in pass 2* at the end names every claim that still needs one.

This chapter answers how to read a data sheet so that its numbers mean something. It explains why a figure without its conditions is worthless, and it goes through frequency response, noise, distortion, impedance, crosstalk, the specifications of digital equipment and the mechanical data, each with what a complete statement contains. It ends with a checklist. Operating levels are in chapter 3, and frequency bands and square-wave tests in chapter 2. How a measurement and a listener can disagree is in chapter 5.

A number needs its conditions

A specification is a measurement result, and a measurement result is only meaningful with the conditions under which it was taken. This gives the best test of any data sheet. A competent technician with standard test equipment should be able to repeat the measurement from the data sheet alone and get the same result. Anything that would stop them, such as an unnamed load, an unknown bandwidth or an unstated level, makes the figure unverifiable.

A data sheet of the kind found on many product pages shows how every line can look plausible while none can be checked.

Line on the sheet	What is missing
Frequency response 20 Hz – 20 kHz	The tolerance in dB and the level at which it was measured
THD < 0.01 %	The level, the frequency, the load and the bandwidth of the measurement
Dynamic range 115 dB	Weighting, bandwidth, and whether it is AES17 or an output noise figure against full scale
Max output +24 dBu	The load and the distortion at which "maximum" was decided
Input impedance 10 kΩ	Whether balanced or unbalanced, and at which frequency
Latency < 1 ms	Which path, at which sample rate, and whether converters are included
Max SPL 135 dB	Peak or continuous, at what distance, with which signal and for how long

A data sheet that gives the conditions for each figure is usually from a manufacturer who expects to be checked. One that gives round numbers without conditions should be read as an advertisement.

Two habits help. The first is to compare devices only on figures measured the same way, ideally to the same standard. The second is to look for the graph behind the number, because a graph shows what a single figure hides.

Frequency response

Three different statements are often confused.

- **Frequency response** is the output level over frequency at a stated level, within a stated tolerance, such as 20 Hz to 20 kHz, +0/−1 dB at +4 dBu. Without a tolerance it says nothing, because every device produces some output at 20 Hz.
- **Frequency range** is the range in which a device does something useful, without a tolerance. It is a marketing figure.
- **Power bandwidth** is the range in which an amplifier reaches its rated output at a stated distortion. It is usually narrower than the small-signal frequency response, because an amplifier can follow a quiet signal at 50 kHz but not a loud one.

For electronic equipment, ±0.5 dB from 20 Hz to 20 kHz is easy to achieve and should be expected. Loudspeakers are a different case. A loudspeaker's response depends on the direction, the distance, the room and the mounting, so its data sheet needs the measuring conditions in detail. A figure such as 45 Hz to 18 kHz at −10 dB says much less than an on-axis graph and a set of off-axis curves. AES2-2012 and IEC 60268-5 describe how loudspeaker components and systems are to be measured, and CTA-2034 describes a set of curves for direct, early-reflected and sound-power response that predicts in-room behaviour well.

A graph is worth more than a printed range, provided its scale is honest. A vertical axis of 50 dB over a short height makes a response with ±3 dB of ripple look flat, and smoothing to one-third octave hides narrow peaks.

Noise

What noise is and how it is shaped

Noise is any unwanted signal that is not related to the wanted one. Its most basic form, thermal noise, comes from the random motion of electrons in every resistor and sets a limit that no amplifier can beat.

- **White noise** has equal energy per hertz. Per octave its energy doubles, so it rises by 3 dB per octave on a fractional-octave analyser, and the octave from 10 to 20 kHz contains half its energy.
- **Pink noise** has equal energy per octave. On a third-octave analyser it reads flat, which is why it is the test signal for system equalization. It is white noise filtered at −3 dB per octave.

A noise figure depends on the bandwidth of the measurement and on the weighting. For white noise, a measurement weighted with the A-curve reads about 2 dB lower than an unweighted one over 20 Hz to 20 kHz, and an unweighted measurement with an 80 kHz bandwidth reads about 6 dB higher. A difference of 8 dB between two data sheets can come from the measuring conditions alone.

Equivalent input noise

The noise of a microphone preamplifier is specified as equivalent input noise (EIN), the noise at the output divided by the gain, so that it can be compared with the signal at the

input. A source resistance must be stated, because the preamplifier cannot be quieter than the resistor at its input. The thermal noise voltage of a resistance R over a bandwidth B at absolute temperature T is given by Boltzmann's constant k.

$$v_n = \sqrt{4kTRB}$$

For 150 Ω, 20 kHz and 290 K, this is 219 nV, or −131.0 dBu. For 200 Ω it is −129.7 dBu. A microphone preamplifier with an EIN of −128 dBu at 150 Ω therefore adds 3 dB to the noise of an ideal source, which is good. A figure of −135 dBu at 150 Ω is below the thermal limit, so it was measured with a shorted input, A-weighted or in a narrower bandwidth, and it cannot be compared with the others.

A complete EIN statement reads like this: −128 dBu, 150 Ω source, 20 Hz to 20 kHz, unweighted, at 60 dB gain. The gain matters, because at low gain the stages after the preamplifier add their own noise and the EIN gets worse.

Output noise

The output noise of a console depends on how many channels are routed to the bus, where the faders and the master sit, whether the hum was included and how the output was terminated. The same console can be described in ways that differ by 20 dB. A useful statement names the routing, the positions of the controls, the bandwidth, the weighting and the load.

Hum, buzz and other noises

Hum is the mains frequency and its harmonics, at 50, 100, 150 Hz and up in Europe, and it gets into a system through earth loops, magnetic fields near transformers and poorly filtered power supplies. Buzz contains much higher harmonics, typically from phase-cut dimmers and switch-mode supplies. Radio frequency that a circuit rectifies turns into crackle or into a radio station in the PA. None of these is usually specified, and all of them depend more on the installation than on the device. Chapter 17 covers the causes and the remedies.

Distortion

Harmonic distortion

A nonlinear device adds harmonics, components at integer multiples of the input frequency. Total harmonic distortion (THD) is their combined level relative to the fundamental, as a percentage or in decibels.

THD	In dB
10 %	−20 dB
1 %	−40 dB
0.1 %	−60 dB
0.01 %	−80 dB
0.001 %	−100 dB

Most analysers measure THD+N, which removes the fundamental with a notch filter and measures everything that is left, including noise and hum. At low levels, THD+N is dominated by noise and rises as the level falls, while at high levels it rises sharply at clipping. A THD+N curve against level shows both, while a single figure shows only one point on it.

A total says nothing about which harmonics are present. Low, even harmonics are much less audible than high, odd ones, and chapter 5 gives an example where a loudspeaker's larger distortion does not hide an amplifier's smaller one. A spectrum of the distortion products tells more than the total.

Intermodulation distortion

Two tones through a nonlinear device produce sum and difference frequencies that are not harmonically related to either, and these intermodulation (IM) products are more audible than harmonics. Two standard methods are in use.

- **SMPTE IM** (also in DIN and IEC variants) uses a low tone, typically 60 Hz, and a high tone, typically 7 kHz, at a level ratio of 4 to 1, and it measures the sidebands around the high tone.
- **CCIF or twin-tone IM** uses two high tones close together, such as 19 and 20 kHz at equal level, and it measures the difference frequency at 1 kHz and other products in the audible range.

Twin-tone tests reveal problems at high frequencies that a THD test at 1 kHz misses, because the harmonics of a 10 kHz tone lie above the audio band and are filtered out of a THD measurement.

Transient intermodulation

Transient intermodulation (TIM) was proposed in the 1970s for amplifiers that distorted on fast transients because their input stage overloaded before the feedback could correct it. It was controversial then and is rarely specified now, because a current amplifier's slew rate lies far above what audio requires. An amplifier delivering 100 W into 8 Ω has a peak voltage of 40 V and needs a slew rate of about 5 V/µs to reproduce a full-power 20 kHz sine.

Impedance

Bridging rather than matching

Every output has a source impedance, and every input has a load impedance. Early telephone and broadcast practice matched them, typically at 600 Ω, which transfers maximum power but halves the voltage, a loss of 6 dB. Modern audio equipment instead uses bridging, where the input impedance is at least about ten times the source impedance, so the voltage arrives almost unchanged.

Source	Load	Loss
600 Ω	600 Ω	−6.02 dB
100 Ω	600 Ω	−1.34 dB

Source	Load	Loss
600 Ω	10 kΩ	−0.51 dB
150 Ω	10 kΩ	−0.13 dB
50 Ω	10 kΩ	−0.04 dB

Line outputs today are typically 50 to 150 Ω and line inputs 10 to 20 kΩ, and a microphone input is 1 to 3 kΩ for a microphone of 150 to 200 Ω. Only some old equipment, such as transformer-coupled outputs that expect a 600 Ω termination and certain vintage effects, still needs a matched load to stay flat. Chapter 9 covers microphone and instrument inputs.

Impedance changes with frequency

An impedance is not a fixed number. A loudspeaker rated 8 Ω may fall to 5 Ω at one frequency and rise to 40 Ω at its resonance, and an amplifier sees all of that. Input and output impedances of electronics also vary, usually rising or falling at the ends of the band through coupling capacitors and cable capacitance. A specification should state the frequency, and a loudspeaker's impedance curve with its minimum is more useful than its nominal rating.

Mismatches

A low impedance load pulls down a source's voltage and can push its output stage into current limiting, as when an amplifier drives 2 Ω or when a headphone amplifier drives a low-impedance headphone. A high source impedance into a long cable loses the high frequencies, because the cable's capacitance forms a low-pass filter with it. Both are reasons why outputs are specified with a minimum load and why a data sheet should give the source impedance of each output.

Crosstalk

Crosstalk is a signal from one channel appearing in another. It arises from capacitive coupling between conductors, from inductive coupling between wires carrying current, from shared ground return paths and from shared power supplies. Capacitive coupling rises with frequency at 6 dB per octave, so a figure of −80 dB at 1 kHz becomes −60 dB at 10 kHz. A useful crosstalk specification therefore gives the figure at several frequencies, or as a curve, and names the channels and the loads.

In cables, crosstalk between pairs in a multicore is small at audio frequencies when the pairs are individually shielded or twisted, and the main risk is a loudspeaker cable or a digital signal run alongside microphone lines. Inside digital equipment, crosstalk is almost absent in the processing and occurs mainly in the analog stages and converters.

Digital equipment

Digital equipment has its own specifications, most of them defined in AES17 for measurement and in AES11 for synchronisation.

- **Dynamic range** per AES17 is measured with a signal at −60 dBFS, so that the noise and distortion are measured together at a low level, and it is usually A-

weighted. A 24-bit word allows 146 dB in theory, and real converters reach 110 to 125 dB, limited by their analog circuits.
- **THD+N** is given in dB relative to full scale, typically at −1 dBFS, and is often better than −100 dB.
- **Maximum level** is the analog level that produces 0 dBFS, such as +18 dBu or +24 dBu. It sets the alignment that chapter 3 explains.
- **Latency** is the delay from analog input to analog output. A buffer of 64 samples takes 1.33 ms at 48 kHz and 0.67 ms at 96 kHz, and the converters add their own filter delay. A complete statement names the path and the sample rate.
- **Jitter** is timing uncertainty in the sample clock. It limits the achievable signal-to-noise ratio at high frequencies. With 1 ns of jitter, a full-scale 20 kHz sine cannot exceed about 78 dB, with 100 ps about 98 dB, and with 10 ps about 118 dB. Modern converters with good clock recovery reach the low picosecond range.
- **Sample rate conversion** introduces its own noise and filter ripple and should be specified by the same AES17 measurements.

A digital processor's frequency response, noise and distortion are often better than any analog circuit around it, so the converters and the analog stages decide the result. A network audio device is specified by the same measures plus the network latency setting, which chapter 15 covers.

Loudspeaker and system specifications

Loudspeaker specifications are where data sheets diverge most. A maximum SPL can be a peak level with a short burst measured at 1 m, a continuous level with pink noise calculated from sensitivity and power, or a measured value with a music-like test signal. The difference can be 10 dB.

- **Sensitivity** is the level at 1 m with 1 W or 2.83 V input. For an 8 Ω loudspeaker, 2.83 V is 1 W, and for a 4 Ω loudspeaker it is 2 W, which makes the 4 Ω sensitivity look 3 dB better.
- **Power handling** per AES2-2012 is measured with a band-limited noise of 6 dB crest factor for two hours, and per IEC 60268-5 with a programme simulation noise. Neither is a recommendation for amplifier power.
- **Maximum SPL** with M-Noise (AES75) uses a test signal designed to resemble music in its crest factor across frequency, so the figure is comparable between manufacturers who use it.
- **Coverage** should be given as the angle between the −6 dB points, horizontal and vertical, and how it changes with frequency. A single figure such as 90° × 40° says nothing about the midrange and bass.
- **Subwoofer output** is described by CTA-2010 as the maximum level at each frequency below a distortion limit.

Chapter 14 returns to loudspeaker data, and chapter 13 to amplifier ratings.

Mechanical data

The 19-inch rack is 482.6 mm wide at the front panel, and one rack unit (U) is 44.45 mm high. A data sheet should give the depth behind the front panel separately from the total

depth with knobs and handles, because the depth of a rack case is usually the limit. Weight should be net, with the packed weight given as well for transport planning. For a road case, add the case and the cables, and for flying loudspeakers, the rigging and its working load limit, which chapter 20 covers.

Checklist for a data sheet

1. **Conditions with every figure.** Level, frequency, bandwidth, weighting, load and gain.
2. **Graphs behind numbers.** Frequency response with tolerance, THD+N against level and frequency, impedance curve, polar data.
3. **Standards named.** AES17 for digital, AES2 or IEC 60268-5 for loudspeakers, AES75 for maximum SPL, CTA-2034 or CTA-2010 where they apply.
4. **Physical limits respected.** An EIN below the thermal limit or a dynamic range beyond the converter's bits is a sign of a different method, not of a better device.
5. **Like with like.** Two devices are compared only on figures measured the same way.
6. **Mechanics.** Depth behind the panel, net weight, power consumption and inrush current.

To verify in pass 2

- The A-weighting and bandwidth effects on white noise (−2 dB, +6 dB at 80 kHz); recompute with the IEC 61672-1 weighting formula and check against AES17.
- The typical EIN figures of current preamplifiers (around −128 dBu at 150 Ω).
- AES2-2012 test signal and duration, and IEC 60268-5 programme simulation noise.
- AES75 (M-Noise) title, year and scope.
- CTA-2034 and CTA-2010 titles, current editions and scope.
- AES17 measurement of dynamic range at −60 dBFS and the weighting it allows.
- The SMPTE and CCIF IM frequencies and ratios; check IEC 60268-3.
- Typical converter dynamic range of 110 to 125 dB in current stageboxes.
- The jitter limit formula and typical jitter values of current converters.
- The 19-inch rack dimensions per IEC 60297 and EIA-310.

Sources

- AES17, measurement of digital audio equipment.
- AES2-2012, specification of loudspeaker components used in professional audio and sound reinforcement.
- AES75, measuring loudspeaker maximum linear sound levels using noise (M-Noise).
- IEC 60268-3, amplifiers, and IEC 60268-5, loudspeakers.
- CTA-2034, standard method of measurement for in-home loudspeakers; CTA-2010, subwoofer output measurements.
- IEC 60297 and EIA-310, rack dimensions.
- Davis, G., Jones, R.: *The Sound Reinforcement Handbook*, 2nd ed., Yamaha / Hal Leonard 1989, §8 (pp. 71–104).

9. Microphones and DI

> **Draft, pass 1.** This chapter was written from general knowledge and checked against reference literature for coverage. Standards, editions and figures are named but have not yet been verified against their sources. The list *To verify in pass 2* at the end names every claim that still needs one.

This chapter answers how a microphone turns sound into a signal, how its pattern, sensitivity and noise are read from its data sheet, how it is powered and connected, and how instruments without a microphone are brought into the system through a DI box. Wireless microphones are in chapter 10, and gain before feedback is in chapter 6.

How microphones work

Every microphone has a diaphragm that moves with the sound pressure, and a transducer that turns that motion into a voltage. The transducer decides most of the microphone's character.

- **Dynamic (moving coil).** A coil attached to the diaphragm moves in a magnetic field and generates a voltage. It needs no power, handles very high levels and survives rough treatment, which makes it the standard for vocals on loud stages and for drums and guitar amplifiers. Its heavier diaphragm responds less to the highest frequencies and fast transients than a condenser.
- **Condenser.** The diaphragm forms one plate of a capacitor, and its motion changes the capacitance against a fixed back plate. The element needs a polarising voltage and an impedance converter, both powered through the cable. Its light diaphragm gives extended high-frequency response and a detailed transient response, and its higher output makes it quieter in use. Condensers are the choice for overheads, acoustic instruments, choirs and speech at a distance.
- **Electret.** A condenser whose polarisation is stored permanently in a charged layer, so it needs power only for its impedance converter. Almost all lavalier and headset microphones, and many measurement microphones, are electrets.
- **Ribbon.** A thin metal ribbon moves in a magnetic field. Its natural pattern is a figure-of-eight, its sound is smooth and its output low. Classic passive ribbons are fragile, and a wiring fault that puts phantom power across the ribbon can damage it. Active ribbons with built-in amplifiers need phantom power and give a normal output.
- **Piezoelectric.** A crystal or ceramic element produces a voltage when it is bent. It survives today in contact pickups on acoustic guitars, violins and drums, with a very high source impedance that needs a suitable input.

The carbon microphone of early telephones has no place in sound reinforcement.

Forms and uses

Form	Typical use	Notes
Handheld	Vocals, speech	Built-in pop protection and handling-noise isolation

Form	Typical use	Notes
Stand or clip	Instruments, choirs, lecterns	Shock mount and windscreen as needed
Lavalier	Speech, theatre	Omni electret, placed on the chest, with a presence lift to make up for the off-axis position
Headset	Speech, singing actors, presenters	Capsule at the mouth corner, the strongest feedback measure for a moving talker
Boundary	Lecterns, conference tables, stage floors	Capsule close to a surface, so the direct and reflected sound arrive together without a comb filter
Shotgun	Film, distant pickup	Directional only at high frequencies, little use on a live stage
Contact pickup	Acoustic instruments	Feedback-resistant, sound differs from an air microphone
Ceiling array	Conference rooms, hybrid events	Many capsules with beamforming that follows the active talker

A lecturer who moves around a lectern microphone changes the level by many decibels with every step, because the inverse square law is steep close to the microphone. A lavalier keeps the distance constant for such a speaker, and a headset is better still where the gain before feedback is short.

Pickup patterns

Most microphones have a first-order pattern, a mix of a pressure element, which responds equally in all directions, and a pressure gradient element, which responds as the cosine of the angle. The sensitivity at angle θ is

$$S(\theta) = A + B\cos\theta, \quad A + B = 1$$

Pattern	A	Null angle	At 90°	At 180°	Directivity index	Distance factor
Omnidirectional	1	none	0 dB	0 dB	0 dB	1.0
Subcardioid	0.7	none	−3.1 dB	−8.0 dB	2.8 dB	1.4
Cardioid	0.5	180°	−6.0 dB	null	4.8 dB	1.7
Supercardioid	0.37	125°	−8.7 dB	−11.4 dB	5.7 dB	1.9
Hypercardioid	0.25	110°	−12.0 dB	−6.0 dB	6.0 dB	2.0
Figure-of-eight	0	90°	null	0 dB	4.8 dB	1.7

The directivity index says how much less of a diffuse sound field, such as reverberation or the sound of a crowd, the microphone picks up than an omnidirectional one. The distance factor says how much further away a directional microphone can be placed for the same ratio of direct to diffuse sound. A hypercardioid at 0.2 m picks up as much room sound as an omnidirectional microphone at 0.1 m.

The null angle decides where a monitor wedge belongs. A cardioid rejects most at the back, so its wedge sits directly behind it, in front of the performer. A supercardioid or hypercardioid rejects most at 110° to 125° and has a rear lobe, so its wedges belong at about 55° to 70° on either side of the line behind the microphone, which usually means

two wedges, one on each side. Placing a wedge for a hypercardioid where a cardioid's would go puts it into the rear lobe.

Real microphones follow these patterns only in the midrange. At low frequencies most directional microphones become less directional, and at high frequencies the size of the capsule narrows the pattern. A pattern graph at several frequencies in the data sheet, or a polar plot at 125 Hz, 1 kHz and 8 kHz, shows how far the microphone keeps its pattern. For feedback, a microphone with smooth off-axis response gains more than one whose pattern is sharper on paper but has peaks off axis.

Frequency response and proximity effect

A vocal microphone usually has a presence peak of a few decibels between 3 and 8 kHz and a roll-off below about 100 Hz. Both are intended. The peak helps the voice cut through, and the roll-off compensates for the proximity effect. A measurement microphone is flat instead, and chapter 16 covers it.

A directional microphone close to a source picks up more bass. The pressure gradient at low frequencies is larger near a point source, because the curvature of the wavefront adds to it. For an ideal cardioid, the boost relative to the far field follows from $|1 + 1/(2jkr)|$, with $k = 2\pi f/c$.

Distance	50 Hz	100 Hz	200 Hz	500 Hz
2 cm	+28.7 dB	+22.7 dB	+16.8 dB	+9.3 dB
5 cm	+20.8 dB	+14.9 dB	+9.3 dB	+3.4 dB
10 cm	+14.9 dB	+9.3 dB	+4.6 dB	+1.1 dB
30 cm	+6.3 dB	+2.6 dB	+0.8 dB	+0.1 dB
1 m	+1.1 dB	+0.3 dB	+0.1 dB	0 dB

These are values for an ideal point source and an ideal capsule, and a real vocal microphone is designed with a rising bass roll-off that takes back much of it at a typical singing distance. Figures of up to 16 dB are commonly quoted. An omnidirectional microphone has no proximity effect, which is one reason lavaliers and headsets are omnis. A singer who uses the proximity effect deliberately gains warmth, and a public speaker who leans in by accident loses intelligibility. A high-pass filter in the channel, set by ear between 80 and 150 Hz, is the remedy.

Peaks in the frequency response matter for a further reason. A peak in the response of any element in the loop is where feedback starts first, which is why a microphone with a smooth response gains more than one with a peaky response and the same pattern.

Sensitivity, noise and maximum level

Sensitivity

Sensitivity is the output voltage for 1 Pa, which is 94 dB SPL, given in mV/Pa or in dBV re 1 V/Pa.

Sensitivity	dBV re 1 V/Pa	At 94 dB SPL	At 120 dB SPL	At 135 dB SPL
1.5 mV/Pa (dynamic vocal)	−56.5	−54.3 dBu	−28.3 dBu	−13.3 dBu
2 mV/Pa	−54.0	−51.8 dBu	−25.8 dBu	−10.8 dBu
5 mV/Pa	−46.0	−43.8 dBu	−17.8 dBu	−2.8 dBu
10 mV/Pa (lavalier)	−40.0	−37.8 dBu	−11.8 dBu	+3.2 dBu
20 mV/Pa (studio condenser)	−34.0	−31.8 dBu	−5.8 dBu	+9.2 dBu

The preamplifier must take the whole range, from a quiet voice on a dynamic microphone at around −60 dBu to a condenser on a kick drum at over +5 dBu. A condenser at 135 dB SPL delivers a level that overloads many microphone inputs at their lowest gain, which is why distortion with close miking usually comes from the preamplifier, not the microphone. A pad in the microphone or at the input solves it.

Self-noise and the noise of a dynamic microphone

A condenser's self-noise is given as an equivalent sound level, such as 13 dB(A), the level of a sound that would give the same output as the microphone's own noise. Below about 15 dB(A) is quiet, and above 25 dB(A) is audible on quiet sources.

A dynamic microphone has almost no self-noise beyond the thermal noise of its coil, but its low output brings the preamplifier's noise into play. With 1.5 mV/Pa and a preamplifier with an EIN of −128 dBu, the signal at 94 dB SPL is 73.7 dB above the noise, which is equivalent to a microphone with a self-noise of about 20 dB. A condenser with 20 mV/Pa into the same preamplifier is limited by its own self-noise instead. For speech at a distance, a condenser is quieter. For a vocal at 5 cm, both are far above any noise.

Maximum level

Maximum SPL is the level at which a condenser's distortion reaches a stated value, usually 0.5 % or 1 % THD. It is limited by the impedance converter, not the capsule, so a pad switch in the microphone raises it. A dynamic microphone has no practical limit on a stage.

Phantom power and connection

Phantom power

Condenser microphones, active ribbons and active DI boxes are powered through the microphone cable. The same voltage is applied to both signal wires through matched resistors, and the current returns through the shield. The audio signal between the wires is unaffected, and a balanced dynamic microphone sees no voltage across its coil.

IEC 61938 defines three variants.

Variant	Voltage	Feed resistors	Current at most
P48	48 V	6.81 kΩ per wire	about 10 mA
P24	24 V	1.2 kΩ per wire	about 10 mA
P12	12 V	680 Ω per wire	about 15 mA

P48 is standard on mixing consoles and stageboxes. With its high feed resistors, the power available is small. A short circuit draws 14 mA, and the most a device can draw is about 169 mW, at 24 V and 7 mA. A microphone or DI box that needs more current than its supply provides loses headroom without warning, which is a common fault on cheap interfaces and battery recorders.

Phantom power is safe for balanced dynamic microphones and active ribbons. It is a risk for a passive ribbon if a cable or adapter has a fault, for an unbalanced source connected through an adapter, and for any device when connections are made with the power on, because the switching surge reaches the output of the device and the input of the preamplifier. Phantom power is switched on after the connection and off before it is broken, with the channel muted.

Balanced connection

A professional microphone has a low source impedance of 150 to 300 Ω and a balanced output on an XLR connector. The input it feeds has a load impedance of 1 to 3 kΩ, at least five to ten times the source impedance, so that the voltage arrives almost unchanged. Balanced means that both wires have the same impedance to ground, so noise induced along the cable is equal on both and cancelled by the differential input. The input's common-mode rejection ratio (CMRR) says how well it does this. Chapter 15 covers cables and connectors.

Accessories

Wind noise is turbulence at the microphone, not sound, and it overloads the low frequencies. A foam windscreen helps indoors and in a light breeze, and a basket with a fur cover helps outdoors. A pop filter or the grille of a vocal microphone stops the blasts of plosives such as p and b. A shock mount isolates the microphone from the vibration of its stand, which a stage floor or a kick drum passes into it. The high-pass filter in the channel removes what is left below the instrument's range.

Digital microphones

AES42 defines a microphone with its own converter and a digital AES3 output, powered with 10 V over the cable, and with remote control of gain and pattern. Such microphones avoid the analog cable and the preamplifier entirely, but they need inputs that support the standard and remain rare on live stages. Network microphones that connect over an audio network such as Dante are another variant, mostly in conference systems.

DI boxes and instrument inputs

A DI (direct injection) box takes the output of an instrument, an amplifier or a playback device and turns it into a balanced, low-impedance signal at microphone level, so that it can run on the same multicore and into the same preamplifiers as the microphones.

Why they are needed

An electric guitar or bass pickup has a source impedance of many kilohms, rising steeply with frequency through its inductance. It needs a load of about 1 MΩ, and a lower load

dulls its tone. An unbalanced output also picks up hum on a long cable. A DI box provides the high input impedance, a balanced output and a ground separation.

Passive and active

- **Passive DI** uses a transformer with a turns ratio of typically 10 to 12 to 1, which lowers the voltage by 20 to 21.6 dB and the impedance by 100 to 144 times. It needs no power, isolates galvanically, and handles high levels well. Its input impedance is lower than that of an active DI, which suits line sources, keyboards and active basses better than passive pickups.
- **Active DI** uses an amplifier with an input impedance of 1 MΩ or more, powered by phantom power or a battery. It suits passive pickups, and its output level is set for a microphone input.

A ground lift switch separates the shield of the output from that of the input. It interrupts a ground loop between the stage equipment and the console, which chapter 17 discusses. A pad of 20 to 40 dB is needed for line-level and loudspeaker-level sources, and some DI boxes take the loudspeaker output of an instrument amplifier directly.

Stereo and playback

Laptops, playback devices and keyboards usually feed a stereo DI with two channels. The ground loop between a laptop on mains power and the console is the most common source of hum on a stage today, and a transformer DI or a ground lift removes it.

Preamplifiers in the console

The microphone preamplifier in the console or stagebox raises signals from around −60 dBu to line level or to the converter's working level, which can need up to about 70 dB of gain. Phono preamplifiers with their RIAA equalization now matter only for DJ turntables and are built into DJ mixers. Chapter 4 covers gain structure and chapter 11 the console.

To verify in pass 2

- The coefficients of the supercardioid (0.37) and hypercardioid (0.25) and their null angles; check textbook definitions (Eargle, *The Microphone Book*).
- The proximity effect model $|1 + 1/(2jkr)|$ for an ideal cardioid, and typical real values.
- Typical self-noise categories (below 15 dB(A) quiet, above 25 dB(A) audible).
- IEC 61938 phantom power values, tolerances and maximum currents for P48, P24 and P12.
- AES42 powering (10 V) and features.
- Typical turns ratios and input impedances of passive DI boxes.
- The 1 MΩ load for passive pickups.
- The maximum preamplifier gain of about 70 dB in current stageboxes.

Sources

- Eargle, J.: *The Microphone Book*, Focal Press.

- IEC 61938, guidance on interconnections and matching values, including phantom power.
- AES42, digital interface for microphones.
- Davis, G., Jones, R.: *The Sound Reinforcement Handbook*, 2nd ed., Yamaha / Hal Leonard 1989, §10.1 to §10.4 (pp. 113–131) and §11.2 (pp. 152–154).

10. Wireless microphones, in-ear monitors and intercom

> **Draft, pass 1.** This chapter was written from general knowledge and checked against reference literature for coverage. Standards, editions and figures are named but have not yet been verified against their sources. The list *To verify in pass 2* at the end names every claim that still needs one.

This chapter answers where wireless audio may transmit in Europe and Austria, how analog and digital systems differ, how several systems are made to work side by side, how antennas and cables decide the range, and how in-ear monitors and crew intercom fit in. The frequency rules have changed more than anything else in this book. Wireless microphones once used VHF and free TV channels, while today's European bands are the result of two rounds of clearing TV spectrum for mobile networks. All frequency ranges and licensing rules below are marked for verification, because they change and differ between countries.

Where wireless audio may transmit

Wireless microphones and in-ear monitors are part of what regulators call programme making and special events (PMSE). In most European bands they are secondary users, which means they may use a frequency only where it does not interfere with the primary service, and they must accept interference from it.

Band	Status in the EU	Typical use
470–694 MHz (UHF TV band)	Secondary use in channels not used by TV in the area, at least until 2030	The main band for professional microphones and in-ear monitors
694–790 MHz (700 MHz band)	Cleared for mobile networks by 2020; a part of its duplex gap open for PMSE in some countries	Older equipment now unusable outside the gap
790–862 MHz (800 MHz band)	Cleared for mobile networks around 2010–2015	Old equipment unusable, except in the duplex gap
823–832 MHz (duplex gap of the 800 MHz band)	Harmonised for PMSE, usually licence-exempt	A few channels, close to mobile uplink
863–865 MHz	Licence-exempt short-range devices	Consumer and entry-level systems, crowded
1785–1805 MHz	Harmonised for PMSE	Professional systems as an alternative to UHF
1880–1900 MHz (DECT)	Licence-exempt	Intercom, some conference microphones
2.4 GHz (ISM)	Licence-exempt, shared with Wi-Fi and Bluetooth	Digital consumer and some professional systems

In the UHF TV band, which channels are free depends on the place. A TV transmitter in Salzburg occupies different channels than one in Vienna, so a touring system needs a

frequency plan for each venue. In Austria, the use of wireless microphones in the TV band is coordinated by the telecommunications authorities, and some bands are covered by a general authorisation while others need an individual assignment. In Germany, the Federal Network Agency issues general authorisations for most PMSE bands, with an individual assignment for larger events.

Equipment bought before the clearing of the 800 and 700 MHz bands may still be in a rental stock or an association's storage. It may no longer be used in the cleared ranges, and using it there interferes with mobile networks and can be prosecuted.

Analog and digital systems

Analog FM

An analog wireless microphone frequency-modulates its carrier with the audio. To gain dynamic range, it compresses the audio before transmission, typically at a ratio of 2 to 1, and expands it in the receiver. The compander is proprietary, which makes systems of different makes incompatible. A transmitter of one make with a receiver of another may work, but it sounds wrong because the compander and the frequency deviation do not match. Analog systems have almost no latency and degrade gradually as the signal weakens, with noise and then crackle before a dropout.

Digital

A digital system converts the audio, codes it and transmits it as data. It needs no compander, so its sound does not change with level, and it is more tolerant of intermodulation, which allows closer spacing of channels. It adds latency, typically 2 to 4 ms for professional systems, which matters for in-ear monitors and when a wireless and a wired microphone pick up the same source. It also fails abruptly. A digital link stays perfect until it loses too many packets and then mutes. Some systems encrypt the audio, which matters for confidential speech at conferences.

Neither type is compatible between makers, and digital systems are often not compatible between product lines of one maker.

Several systems side by side

Intermodulation

When two or more transmitters are close together, for example on performers who stand next to each other or in a rack of in-ear transmitters, their signals mix in the output stages and in the receivers' input stages. The result is new signals at frequencies that are combinations of the original ones. The strongest are the third-order products $2f_1 - f_2$ and $2f_2 - f_1$ for every pair, and $f_1 + f_2 - f_3$ for every triple.

With carriers at 606.0, 606.4 and 606.8 MHz, equally spaced by 400 kHz, the third-order products fall at 605.2, 605.6, 606.0, 606.4, 606.8, 607.2 and 607.6 MHz. Each of the three carriers is hit by a product of the other two, so equal spacing is the worst plan. Moving the third carrier to 607.2 MHz moves all products off the carriers.

The number of products grows quickly with the number of systems.

Systems	Products $2f_1 - f_2$	Products $f_1 + f_2 - f_3$
4	12	12
8	56	168
16	240	1680
32	992	14 880

Beyond a handful of systems, a frequency plan has to be calculated, which is what the coordination software of the microphone makers and independent tools do. They take the free TV channels, the products up to fifth order, the required spacing for each model and the systems already in use, and they produce compatible groups.

Scanning and coordination

A receiver or a scanner measures the RF spectrum at the venue, and the coordination software places the channels in the quiet parts. The scan is repeated on the day of the show with the other wireless users present, because a TV transmitter, an OB van or the next stage at a festival changes the picture. At a larger event, one person coordinates all wireless users, and every system that arrives is entered into one plan.

Range, antennas and cables

What limits the range

Radio signals weaken with distance in the same way as sound in a free field, and the free-space path loss in decibels for a distance d in kilometres and a frequency f in megahertz is

$$\mathrm{FSPL} = 20\lg d + 20\lg f + 32.44$$

Distance	600 MHz	1800 MHz	2.44 GHz
10 m	48.0 dB	57.5 dB	60.2 dB
30 m	57.5 dB	67.1 dB	69.7 dB
50 m	62.0 dB	71.5 dB	74.2 dB
100 m	68.0 dB	77.5 dB	80.2 dB

A transmitter of 10 to 50 mW, or 10 to 17 dBm, therefore reaches a stage-sized range comfortably in free space. The problems come from the rest. The human body absorbs strongly at these frequencies, so a beltpack on the back of a performer who faces away from the antenna can lose 10 to 20 dB. Higher bands lose more to distance and to bodies.

Multipath causes dropouts. The signal reaches the receiver by several paths, directly and reflected from walls, trusses and the floor, and at some positions the paths cancel. The cancellation zones are about half a wavelength apart, which is 25 cm at 600 MHz, so a performer who takes a step moves from a dead spot to a good one.

Diversity

A diversity receiver has two antennas at different positions and uses whichever receives the better signal at each moment, or combines both. It is unlikely that both are in a cancellation zone at once. The antennas are placed at least a quarter wavelength apart, and in practice one metre or more for wide-area coverage.

Antennas

Each receiver comes with two small whip antennas. Their length matches a quarter wavelength, about 10.8 cm at 694 MHz and 15.9 cm at 470 MHz. For more than a few systems, or when the receivers are not at the stage, an antenna distribution system is used. It takes two antennas, amplifies their signals and splits them to all receivers.

Directional antennas, such as log-periodic paddles and helical antennas, gain several decibels towards the stage and reject signals from behind. High-gain beams with a very narrow pattern would need to follow the performer and are avoided, but the moderate directivity of a paddle or a helical aimed at the performance area is now standard. Helical antennas receive circular polarisation, which suffers less when a transmitter is held at an angle.

Cables

Antenna cable loses signal with length and frequency, and the loss comes before the receiver's input stage, so it adds directly to the noise. Placing the receivers near the stage and running the audio over a long line instead avoids that loss, and a networked receiver placed at the stage is the modern form of it.

Cable type, loss per 100 m at about 600 MHz	10 m	20 m	30 m
RG58, about 35 dB	3.5 dB	7.0 dB	10.5 dB
RG213, about 14 dB	1.4 dB	2.8 dB	4.2 dB
Low-loss 10 mm, about 12 dB	1.2 dB	2.4 dB	3.6 dB

RG58 is acceptable for a few metres only. A booster at the antenna makes up for a long cable but also amplifies interference, so it is set to compensate the cable loss and no more. All cables, splitters and antennas are 50 Ω.

In-ear monitors

An in-ear monitor (IEM) system is a wireless link in the other direction, from a transmitter at the monitor position to a bodypack receiver worn by the performer. Analog systems use stereo FM with a multiplex signal like FM radio, and digital systems are spreading.

Several IEM transmitters in one rack produce strong intermodulation with each other, because their outputs are close together and at full power. An active combiner merges them into one antenna with isolation between the inputs. IEM frequencies are placed in the frequency plan together with the microphones, often in a separate part of the band, so that the strong IEM carriers do not land products on the weak microphone signals at the receiver.

The latency of a digital IEM link adds to the latency of the console. A singer hears their own voice through the bone of the skull at once and through the in-ear mix with a delay, and more than a few milliseconds of total delay can be noticed as a comb filter or a doubling. Chapter 18 covers monitor mixing and in-ear levels, and chapter 5 the risk to hearing.

Intercom

Wireless intercom began as a specialised tool for TV and theatre crews, with belt packs, base stations and the distinction between simplex, half duplex and full duplex. Today intercom has become a system in its own right at most events beyond the smallest.

- **Wired partyline.** A two-wire or four-wire system over microphone cable, with belt packs and speaker stations. All stations on a channel hear each other. Robust and simple, still common.
- **Wireless.** Belt packs that work full duplex, so that a crew member talks without pressing a button. DECT-based systems at 1.9 GHz are licence-exempt in Europe and handle many belt packs, and other systems use 2.4 GHz or 5 GHz.
- **Matrix and IP.** A central matrix routes any station to any other, and IP-based systems carry intercom over the same network as the audio, often using AES67 for interoperability with other audio devices. Chapter 15 covers networks.

At an event, a minimum intercom connects the front-of-house position, the monitor position and the stage manager. Where a crew depends on it for safety, such as for flown loads or pyrotechnics, a second, independent path is planned.

Batteries and practice

Batteries decide more shows than any specification. Alkaline AA cells give the longest runtime per cell in most transmitters but are used once. NiMH cells and the makers' lithium packs are rechargeable, and the packs report their remaining runtime to the receiver, which shows it at the console. A routine that has served many crews is to start every show with fresh or fully charged cells, to note the time of insertion, and to watch the runtime display during the show.

Further practice that keeps wireless reliable:

1. **Plan before the venue.** Check the free TV channels for the location, plan with coordination software and bring spare frequencies.
2. **Scan on site** with all systems of all users switched on.
3. **Walk the stage** with a transmitter before the show, listening and watching the receiver's RF meters for dropouts.
4. **Antennas in line of sight** of the performance area, above the heads of the audience, away from metal and LED walls, which radiate broadband interference.
5. **Spares ready.** A spare handheld with fresh batteries, set up and tested, at the side of the stage.

To verify in pass 2

- Every band in the table, its status in the EU and in Austria, and the end date of PMSE use in 470–694 MHz; check Decision (EU) 2017/899, ECC Report 204 and ERC Recommendation 70-03 Annex 10, and the Austrian frequency usage plan (FNV) and RTR information.
- The licensing rules in Austria (general authorisation or individual assignment by band) and the German BNetzA general authorisations.
- The PMSE parts of the 700 MHz duplex gap (around 738–753 MHz) and their availability in Austria.
- The typical latency of digital systems (2 to 4 ms) and of digital IEM systems.
- The 10 to 20 dB loss through a performer's body at UHF.
- The cable loss figures per type at 600 MHz.
- The DECT intercom band and the use of AES67 in IP intercom systems.

Sources

- Decision (EU) 2017/899 on the use of the 470–790 MHz band in the Union.
- CEPT ECC Report 204 and ERC Recommendation 70-03.
- Austrian Frequenznutzungsverordnung (FNV), RTR.
- AES67, high-performance streaming audio-over-IP interoperability.
- Davis, G., Jones, R.: *The Sound Reinforcement Handbook*, 2nd ed., Yamaha / Hal Leonard 1989, §10.5 and §10.6 (pp. 132–148).

11. Mixing consoles

> **Draft, pass 1.** This chapter was written from general knowledge and checked against reference literature for coverage. Standards, editions and figures are named but have not yet been verified against their sources. The list *To verify in pass 2* at the end names every claim that still needs one.

This chapter answers what a mixing console does with its inputs, how its channels, busses and outputs are organised, how a digital console with a stagebox differs from an analog desk, how one set of microphones feeds several consoles, and where the front-of-house console belongs. Gain structure is in chapter 4, microphone preamplifiers and DI boxes are in chapter 9, and monitor mixing is in chapter 18.

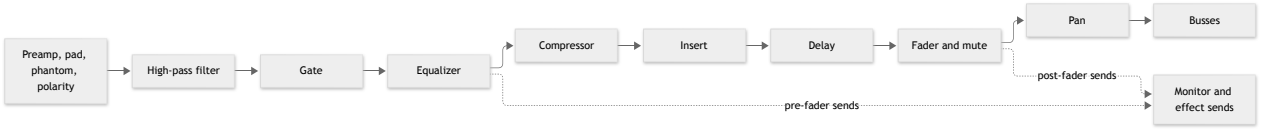
What a console does

A console takes many inputs, adjusts each one in level, tone and dynamics, and combines them into a smaller number of mixes. The same inputs usually feed several mixes at once. The main mix for the audience, a monitor mix for each performer, a recording or a stream, and a feed for a broadcaster all want different balances of the same sources.

Consoles used to be distinguished by the mix they make, such as house, monitor, recording and broadcast consoles. The distinction remains in how the console is used, but a digital console of moderate size can make all of these mixes at once, and at small events one console and one engineer often do.

The channel strip

Each input passes through a chain of processing in a fixed order. The order matters, because a change early in the chain affects everything after it. A typical modern channel looks like this.



- **Preamp section.** Gain, pad, phantom power and a polarity switch, followed by the converter in a digital console. The gain is set here and nowhere else, as chapter 4 explains.
- **High-pass filter.** Removes rumble, handling noise and proximity effect below the source's range. It is the most useful single control on a vocal channel.
- **Gate and compressor.** Dynamics processing per channel, which chapter 12 covers. On many digital consoles the order of gate, equalizer and compressor can be changed.
- **Insert.** A send and return point for external processing.
- **Delay.** Aligns a source in time with others, for example a close microphone on a guitar amplifier with the microphones further away, or a source on stage with its

own direct sound in the hall.
- **Fader, mute and pan.** The fader sets the channel's contribution to the main mix, and the pan places it between left and right.

Sends

Sends feed the auxiliary busses that make monitor mixes and effect returns. A pre-fader send takes the signal before the fader, so a monitor mix does not change when the engineer moves the channel fader for the audience. A post-fader send follows the fader, which is right for reverb and other effects that should keep their proportion to the dry signal. On most consoles each send can be switched between the two, and the tap point of the pre-fader send can be set before or after the equalizer and dynamics.

Panning

Pan places a mono source between two outputs. A pan pot that only turned one side down would make a centred source louder in the room than a source panned hard to one side, because both loudspeakers would play it at full level. The pan law reduces both sides at the centre to compensate.

Pan law at the centre	Level of a centred source in a mono sum, relative to a hard-panned one
−3 dB	+3.0 dB
−4.5 dB	+1.5 dB
−6 dB	0 dB

A −3 dB law keeps the acoustic power constant across the stereo field, which suits a room where both loudspeakers are heard. A −6 dB law keeps the electrical sum constant, which suits a mono sum or a broadcast. Many consoles use −3 dB, some allow a choice. In a live system that covers most of the audience from one side, the pan law matters less than the coverage, and many engineers mix vocals and bass in mono and use the stereo field only for effects.

Busses, groups, matrices and DCAs

The outputs of the channels are combined on busses. A console offers several kinds.

Bus	What it carries	Typical use
Main or master	The main mix, left and right, often with a mono or centre bus	The PA
Aux	Individual mixes from the sends	Monitors, effects, recording feeds
Group or subgroup	A summed group of channels, with its own processing and fader	Drums, vocals, a choir, compressed together
Matrix	Mixes of busses rather than channels	Delay loudspeakers, front fills, broadcast and recording feeds, each with its own level and delay
DCA or VCA	No audio, only control	One fader that moves the level of several channels without summing them

A DCA (digitally controlled amplifier) group, the successor of the VCA groups of analog consoles, lets one fader ride a set of channels while each keeps its own sends, processing

and routing. It is the usual way to give an engineer eight faders for the whole show.

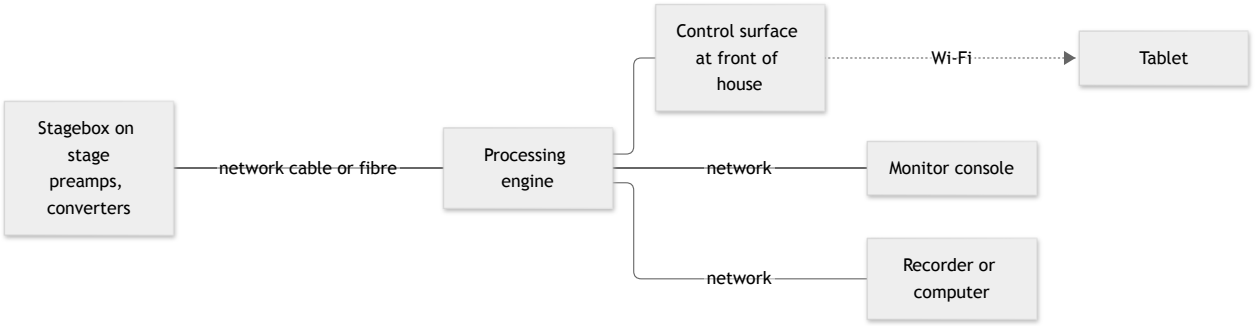
The noise of a summed bus rises with the number of channels routed to it, by $10 \lg N$ when their noise is independent. Sixteen channels add 12 dB and 48 channels 16.8 dB to the noise of one. Analog consoles therefore offered switches to unassign unused channels, and what is not used is not routed. In a digital console the summing is noiseless, and only the analog inputs contribute.

Meters

Analog consoles showed levels on a VU meter, which reads an average, or a peak meter. A digital console replaces both with a meter in dBFS that shows the sample peak and often a slower average, and it adds a gain reduction meter for every dynamics processor. Two more meters have become common. A true-peak meter shows the level between the samples, which matters at the output to a converter or a stream, and a loudness meter in LUFS shows the perceived loudness of a mix, which broadcasters and streaming platforms specify. Chapter 3 explains all three.

Digital consoles and stageboxes

The largest change in recent decades is that the console no longer processes analog audio. A digital console consists of a control surface, a processing engine and input and output units, which may be in one box or spread out.



The stagebox sits on the stage with the microphone preamplifiers and converters, and a single network or fibre cable replaces the analog multicore. The preamplifiers are controlled remotely from the console. Chapter 15 covers the networks.

Features that analog consoles lacked change how a show is run.

- **Scenes and snapshots.** The console stores the complete state, or a chosen part of it, and recalls it at a button press, for each song, each act or each scene of a play.
- **Recall safe and scope.** Channels or parameters marked as safe are not changed by a scene recall, so that a scene for a song does not reset the preamplifier gain or the main equalization.
- **Remote control.** A tablet controls the console over Wi-Fi, which lets the engineer walk the room or stand at the monitor position, and lets performers set their own monitor mixes.
- **Virtual sound check.** The console records every input to a computer over the network, and plays the recording back through the channels later, so the mix can be

refined without the band.
- **Latency compensation.** Each path through a digital console takes a little time, and an insert or a plug-in adds more. The console delays the other paths to match, so that sources that are mixed together stay aligned. Typical latency from analog input to analog output is 1 to 3 ms, which chapter 15 discusses together with networks.

Choosing a console's size

Counting inputs, mixes and outputs is the first step in choosing a console. A practical count includes every microphone and DI on the input list, the returns from effects and playback, a few spares, and every monitor mix, matrix output and recording feed. Digital consoles often have more mixing channels than local inputs, and the stagebox decides how many sources can be connected.

Balancing with transformers or electronic inputs

A balanced input can use a transformer or an electronic differential amplifier. Both reject the noise that is common to the two wires of a balanced line. The arguments are now mostly settled.

- **Electronic inputs** are cheap, small, flat in frequency and free of the low-frequency saturation of a small transformer. They are standard on every console and stagebox.
- **Transformers** provide galvanic isolation, which breaks ground loops and protects against fault voltages between two systems, and they have a high common-mode rejection that does not depend on the balance of the source. They are used where isolation matters, in splitters, DI boxes and broadcast feeds.

Transformers have one safety advantage. A transformer between two systems that are fed from different mains supplies prevents a fault current from flowing along the audio cable. Chapter 17 covers earthing and electrical safety.

Splitting microphone signals

When several consoles need the same microphones, typically front of house, monitors and a recording or broadcast, the signals are split. Four methods are in use.

Method	How it works	Strengths	Weaknesses
Y-split (direct)	Parallel wiring of the inputs	Cheap	Phantom power and ground from both consoles, hum, risk
Transformer split	A transformer with one primary and two or three secondaries	Isolation, no power needed	Weight, cost, each extra secondary slightly loads the microphone
Active split	A splitter amplifier with several isolated outputs	Many outputs, gain for long lines	Needs power, adds noise and a point of failure
Digital split	One stagebox feeds several consoles over a network	No extra hardware, any number of consoles	One console must own the preamp gain, or gain compensation is needed

With a transformer split, the direct output goes to the console that supplies phantom power, usually monitors, and the isolated outputs go to the others, whose phantom power

is switched off. With a digital split, the console that owns the preamplifier gain changes the level for everyone. Gain compensation, or a fixed gain agreed before the show, keeps one engineer's change from upsetting the other's mix, which chapter 4 explains in Split systems and gain sharing.

Sub-mixers, feeds and cascades

A sub-mixer combines a group of sources, such as a keyboard rig or a drum kit, and sends a few outputs to the main console. Each stage should take part of any level reduction needed, so that faders keep a useful range of travel and nothing runs at a level where its noise dominates. In a digital system, the same idea applies to cascading two consoles over a network, and to playback computers that send a stereo mix.

Feeds in the other direction also need care. A broadcaster or a stream needs a mix that is balanced differently from the room, because the room adds the acoustic sound of the drums and amplifiers that the PA leaves out. A separate matrix or aux with its own processing and a loudness meter is the usual source. A mix-minus feed sends a remote participant everything except their own voice, so that they do not hear themselves with a delay, which matters for remote guests and for hybrid events.

Where the consoles go

Front of house

The engineer mixes what they hear. The house console therefore belongs where the sound represents what most of the audience hears. In practice this means the following.

- **In the audience, on the axis of the system.** Between the left and right systems, at about two thirds of the depth of the audience. Off-centre positions shift the image and the balance between the sides.
- **Not under a balcony, not at the back wall and not in a booth.** Under a balcony, the high frequencies of the main system are shadowed. At the back wall, the reflection adds bass. In a booth with a window, the engineer hears the booth.
- **Not too far.** At 30 m the sound of the stage arrives about 87 ms after it is played, and at 40 m about 117 ms, which makes it harder to cue and to judge timing. Much beyond 30 to 35 m, a position closer to the stage with a check at the back is preferable.
- **With a clear sight line to the stage** and a cable route that does not cross escape routes or is covered where it does.

At events where the production cannot give up a central position, a position off-axis is used with care. The engineer then walks the room and checks the mix, which a tablet makes easy.

Monitors

The monitor console sits at the side of the stage, where the engineer can see the performers and their signals and can hear the wedges at roughly the performers' level. With in-ear monitors, the engineer listens on the same type of earpiece. Chapter 18 covers monitor mixing.

To verify in pass 2

- The pan laws used by current consoles and whether −3 dB is the most common default.
- Typical analog-to-analog latency of current digital consoles (1 to 3 ms).
- The recommended maximum distance of front of house from the stage (about 30 to 35 m); check rider practice and AES or manufacturer guidance.
- The loading effect of extra secondaries in transformer splitters; check splitter data sheets.
- The term DCA and its relation to VCA groups across manufacturers.

Sources

- AES67, for network audio between stageboxes, consoles and recorders.
- Davis, G., Jones, R.: *The Sound Reinforcement Handbook*, 2nd ed., Yamaha / Hal Leonard 1989, §11.1 to §11.6 (pp. 149–171), §11.8 (pp. 175–177), §11.10 (pp. 182–185) and §11.12 (pp. 189–192).

12. Signal processing

> **Draft, pass 1.** This chapter was written from general knowledge and checked against reference literature for coverage. Standards, editions and figures are named but have not yet been verified against their sources. The list *To verify in pass 2* at the end names every claim that still needs one.

This chapter answers what the processors between the console and the amplifiers do, and what they cannot do. It covers equalizers, delay, reverb, dynamics processors, loudspeaker management, feedback suppression, automatic mixing and the modulation effects. Filters, slopes and phase are explained in chapter 2, compression as range management in chapter 4, and the measurement that guides equalization in chapter 16.

Each of these functions used to be a separate box in a rack. Today almost all of them run inside the digital console or a loudspeaker processor, often several times per channel. What they do has changed less than where they sit.

Equalizers

Kinds of filter

- **Shelving filters** raise or lower everything above or below a corner frequency by a set amount, like the bass and treble controls of a hi-fi amplifier. They suit broad tonal changes.
- **Parametric filters** raise or lower a band around a centre frequency, with three controls for frequency, gain and bandwidth. The bandwidth is given as Q or in octaves, and a Q of 1.41 is one octave and 4.32 one third of an octave, as chapter 2 explains. Four to eight parametric bands per channel are standard on digital consoles.
- **High-pass and low-pass filters** remove everything beyond a corner frequency, with a slope of 6 dB per octave per order. They are covered in chapter 2.
- **Graphic equalizers** have a fixed set of bands, each with a slider, so the positions of the sliders draw a rough picture of the correction. A third-octave graphic has 31 bands on the ISO 266 centre frequencies from 20 Hz to 20 kHz.

Older designs include sweep equalizers, which have an adjustable frequency but a fixed bandwidth, and the so-called paragraphic, which is a set of parametric bands. Both are forms of the parametric filter today.

Graphic equalizers in detail

The bands of a graphic overlap. Two behaviours are common.

- **Constant Q.** Each band keeps its bandwidth whatever its gain, so a small boost is as narrow as a large one. Neighbouring bands then combine with ripple between their centres.
- **Proportional Q.** The bandwidth narrows as the gain rises, so a small boost is broad and gentle and a large cut is narrow. Neighbouring bands combine more smoothly at small settings.

Neither shape matches the slider picture exactly. When two adjacent sliders are both raised by 6 dB, the response between them rises by more or less than 6 dB depending on the design, and a single slider at full cut narrows or widens differently from its neighbours. The graphic shows the intent, while the response curve on a digital console shows the result, which is the one to trust.

What equalization can and cannot fix

An equalizer changes the level of a band for everybody who hears the system. It fixes problems that are the same at every seat, such as the tonal balance of a loudspeaker, the build-up of low frequencies from several boxes and the air absorption at a distance. It cannot fix a problem that differs from seat to seat.

- **Comb filters** from two sources or a reflection put notches at frequencies that change with the position, as chapter 2 shows. A boost at a notch at one seat fills nothing there, because the notch is a cancellation, and adds a peak at the next seat.
- **Room modes** below the Schroeder frequency differ by position too, as chapter 7 shows.
- **Gaps between the ways of a loudspeaker** at the crossover come from the alignment of the drivers. An equalizer can lift the band, but the cancellation stays, and it moves with the listening angle.
- **Reverberation** is energy that the room returns late. An equalizer that lowers a band lowers the direct sound in that band too.

Why tuning with an analyser that shows only level misleads is covered in chapter 5. Modern transfer-function measurement averages several positions and shows the phase and the coherence. Corrections where the coherence is low address reflections and are mostly wasted. Chapter 20 describes system tuning.

Linear phase and dynamic equalizers

A conventional equalizer filter is minimum phase, so it changes the phase together with the level, which chapter 2 explains. An FIR filter can be linear phase, changing the level without the phase, at the cost of latency. In loudspeaker processors, FIR filters are used to correct the phase response of a loudspeaker and to build crossovers without phase shift.

A dynamic equalizer applies a band's boost or cut only when the level in that band exceeds a threshold. It suits problems that come and go, such as a harsh resonance on loud notes of a singer, a boomy note on a bass guitar or sibilance.

Delay

A delay holds the signal back for a set time. Its uses in sound reinforcement are alignment and effect.

Alignment

Sound in air travels 0.343 m per millisecond. A delay that matches a distance brings two sources into time.

Distance	Delay
1 m	2.92 ms
3 m	8.75 ms
10 m	29.15 ms
20 m	58.31 ms
40 m	116.62 ms

- **Delay loudspeakers.** A loudspeaker 40 m from the main system receives a delay of about 117 ms so that its sound arrives with the main system's. A further few milliseconds, often 5 to 15 ms, make the main system arrive first, so that the precedence effect keeps the image on the stage, as chapter 5 explains. Chapter 20 sets the times by measurement.
- **Front fills and subwoofers.** The front fills along the stage edge and a subwoofer array on the ground are aligned to the main system at the seats where they overlap.
- **Microphones on one source.** A microphone close to a guitar cabinet and another further away are aligned in the console to avoid a comb filter in the mix.

History

Tape echo and analog bucket-brigade delays have given way to digital ones. All delay is digital today, with any length, no loss of quality and a resolution of one sample, which is 21 µs at 48 kHz.

Reverb

Reverb adds the decay of a room to a dry signal. In a live system it gives close-miked vocals and instruments the sense of space that they lose by being picked up from a few centimetres away.

Reverb was once made physically, in rooms, ducts, springs and steel plates. Two digital methods are in use today.

- **Algorithmic reverb** builds the decay from networks of delays and filters. It offers parameters such as the decay time, the pre-delay before the reverb starts, the size, the damping of high frequencies and the density of the early reflections. Simulations of plates and springs are among its presets.
- **Convolution reverb** applies a measured impulse response of a real room or device. It sounds like that room, but it offers fewer controls and needs more processing.

In a live room, the reverb of the system adds to the room's own reverberation. A reverberant hall needs less of it, and a reverb with a long tail on a vocal in a hall with 2 s of reverberation mostly reduces intelligibility. A pre-delay of 20 to 60 ms keeps the start of each word dry, and a high-pass filter on the reverb return keeps it out of the low frequencies that the room already has plenty of.

Dynamics processors

Compressor and limiter

A compressor reduces the level of the signal above a threshold by a ratio. With a threshold of −20 dB and a ratio of 4 to 1, a signal 12 dB over the threshold comes out 3 dB over it.

Input	Output	Gain reduction
−30 dB	−30 dB	0 dB
−20 dB	−20 dB	0 dB
−12 dB	−18 dB	6 dB
−8 dB	−17 dB	9 dB
0 dB	−15 dB	15 dB

The controls are the same on every compressor.

- **Threshold and ratio** set where and how strongly it acts. A ratio of 10 to 1 or more acts as a limiter.
- **Attack** sets how quickly the gain falls when the signal rises above the threshold, typically from under a millisecond to about 100 ms. A slower attack lets the start of a note through, which keeps punch.
- **Release** sets how quickly the gain returns, typically from 50 ms to over a second. A short release pumps audibly, a long one holds the level down after a loud passage.
- **Knee** sets whether the ratio applies suddenly at the threshold (hard knee) or gradually around it (soft knee).
- **Make-up gain** restores the level lost to compression.

Newer forms

- **Look-ahead limiter.** The processor delays the signal by a few milliseconds and reacts to what is coming, so it can catch a peak completely. It is standard in loudspeaker processors, where a missed peak can damage a driver.
- **True-peak limiter.** It limits the level between the samples, which matters before a converter or a lossy codec.
- **Multiband compressor.** Splits the signal into bands and compresses each separately, so a loud bass note does not pull down the voice.
- **De-esser.** Compresses only the sibilant band, typically between 4 and 10 kHz, when it exceeds a threshold.
- **Sidechain.** The detector listens to a filtered or a different signal from the one it controls, which is how a de-esser and a ducker work.

Gate, expander and ducker

A gate mutes a signal below a threshold, and an expander lowers it by a ratio. On a drum kit, a gate on each tom removes the bleed from the rest of the kit, and on a noisy instrument it removes the hum between notes. A gate needs a range, so that it lowers rather than mutes when that sounds more natural, and a hold time and hysteresis, so that it does not chatter at the threshold. A gate can also be keyed from another source, such as a kick drum gating a bass tone, on any console with sidechain inputs.

A ducker lowers one signal when another is present, such as the background music when a presenter speaks.

Loudspeaker management

A loudspeaker management processor sits between the console and the amplifiers and turns a mix into what each amplifier channel needs. It combines several functions that were once separate boxes.

- **Crossover.** Splits the signal into the bands for subwoofers, low, mid and high drivers, with chosen filter types and slopes, as chapter 2 explains.
- **Delay.** Aligns the drivers within a box and the boxes within a system.
- **Equalization.** Corrects the loudspeaker's response and the system's response in the room.
- **Limiters.** Protect each driver and amplifier, with thresholds set from the amplifier's gain and the driver's power rating, as chapter 4 shows. A peak limiter protects against clipping and excursion, and an RMS limiter against heat over time.
- **Polarity and gain** for each output.

Many amplifiers now contain this processing, and loudspeaker manufacturers supply presets that set it for their boxes. A preset from the manufacturer is usually the better start than one built by ear.

Feedback suppression and automatic mixing

Feedback suppressors

A feedback suppressor detects a frequency that rings, by its steady and growing sine-like character, and places a narrow notch on it. Fixed notches are set during the ring-out before the show, and dynamic ones follow new feedback during it and are released when it stops. A good suppressor gains a few decibels of gain before feedback with little audible change. A suppressor that is allowed too many notches cuts into the programme and can mistake a sustained note for feedback. Chapter 7 describes the manual ring-out.

Automatic mixers

An automatic mixer opens the microphone of whoever is talking and lowers the others. The most common method, gain sharing, distributes a constant total gain among the microphones in proportion to their levels. A single talker gets full gain, and two talkers at equal level each get −3 dB, so the system behaves as if one microphone were open. The gain before feedback that chapter 6 loses with the number of open microphones, 3 dB per doubling, is gained back. Automatic mixers are standard for panel discussions, conferences and council meetings, and many digital consoles have one built in.

Effects

Flangers, phasers and exciters remain in use as effects rather than corrections.

- **Flanger and chorus.** Mix the signal with a copy that is delayed by a few milliseconds, with the delay slowly varied. A delay of 1 ms places the first notch of

the resulting comb filter at 500 Hz and one of 5 ms at 100 Hz.
- **Phaser.** Mixes the signal with a copy passed through all-pass filters, which shift the phase and produce fewer, irregularly spaced notches.
- **Exciter.** Adds generated harmonics to the high frequencies to make a signal brighter and more present. Used sparingly on dull sources, it can help a voice through a dense mix.
- **Pitch correction and harmonisers** process vocals in real time and are common in some styles of music.

In the SdWa5 rig

The rig's DSP 8x8 was measured on the bench after the MARK Salzburg event, and three findings illustrate this chapter. Its limiters act on the peak, and their threshold holds a 100 Hz signal about 2.2 dB higher than a 1 kHz one. Its compressors produce an image of the signal mirrored around 24 kHz while they reduce gain, so a bus compressor that works all the time colours the top end, while a limiter that catches the odd peak costs nothing. The automatic equalization of one Tecnare top left a hollow of 8.5 dB at 2.5 kHz between its two ways, which the crossover rather than the equalizer has to close.

The findings and their consequences for the rig are in On the bench, 2026-09-29. The raw measurements are in B3, limiter and threshold and Auto EQ and gain rider on the device, and the tools that took them in Gain rider, compressor, auto EQ and measurements.

To verify in pass 2

- The behaviour of constant-Q and proportional-Q graphic equalizers when adjacent bands are combined; check Rane and other manufacturer notes.
- The typical ranges of attack (under 1 ms to 100 ms) and release (50 ms to over 1 s).
- The precedence offset of 5 to 15 ms for delay loudspeakers.
- The de-esser band of 4 to 10 kHz.
- Gain-sharing automatic mixing (Dugan) and its constant total gain.
- The detection methods of feedback suppressors.

Sources

- ISO 266, preferred frequencies.
- Dugan, D.: Automatic microphone mixing, *JAES* 23, 1975.
- SdWa5: Signal chain, section *On the bench, 2026-09-29*; Measurements; Automation.
- Davis, G., Jones, R.: *The Sound Reinforcement Handbook*, 2nd ed., Yamaha / Hal Leonard 1989, §14 (pp. 244–280).

13. Power amplifiers

> **Draft, pass 1.** This chapter was written from general knowledge and checked against reference literature for coverage. Standards, editions and figures are named but have not yet been verified against their sources. The list *To verify in pass 2* at the end names every claim that still needs one.

This chapter answers how much voltage, current and power an amplifier delivers into a loudspeaker, how its ratings are to be read, how amplifiers and loudspeakers are matched, how much a mains circuit can feed, and how the cable between them and the constant-voltage systems of installations fit in. Setting the amplifier's gain against the processor's limiters is in chapter 4, and the loudspeakers themselves are in chapter 14.

Voltage, current and power

An amplifier is a voltage source. It turns a small input voltage into a large output voltage, and the loudspeaker's impedance decides how much current flows and therefore how much power is delivered. Ohm's law and the power law tie the four quantities together.

$$V = IR, \qquad P = VI = \frac{V^2}{R} = I^2 R$$

For a sine wave, the RMS values give the average power. The table gives the voltage and current an amplifier must deliver for a given power into three loads.

Power	8 Ω	4 Ω	2 Ω
100 W	28.3 V, 3.5 A	20.0 V, 5.0 A	14.1 V, 7.1 A
250 W	44.7 V, 5.6 A	31.6 V, 7.9 A	22.4 V, 11.2 A
500 W	63.2 V, 7.9 A	44.7 V, 11.2 A	31.6 V, 15.8 A
1000 W	89.4 V, 11.2 A	63.2 V, 15.8 A	44.7 V, 22.4 A
2000 W	126.5 V, 15.8 A	89.4 V, 22.4 A	63.2 V, 31.6 A

An ideal amplifier with a fixed maximum voltage of 40 V RMS would deliver 200 W into 8 Ω, 400 W into 4 Ω and 800 W into 2 Ω. Real amplifiers fall short of the doubling at low impedances, because their power supply and output stage limit the current. The ratio between the 8 Ω and 4 Ω ratings shows how stiff an amplifier is.

A loudspeaker's impedance is not a resistor. It varies with frequency, falling below the nominal value at some frequencies and rising far above it at resonance, and it has a reactive part that shifts current and voltage apart in phase. An amplifier that drives a nominal 4 Ω load may see 2.5 Ω at one frequency, and the reactive load makes the output stage dissipate more than a resistor of the same value would. Chapter 5 explains why static tests on a resistor miss this.

How amplifiers are built

Output stages

- **Class AB.** The traditional linear output stage. Its efficiency is about 50 to 60 % at full power and much lower at the average levels of music, so it turns most of its input into heat.
- **Class G and H.** Linear stages with two or more supply voltages, or a supply that tracks the signal, which raise the efficiency at average levels. They were the standard for high-power touring amplifiers before class D.
- **Class D.** The output transistors switch on and off at several hundred kilohertz, and a filter recovers the audio from the pulse width. Efficiency reaches about 80 to 90 % at high power, so a 4-channel amplifier of several kilowatts fits into two rack units and weighs a fraction of a linear one. Class D is now the standard for professional amplifiers.

Power supplies

A linear supply with a mains transformer is heavy and draws its current in short peaks. A switch-mode supply (SMPS) is much lighter. With power factor correction (PFC), it draws a sine-shaped current from the mains, which lets more real power through a given breaker and reduces the load on generators. Many amplifiers can also limit their own mains current, so that they do not trip a breaker on a loud passage.

Processing and networks

Most current touring amplifiers include a loudspeaker processor with crossover, equalization, delay and limiters for each output, as chapter 12 describes, and a network connection for audio, control and monitoring. The monitoring reports load impedance, temperature, mains voltage and clip and limit events for each channel, which turns the amplifier rack into a source of live diagnostics.

Ratings

What a power rating means

A power rating is meaningful only with its conditions, as chapter 8 explains. The conditions are the load, the frequency range, the distortion at which the power was measured, how many channels were driven at once and for how long.

- **Continuous power** is what the amplifier delivers with a sine wave, all channels driven, for a stated time.
- **Burst or peak power** is what it delivers for a short time, such as with a burst signal of a few milliseconds, before the supply sags. It is larger and closer to what music asks for, but it is not comparable between manufacturers who use different bursts.
- **Power bandwidth** is the frequency range over which the rated power is reached, as chapter 8 explains.

In the United States, the FTC rule on amplifier power ratings has required a preconditioning period and continuous ratings for consumer amplifiers since the 1970s. CTA-2006 is a later test method for car audio amplifiers. Professional amplifiers are not

covered by either, and their data sheets give whatever the manufacturer chooses. A sheet that states the burst signal, the duration, the load and the number of channels driven can be compared with another that does the same.

Bridging

Two channels of an amplifier can drive one loudspeaker in bridge mode. One channel receives the signal inverted, and the loudspeaker is connected between the two hot outputs, so it sees twice the voltage. Into the same load, the power quadruples in theory. Each channel, however, sees half the load impedance, so a bridged amplifier into 4 Ω loads each channel as if it were driving 2 Ω. A bridged amplifier is therefore never connected to a load below its rated bridged minimum.

Gain and input sensitivity

An amplifier is specified either by its voltage gain in decibels or by its input sensitivity, which is the input level that drives it to rated power. The two describe the same thing. An amplifier rated 1000 W into 8 Ω delivers 89.4 V at full power, which needs 2.25 V or +9.2 dBu at its input with 32 dB of gain, and +15.2 dBu with 26 dB. A fixed voltage gain across all amplifiers of a system makes the processor's limiter thresholds the same for every channel, which chapter 4 uses.

Clipping and protection

When the signal asks for more voltage than the supply provides, the output clips. The peaks are flattened, and the signal gains harmonics at high frequencies. Clipping can damage a tweeter even when the amplifier is small. The clipped signal sends energy into the high-frequency driver that the programme did not have, and a small, heavily clipping amplifier can burn a tweeter that a large amplifier running cleanly would not.

A limiter before the amplifier, set to the amplifier's clip level and the loudspeaker's limits, prevents both. Most amplifiers have a clip limiter of their own, and the processor adds the driver protection, as chapter 4 shows.

Amplifiers also protect themselves, with current limits, thermal sensors and fans, DC protection that disconnects the loudspeaker if a fault puts direct current on the output, and a soft start that limits the inrush current at power-on.

Matching amplifiers to loudspeakers

Power

A loudspeaker's power rating says how much it survives under a stated test signal, which chapter 14 explains. An amplifier is usually chosen with more power than that rating, because a larger amplifier that runs below clipping is safer than a smaller one that clips. A common rule is an amplifier rated at about 1.5 to 2 times the loudspeaker's continuous rating, with the processor's limiters set to protect the loudspeaker. The manufacturer's recommendation for a given box overrides any rule.

Every 3 dB of level costs twice the power, and every 10 dB ten times. A loudspeaker with a sensitivity of 98 dB at 1 W and 1 m reaches $L = L_{1W} + 10 \lg P - 20 \lg d$, so 125 dB at 1

m with 500 W, and 99 dB at 20 m. Doubling the amplifier power to 1000 W adds 3 dB. More level beyond that point comes from more boxes or more efficient ones, not from more amplifier.

Impedance

Loudspeakers connected in parallel lower the impedance, and in series they raise it.

- Two 8 Ω boxes in parallel make 4 Ω, four make 2 Ω.
- Two 8 Ω boxes in series make 16 Ω, which is rarely useful because each box's impedance curve then affects the other's response.

Most professional amplifiers are stable down to 2 Ω, but they deliver their best performance at 4 Ω and above, and many manufacturers recommend no more than two boxes per channel.

The cable and the damping factor

The damping factor is the ratio of the load impedance to the source impedance the loudspeaker sees. An amplifier's output impedance is very small, often a few milliohms, so the cable decides it. The resistance of a copper cable is $R = \rho \cdot 2L / A$, with $\rho = 0.0175\ \Omega\,\mathrm{mm^2/m}$, the length L one way and the cross-section A.

Cable	Length	Resistance	Loss into 8 Ω	Loss into 4 Ω	Damping factor at 4 Ω
1.5 mm²	20 m	0.47 Ω	−0.49 dB	−0.96 dB	8.4
2.5 mm²	20 m	0.28 Ω	−0.30 dB	−0.59 dB	13.8
2.5 mm²	50 m	0.70 Ω	−0.73 dB	−1.40 dB	5.6
4 mm²	20 m	0.18 Ω	−0.19 dB	−0.37 dB	21.6
4 mm²	50 m	0.44 Ω	−0.46 dB	−0.90 dB	8.9

The loss in level is small, but the cable resistance adds to the loudspeaker's own and changes the tuning of a subwoofer, which damps its resonance less. Keeping the amplifiers close to the loudspeakers and the cables short and thick matters more than an amplifier's damping factor of several hundred. Chapter 15 covers loudspeaker cables.

Mains power

A single-phase circuit at 230 V with a 16 A breaker delivers at most 3680 W, and one with 10 A delivers 2300 W. An amplifier of 85 % efficiency could turn that into about 3100 W of continuous output, so the mains circuit, not the amplifier, is often the limit of a rack.

Music does not draw continuous power. Its average lies 6 to 12 dB below its peaks, so a channel that reaches its rated power on the peaks draws a quarter to a sixteenth of it on average. Manufacturers state the mains current at one eighth of rated power as a typical programme figure. Eight channels of 1000 W at one eighth power deliver 1000 W in total, and with 80 % efficiency draw about 1250 W, which fits easily on a 16 A circuit. Heavily compressed programme and limiters that work all the time raise the average towards a quarter, which halves the margin.

Large systems therefore spread their amplifiers over the three phases of a CEE supply, and they switch the amplifiers on one after another, because their combined inrush current can trip a breaker before any programme plays. Chapter 17 covers mains distribution and safety.

Constant-voltage systems

Installations with many loudspeakers far apart, such as public address in a building, background music and voice alarm, use a constant-voltage distribution. The amplifier delivers a fixed voltage at full power, 100 V in Europe and 70.7 V in the United States, and each loudspeaker has a transformer with taps for the power it should draw. A tap of P watts at 100 V presents $Z = 100^2 / P$, which is 1000 Ω for 10 W and 2000 Ω for 5 W.

The loudspeakers are connected in parallel, and the sum of their taps must stay below the amplifier's rating. A 500 W amplifier at 100 V can drive a total of 500 W of taps, which is a minimum load of 20 Ω. The high voltage and low current allow long, thin cables, and the taps make the level of each loudspeaker independent of how many are connected.

Voice alarm systems for evacuation are constant-voltage systems with requirements of their own, under EN 54 for their components and the national installation rules, including monitored lines, battery backup and a minimum intelligibility. Their planning and installation are a specialist field outside the scope of an event system.

In the SdWa5 rig

The rig's mains plan, which places four amplifiers and two DSPs on the three phases of a 16 A CEE supply, is in Mains. It shows that the breaker is the tightest limit of the rig, that the phases trip at about a third of the limiters' maximum, and that the amplifiers are switched on one at a time. The choice of 32 dB of gain on the MM14K amplifiers and 26 dB on the TIP10000q follows from the DSP's ceiling and the limiter thresholds, as Why 32 dB and 26 dB explains. The mains figures there rest on assumed efficiency and programme fractions, which the rig document names as such.

To verify in pass 2

- The efficiency of class AB (50 to 60 %) and class D (80 to 90 %) amplifiers at rated power.
- The FTC rule (16 CFR Part 432), its preconditioning and its scope; CTA-2006 title and scope.
- The rule of 1.5 to 2 times the loudspeaker's continuous rating; check loudspeaker manufacturers' guidance.
- The one-eighth power convention for mains current in amplifier data sheets (IEC 62368-1 or manufacturer practice).
- The resistivity of copper (0.0175 Ω mm²/m at 20 °C).
- EN 54-16 and EN 54-24 for voice alarm components, and the Austrian TRVB and German DIN VDE 0833-4 for installation.

Sources

- 16 CFR Part 432, power output claims for amplifiers utilized in home entertainment products.
- CTA-2006, testing and measurement methods for mobile audio amplifiers.
- EN 54, fire detection and fire alarm systems, parts 16 and 24.
- SdWa5: Signal chain, sections *Mains* and *Why 32 dB and 26 dB*.
- Davis, G., Jones, R.: *The Sound Reinforcement Handbook*, 2nd ed., Yamaha / Hal Leonard 1989, §12 (pp. 193–208).

14. Loudspeakers

> **Draft, pass 1.** This chapter was written from general knowledge and checked against reference literature for coverage. Standards, editions and figures are named but have not yet been verified against their sources. The list *To verify in pass 2* at the end names every claim that still needs one.

This chapter answers how a loudspeaker turns power into sound, why a box of a given size has the bass and the directivity it has, how drivers are combined with crossovers, how boxes are combined into point-source systems, line arrays and subwoofer arrays, how a loudspeaker data sheet is read, and how loudspeakers fail. Amplifiers are in chapter 13, and filters and crossover alignments in chapter 2.

Drivers

Almost every loudspeaker in sound reinforcement uses the electrodynamic principle. A voice coil in the gap of a magnet carries the signal current and moves, and it drives a diaphragm that moves the air.

- **Cone drivers** move a paper or composite cone, from small midrange drivers to 21-inch subwoofer drivers. Their long-throw versions move the cone by tens of millimetres at full power.
- **Compression drivers** drive a light dome diaphragm into a small chamber, which opens through a narrow throat into a horn. The compression raises the efficiency strongly, to the point that a compression driver on a horn is far louder than a cone driver of the same power, and it is the standard for high frequencies in professional systems.
- **Ribbon and air-motion transformer (AMT) drivers** use a thin folded or flat membrane, and they appear as high-frequency drivers in some systems.

Piezoelectric tweeters, which were cheap and needed no crossover, are not used in professional systems today. Neodymium magnets, much stronger per kilogram than ferrite, have cut the weight of drivers by half or more and made flown arrays practical.

Directivity

A driver radiates in all directions when it is small compared with the wavelength, and it concentrates its output forward as the wavelength becomes comparable to its size. As a rule, the pattern starts to narrow where the circumference equals the wavelength, $ka = 1$, and it is narrow where the diameter equals the wavelength.

Driver	Effective diameter	Starts to narrow ($ka = 1$)	Narrow ($\lambda = d$)
1-inch dome	2.5 cm	4.4 kHz	13.7 kHz
6.5-inch cone	14 cm	780 Hz	2.5 kHz
12-inch cone	26 cm	420 Hz	1.3 kHz
15-inch cone	33 cm	330 Hz	1.0 kHz
18-inch cone	40 cm	270 Hz	860 Hz

A 15-inch driver in a two-way box therefore beams above about 1 kHz, and a crossover to the horn below that point keeps the pattern consistent. A subwoofer of any size is practically omnidirectional over its range. Directivity at low frequencies needs size or several sources, which is the reason for the subwoofer arrays later in this chapter.

Horns and waveguides

A horn loads a driver acoustically and shapes its coverage. It controls the pattern down to roughly the frequency at which its mouth is one wavelength wide, so a mouth of 0.5 m controls to about 690 Hz and one of 1 m to about 340 Hz. Exponential horns, whose pattern narrowed with rising frequency, gave way in the 1980s to constant-directivity horns, which hold a stated coverage over a wide band. Today's waveguides continue that line. They are shallow horns optimised by simulation for smooth coverage, often with asymmetric patterns that throw further at the top of the pattern than at the bottom.

A horn whose throat is driven hard adds distortion, because air is not linear at very high pressure. This is part of why a compression driver sounds harsh when it is pushed close to its limits.

Enclosures and bass

A cone driver on its own cancels its bass, because the front and the back of the cone radiate in opposite polarity and the sound from the back wraps around. An enclosure separates the two.

- **Sealed box.** The air in the box acts as a spring, and the response falls by 12 dB per octave below the resonance. It is compact and has a good transient response.
- **Vented box (bass reflex).** A port, tuned with the air volume to a frequency near the bottom of the range, adds output there and falls by 24 dB per octave below it. It is the most common design for tops and subwoofers.
- **Bandpass.** The driver sits inside one or two chambers and radiates only through a port, which gives high output in a narrow band.
- **Horn-loaded bass.** A horn, usually folded to fit into a box, raises the efficiency of a bass driver. A real low-frequency horn needs a mouth of several metres, so a folded subwoofer is a compromise, and several of them side by side act as a larger mouth.

The behaviour of a driver in a box can be predicted from a few measured parameters, the Thiele-Small parameters, such as the resonance frequency, the moving mass, the compliance and the losses. Box design software uses them.

One law limits every design. For a direct radiator, the efficiency is proportional to the box volume and to the cube of the lowest frequency, which is often called Hofmann's iron law. To halve the lowest frequency at the same efficiency, the box needs eight times the volume. In the same box, halving the lowest frequency costs a factor of eight in efficiency, or 9 dB. A small box with deep bass is possible, but only by giving up level, which is why loudspeakers for events are large.

Crossovers

A crossover splits the signal so that each driver receives only the band it reproduces well.

- **Passive crossovers** sit inside the box, after the amplifier. They are simple to connect but waste power in their components, cannot be adjusted, and place the amplifier's output impedance behind a network that weakens its control of the bass driver.
- **Active crossovers** split the signal before separate amplifier channels. Their advantages are more headroom because each amplifier handles only one band, efficiency, better damping, and lower intermodulation because a clipping bass channel does not distort the high frequencies.

A digital processor or an amplifier with DSP now provides the crossover along with delay to align the drivers in time, equalization for each way and limiters for each driver, as chapter 12 describes. A Linkwitz–Riley alignment, with two filters of the same order that sum flat in level and in phase, is the usual choice, and chapter 2 explains it. FIR filters can build crossovers with linear phase, at the cost of latency. Loudspeaker manufacturers supply presets that contain all of this for their boxes, and a box run without its preset sounds and behaves differently from its data sheet.

Many current boxes are self-powered, with the amplifier and the processing built in.

Combining boxes

Point-source systems and clusters

A single box or a small cluster of boxes, each covering its own angle, acts as a point source and loses 6 dB per doubling of distance, as chapter 6 explains. Where the coverage of two boxes overlaps, they interfere and produce a comb filter, so a well-built cluster splays its boxes so that their patterns meet at about their −6 dB points. A common estimate for clusters, between 3 and 6 dB more level per doubling of the number of boxes, assumes some overlap. Coaxial boxes, with the horn in the centre of the bass driver, behave like one source and are common as monitors and fills.

Line arrays

A line array is a vertical column of identical boxes, close enough together that they act as one source over most of their range. Its advantage is control of the vertical pattern, which can be shaped by the angles between the boxes and by processing so that the level stays even from the front to the back of the audience. Chapter 6 shows that a line source loses only 3 dB per doubling of distance in its near field.

The boxes act as one source only where their spacing is less than half a wavelength. A box of 0.3 m height couples coherently up to about 570 Hz with conventional drivers, which is why line-array boxes use waveguides that form a flat wavefront at high frequencies across the whole height of the box. Criteria for when a column of sources behaves as a continuous line have been published as the wavefront sculpture technology (WST) criteria, which require that the flat wave sources fill most of the array's height. At low frequencies, a line array is shorter than the wavelength and becomes a point source.

Line arrays are designed in the manufacturer's prediction software, which chapter 20 discusses, because the angles, the trim height and the processing interact.

Steerable columns

A column of small drivers, each with its own amplifier and processing, can steer and shape its vertical beam electronically. Such columns serve speech in reverberant spaces such as churches, stations and halls, where they place the sound on the audience and keep it off the ceiling, which chapter 7 explains.

Subwoofer arrays

Subwoofers are omnidirectional on their own, but several together can be arranged to shape their pattern.

Coupling

Subwoofers placed close together, within a fraction of a wavelength, add coherently in all directions and gain 6 dB per doubling of their number at low frequencies. A floor and a wall add further gain, as chapter 7 shows.

Left and right or centre

Subwoofers placed left and right of the stage, as most systems were built for decades, produce a comb filter across the audience. On the centre line both arrive together, and the sum is loud, which is called the power alley. Off the centre line the path difference creates cancellations.

Listener off the centre line (subwoofers 10 m apart, listener at 30 m)	Path difference	First cancellation
1 m	0.33 m	522 Hz
2 m	0.66 m	261 Hz
5 m	1.62 m	106 Hz
10 m	3.13 m	55 Hz

A few metres off the centre line, the first cancellation falls into the subwoofer range, and the bass changes from seat to seat. A central array, either a line of subwoofers along the front of the stage or a flown array, gives the same bass across the audience and is preferred wherever the stage allows it.

Arcs and delay-shaded lines

A straight line of subwoofers along the stage narrows the pattern in the horizontal plane and sends side lobes to the sides. Placing the boxes on an arc, or electronically delaying the outer boxes so that the line behaves like an arc, widens the pattern to cover a wide audience.

Cardioid arrays

A cardioid array suppresses the sound behind the subwoofers, which keeps the stage quieter and helps feedback, the monitor mix and the neighbours behind the stage.

- **Gradient arrays** use a box facing backwards, or a box behind the front ones, delayed by the distance between them and polarity-inverted, so that its sound

cancels the front boxes' sound behind the array and adds in front. With 0.6 m between the fronts of the boxes, the delay is 1.75 ms.
- **End-fire arrays** place several boxes in a row from front to back, spaced a quarter wavelength apart at a design frequency and delayed so that they add in front and cancel behind. For 80 Hz, the spacing is 1.07 m and the delay per box 3.12 ms.

Both reach well over 10 dB of rear rejection over most of the subwoofer range when they are set up by measurement. Chapter 6 gives the basic delay for a pair, and chapter 20 the measurement.

Reading a loudspeaker data sheet

Chapter 8 explains why data sheets need conditions. For loudspeakers they need more than for any other device.

- **Frequency response.** A curve with a tolerance and the measurement conditions, preferably on axis and at several off-axis angles, and with the processing preset named.
- **Sensitivity.** The level at 1 m with 1 W or with 2.83 V. Into 8 Ω both are 1 W, but into 4 Ω 2.83 V is 2 W, which makes the 4 Ω box look 3 dB more sensitive than it is per watt.
- **Power handling.** Per AES2-2012, with band-limited noise of 6 dB crest factor for two hours, or per IEC 60268-5 with programme simulation noise. It is a survival rating, not a recommendation for amplifier power.
- **Maximum SPL.** Peak, continuous or measured with M-Noise (AES75), which chapter 8 explains.
- **Impedance.** A curve with its minimum, which is what the amplifier sees.
- **Coverage.** Horizontal and vertical angles between the −6 dB points, and a plot of how they change with frequency. A full balloon of the directivity in a format such as GLL or CLF lets prediction software model the box.

How loudspeakers fail

The ways loudspeakers fail have not changed in decades.

- **Over-excursion.** At a constant level, the cone excursion needed rises by a factor of four for every octave down, so 50 Hz needs four times and 25 Hz sixteen times the excursion of 100 Hz. Below the tuning frequency of a vented box, the port no longer loads the driver, and the cone moves freely until it hits its limits. A high-pass filter at or just below the tuning frequency prevents this, and it belongs in every processor preset.
- **Heat.** The voice coil turns most of the power into heat, and its resistance rises with temperature, which lowers the output as the coil heats up, called power compression. Sustained power beyond the rating burns the coil.
- **Clipping.** A clipping amplifier sends high-frequency energy to the tweeter that the programme did not contain, as chapter 13 explains.
- **Feedback, DC and drops.** A sustained feedback tone concentrates power in one band. A fault that puts DC on the output pushes the cone to one side and heats the coil. A box that is dropped can shift its voice coil.

- **Intermodulation.** A cone that moves a long way for the bass modulates the midrange it also reproduces, a form of Doppler distortion, which is another reason to keep large excursions in separate subwoofers.

Limiters in the processor, set from the driver's ratings, prevent most of these. A defective box usually shows itself by an impedance curve that differs from its siblings, which chapter 16 covers.

In the SdWa5 rig

Most SdWa5 cabinets are DIY builds of commercial designs, so their weight, dimensions and rigging points can differ from the original's data sheet. What drifts between a clone and its original lists what to measure first, and the rest of that page describes how a cabinet is measured for the rig's model.

To verify in pass 2

- The $ka = 1$ and $\lambda = d$ rules for the onset of beaming and the effective diameters used.
- The rule that a horn controls its pattern down to where its mouth is one wavelength.
- Hofmann's iron law and its statement in terms of volume and cut-off cubed.
- The WST criteria (Heil and Urban, *JAES* 2003) and the fill-ratio condition.
- The rear rejection of gradient and end-fire arrays (over 10 dB).
- AES2-2012 test signal and duration; IEC 60268-5 programme simulation noise; AES75.
- GLL (EASE) and CLF (Common Loudspeaker Format) as directivity formats.

Sources

- Thiele, A. N.: Loudspeakers in vented boxes, *JAES* 19, 1971; Small, R. H.: Vented-box loudspeaker systems, *JAES* 21, 1973.
- Heil, C., Urban, M.: Sound fields radiated by multiple sound source arrays, AES convention paper, 1992; Urban, M., Heil, C., Bauman, P.: Wavefront sculpture technology, *JAES* 51, 2003.
- AES2-2012, AES75, IEC 60268-5.
- SdWa5: Measuring a cabinet.
- Davis, G., Jones, R.: *The Sound Reinforcement Handbook*, 2nd ed., Yamaha / Hal Leonard 1989, §13 (pp. 209–243).

15. Cables, connectors and networks

> **Draft, pass 1.** This chapter was written from general knowledge and checked against reference literature for coverage. Standards, editions and figures are named but have not yet been verified against their sources. The list *To verify in pass 2* at the end names every claim that still needs one.

This chapter answers which cables and connectors carry analog signals, loudspeaker power and mains power on a sound system, how thick a loudspeaker cable must be, and how digital audio travels over dedicated links and over networks. Grounding and hum are in chapter 17, and the levels the cables carry in chapter 3.

Most failures of a sound system are failures of a cable or a connector, and the cheapest reliability measure on a stage is still a spare for every cable type and a cable tester.

Analog signal cables

Shielding

Microphone and line signals are small and will be amplified, so any interference that enters the cable is amplified with them. A shield around the conductors intercepts electric fields and drains them to ground. Three kinds are common.

- **Braided shields** cover 85 % to 95 % of the conductors, are robust and are the usual choice for cables that are plugged and coiled every day.
- **Spiral (served) shields** are more flexible, but the spiral opens when the cable is bent, which lowers the coverage and can make the cable microphonic, so that handling it produces noise.
- **Foil shields** cover fully and are thin, but they break under repeated flexing, and they suit fixed installation.

A shield does little against magnetic fields from transformers, dimmers and mains cables. The defence against those is a twisted pair, because the field induces opposite voltages in successive twists, and a star-quad cable, with two pairs twisted together and the opposite conductors connected, rejects magnetic hum further. Keeping signal cables away from mains and dimmer cables, and crossing them at a right angle where they must meet, still helps.

Capacitance

Every cable has capacitance between its conductors and the shield, typically around 100 pF per metre for microphone cable. Together with the output impedance of the source it forms a low-pass filter.

Source	Cable	Upper limit (−3 dB)
Line output, 100 Ω	100 m at 100 pF/m	159 kHz
Old transformer output, 600 Ω	100 m at 100 pF/m	27 kHz
Source of 10 kΩ	100 m at 100 pF/m	1.6 kHz

Source	Cable	Upper limit (−3 dB)
Passive pickup, about 47 kΩ	5 m at 100 pF/m	6.8 kHz

A modern low-impedance output drives a long cable without loss. A high-impedance source, such as a passive guitar pickup, loses treble already on a few metres, which is why such instruments go into a DI box close to the source, as chapter 9 explains. The pickup's own inductance makes the real behaviour a resonance rather than a plain roll-off, but the conclusion is the same.

Balanced and unbalanced

An **unbalanced** connection carries the signal on one conductor and returns it through the shield. Any voltage between the grounds of the two devices, and any interference on the shield, adds to the signal. Unbalanced connections suit short runs between devices on the same power outlet, which is the domain of consumer equipment at −10 dBV.

A **balanced** connection carries the signal on a twisted pair, with the shield separate. The receiving input amplifies only the difference between the two conductors, so interference that reaches both equally cancels. What matters is that both conductors have the same impedance to ground at the source and at the input, not that both carry a mirror image of the signal. An output that drives only one conductor and holds the other at ground through the same impedance is balanced in this sense. Professional line and microphone connections are balanced, and they work over hundreds of metres.

Modern equipment uses voltage bridging, with an output impedance of 50 Ω to 200 Ω and an input impedance of 10 kΩ or more. The 600 Ω matching of the telephone era is needed only for old transformer equipment and long telephone lines. Connecting a +4 dBu output to a −10 dBV input, or the reverse, gives an 11.8 dB level mismatch, as chapter 3 computes. Interface boxes, or the trim of the receiving input, correct it, and an unbalanced output into a balanced input needs the cold conductor tied to ground at the source end.

Multicores and stage boxes

A multicore (snake) carries many balanced pairs from the stage to the console, each pair individually shielded, and ends in a stage box with XLR connectors. On most systems today the stage box is digital, with the microphone preamplifiers on the stage and a single network or fibre cable to the console, as chapter 11 describes. Analog multicores remain in use on small systems and as a backup.

Strain relief and care

The cable clamp of the connector must hold the jacket, so that a pull never reaches the solder joints. Cables are coiled over-under so that they lie flat and do not twist, and a cable that crackles when it is moved is replaced rather than repaired on the night.

Connectors

Connector	Use	Notes
XLR-3	Microphone, line, AES3	Locks; pin 1 shield, pin 2 hot, pin 3 cold
XLR-5	DMX512 lighting control, stereo intercom	Not for audio between unknown devices
6.35 mm TRS (jack)	Balanced line, insert points, headphones	Tip hot, ring cold, sleeve shield
6.35 mm TS	Unbalanced instrument and line	Shorts briefly when plugged
3.5 mm TRS	Consumer stereo, laptops, phones	Unbalanced stereo
RCA (phono)	Consumer line, S/PDIF	Unbalanced, no lock
speakON NL2, NL4, NL8	Loudspeaker power	Locks; 2, 4 or 8 poles for one to four ways
powerCON, powerCON TRUE1	Mains to devices	Locks; TRUE1 may be plugged under load
etherCON	Ethernet in rugged housing	Locks over a standard RJ45 plug
CEE (IEC 60309)	Mains distribution	Blue 230 V, red 400 V three-phase; 16 A, 32 A, 63 A
Optical (opticalCON, LC, ST)	Fibre links	Clean caps on every unused end

The XLR connector has been the standard for professional audio for decades. It locks, and pin 1 makes contact first, so the grounds connect before the signal does. The polarity of pin 2 and pin 3 was not uniform in the 1980s, and some American equipment used pin 3 as hot. Pin 2 hot has since been standardized internationally, and equipment built to the older convention is now rare, but a polarity tester (chapter 16) settles any doubt.

The jack plug lacks a lock, and its tip and ring touch the sleeve contact while it is inserted, which can cause loud bangs and short an amplifier output. Jack plugs on loudspeaker cables are therefore avoided, and speakON has replaced them. A speakON NL4 cable carries pins 1+ and 1− for a single box, and pins 2+ and 2− for the second way of a biamped box or for a second box. The convention for which pins carry which way varies between manufacturers, so the pinout of an NL4 or NL8 cable is checked against the box before use.

The pin-1 problem

A device that connects the cable shield (pin 1) to its internal signal ground, rather than directly to the chassis where the cable enters, lets current on the shield flow through its signal circuits. This produces hum and buzz that no balanced input can reject, because it is created inside the device. The effect was described in 1995 and is the subject of AES48, which requires the shield to be bonded to the chassis at the connector. Chapter 17 explains how to find it.

Loudspeaker cables

Loudspeaker cables carry large currents into low impedances, so their resistance matters and their shielding does not. Both conductors count, so the resistance of a cable of length L and cross-section A is $R = 2\rho L / A$, with $\rho = 0.0175\ \Omega\,\mathrm{mm^2/m}$ for copper. The loss in level is $20\,\lg \frac{Z}{Z+R}$ for a load impedance Z.

Cross-section	Length	Resistance	Loss into 8 Ω	Loss into 4 Ω
1.5 mm²	20 m	0.47 Ω	0.5 dB	1.0 dB
2.5 mm²	20 m	0.28 Ω	0.3 dB	0.6 dB
2.5 mm²	50 m	0.70 Ω	0.7 dB	1.4 dB
4 mm²	20 m	0.18 Ω	0.2 dB	0.4 dB
4 mm²	50 m	0.44 Ω	0.5 dB	0.9 dB
6 mm²	50 m	0.29 Ω	0.3 dB	0.6 dB

A loss of less than 0.5 dB is a fair practical limit. With 2.5 mm², it allows about 34 m into 8 Ω and 17 m into 4 Ω, and with 4 mm² about 54 m and 27 m. A subwoofer at 2 Ω, or two boxes in parallel on one cable, halves these lengths again. The cable resistance also lowers the damping that the amplifier exerts on the bass driver, as chapter 13 shows, which is a second reason for the thicker cable on long subwoofer runs, or for placing the amplifiers next to the boxes.

The capacitance and inductance of unsuitable cable, such as coaxial or thin multicore cable, can roll off the response. Proper loudspeaker cable with two heavy conductors avoids this.

Loudspeaker polarity is kept by convention, with pin 1+ to the positive terminal of the amplifier, and every cable is checked once with a polarity tester or by measurement before it goes into stock.

Digital audio links

Digital audio removes the noise and loss of long analog runs and carries many channels on one cable.

Format	Medium	Channels at 48 kHz	Notes
AES3 (AES/EBU)	110 Ω twisted pair, XLR	2	Self-clocking; 75 Ω coaxial variant AES-3id
S/PDIF	Coaxial RCA or optical	2	Consumer variant of AES3
ADAT	Optical	8	Common on interfaces; 4 channels at 96 kHz
MADI (AES10)	Coaxial or fibre	64	Point to point, also 56-channel mode
AES50	Ethernet cable, point to point	48	Not a network; carries its own clock
Dante	Ethernet network	Up to hundreds	Proprietary, most widely used; AES67 mode

Format	Medium	Channels at 48 kHz	Notes
AES67	Ethernet network	Depends on network	Interoperability standard between networks
AVB / Milan	Ethernet network, AVB switches	Depends on network	IEEE 802.1 standards; Milan profile for audio
SMPTE ST 2110-30	Ethernet network	Depends on network	Broadcast, based on AES67

A MADI link with 64 channels of 24 bits at 48 kHz carries about 74 Mbit/s of audio. A gigabit network carries hundreds of such channels, which is why one network cable now replaces a multicore.

Clocking

Every digital device samples at a rate set by a clock, and two devices that exchange audio must share that clock, or samples are lost or repeated at regular intervals, heard as clicks. Point-to-point links such as AES3 and MADI carry the clock in the signal or alongside it on a word clock cable, and one device is the master. Networks distribute the clock with the Precision Time Protocol (PTP, IEEE 1588), which elects one device as the leader and synchronises all others to it. Dante uses PTP version 1, and AES67 and Milan use version 2, which matters when they are combined.

Latency

Each link and each conversion adds latency. A network such as Dante adds a set latency per receiver, chosen between about 0.25 ms and a few milliseconds depending on the number of switches in the path. One millisecond is 48 samples at 48 kHz and equals 0.34 m of sound travel, so a few milliseconds in the signal chain are small next to the distances in a venue, but they add up across a console, a processor and the network, and they matter for in-ear monitoring, as chapter 18 explains.

Networks in practice

Audio networks use standard Ethernet equipment, but they need more care than an office network.

- **Dedicated switch or VLAN.** Audio runs on its own switches, or on a separate VLAN, so that other traffic cannot delay it. Lighting and video may share a switch through VLANs, not as a flat network.
- **Quality of service.** The switch prioritises the clock (PTP) and audio packets over other traffic, with the DSCP values the protocol documents.
- **Energy-efficient Ethernet off.** EEE (IEEE 802.3az) lets ports sleep and disturbs the clock, and it is disabled on every switch port that carries audio.
- **Multicast.** Flows to several receivers are sent as multicast, and IGMP snooping keeps them from flooding every port.
- **Redundancy.** Dante and similar protocols can run a primary and a secondary network in parallel, on separate switches and cables, so that a failed cable does not interrupt the audio.

- **Length.** Copper Ethernet runs up to 100 m per segment. Longer runs, and runs that need electrical isolation, use fibre, with multimode fibre for some hundreds of metres and single-mode fibre for kilometres.
- **Documentation.** A diagram of the network with IP addresses, switch ports and device names, kept with the system, saves an hour on every fault.

USB, Bluetooth and computers

Computers join the system through USB audio interfaces, through the console's own USB or network card, or through a software implementation of the network protocol. Class-compliant USB audio needs no driver on most operating systems.

Bluetooth audio adds a latency of roughly 100 ms to 300 ms, which corresponds to 34 m to 100 m of sound travel. It is acceptable for background music from a phone, but not for any signal that must be in time with live sound or a screen. Its codecs reduce the data, and its range and reliability on a crowded stage are poor, so a cable or a dedicated wireless system is preferred wherever it matters.

In the SdWa5 rig

The connections of one event, MARK Salzburg, list every device and cable of the rig. Which output socket of an amplifier feeds which way of a two-way box over a single speakON cable was found on the bench by measurement, as S3, pinout and high-pass check records, which is the pin-convention problem of the connector section in practice.

To verify in pass 2

- Typical capacitance of microphone cable (about 100 pF per metre) and of star-quad cable.
- Pin 2 hot in IEC 60268-12 and AES14, and when the American convention changed.
- AES48 and the 1995 description of the pin-1 problem (Muncy, *JAES* 43).
- powerCON TRUE1 rated for connection under load; CEE colours and ratings per IEC 60309.
- Channel counts of ADAT, MADI (56 and 64), AES50 at 48 kHz.
- Dante latency settings, its PTP version, and the AES67 and Milan PTP versions.
- The EEE effect on PTP and the switch recommendations of Dante and Milan.
- Multimode and single-mode fibre ranges for the common transceivers.
- Bluetooth latency of current codecs, including LE Audio.
- The speakON NL4 pin convention for two-way boxes across manufacturers.

Sources

- AES3, AES10 (MADI), AES14, AES48, AES50, AES67; IEC 60268-12; IEC 60309; IEEE 1588; IEEE 802.1 (AVB); SMPTE ST 2110-30.
- Muncy, N.: Noise susceptibility in analog and digital signal processing systems, *JAES* 43, 1995.
- SdWa5: signal chain, measurements.
- Davis, G., Jones, R.: *The Sound Reinforcement Handbook*, 2nd ed., Yamaha / Hal Leonard 1989, §15 (pp. 281–302), §17.3 (pp. 325–327), §18.4 (pp. 355–358).

16. Measurement

> **Draft, pass 1.** This chapter was written from general knowledge and checked against reference literature for coverage. Standards, editions and figures are named but have not yet been verified against their sources. The list *To verify in pass 2* at the end names every claim that still needs one.

This chapter answers which instruments a sound system needs, what each of them can and cannot show, and how the measurement of a loudspeaker or a room is set up so that its result can be trusted. Tuning a system with these measurements is the subject of chapter 20, and the transfer function behind most of them is explained in chapter 2.

The classic test bench held a volt-ohm meter, a sine wave oscillator, an oscilloscope, a phase tester, a sound level meter, a real-time analyzer and a loudness monitor. Most of them still exist, but a laptop with an audio interface, a measurement microphone and free software now does the work of the larger part of that list, and does some of it far better.

Electrical test equipment

The multimeter

A digital multimeter that reads true RMS, rather than an average scaled for sine waves, measures AC voltages correctly on any waveform. It checks mains voltage, battery voltage, the DC on a microphone line that carries phantom power, and the continuity of a cable.

Its resistance range also checks a loudspeaker line before the amplifier is connected. It reads the DC resistance of the voice coils, which is lower than the nominal impedance, typically by 20 % to 30 %, so a nominal 8 Ω box reads around 6 Ω.

Line	Nominal	What the meter shows
One 8 Ω box	8 Ω	About 6 Ω
Two 8 Ω boxes in parallel	4 Ω	About 3 Ω
Four 8 Ω boxes in parallel	2 Ω	About 1.5 Ω
Four boxes, one not connected	2.67 Ω	About 2 Ω, a third higher than expected
Short circuit	—	Near 0 Ω

A reading that differs from the sibling lines finds a missing box, a short or a burnt voice coil before power is applied. A constant-voltage line (chapter 13) cannot be checked this way, because its transformers read almost as a short at DC. It needs an impedance meter that measures at 1 kHz, and its reading should match $Z = U^2/P$ for the connected taps, so 100 Ω for ten loudspeakers at 10 W on a 100 V line.

Cable and polarity testers

A cable tester checks every pin of XLR, jack, speakON and network cables for continuity, shorts and swapped conductors in seconds. It is the most useful instrument on a stage per euro spent.

A polarity tester, often called a phase tester, sends a pulse of known polarity into a device or a loudspeaker and reports whether it comes back with the same sign. It has its traps. It is sensitive to level, and it fails on anything that changes phase with frequency. A multi-way box with a crossover reverses the polarity of one way on purpose in some designs, and the phase shift of the crossover filters makes the pulse arrive smeared, so the tester's answer for the whole box is close to meaningless. It is reliable for cables, for single drivers and for checking that all boxes of one type agree with each other. The polarity of a multi-way system is settled by a transfer-function measurement, described below.

Generators and the oscilloscope

A signal generator is now an app or a function of the measurement software, and the console and the processor often contain one. Sine tones find rattles and buzzes, sweeps measure responses, and pink noise is the standard signal for listening and for real-time analysis, because it carries equal energy per octave like music does.

The oscilloscope, once the central bench instrument, has become a specialist tool. A USB oscilloscope or the scope view of the measurement software shows clipping, oscillation and DC offset, which is mostly what a sound system needs. Square-wave tests are covered in chapter 2.

Sound level meters

A sound level meter measures the sound pressure level with a defined frequency weighting and time weighting, as chapter 3 explains. IEC 61672-1 defines two accuracy classes. Class 1 meets tighter tolerances and is required where the measurement has legal consequences, such as noise limits at the neighbours or audience exposure. Class 2 is adequate for setting up a system. An acoustic calibrator per IEC 60942, which produces 94 dB or 114 dB at 1 kHz, checks the meter before and after each measurement.

The meter must read the quantity the limit refers to. Exposure limits refer to an equivalent continuous level such as L_{Aeq} over a stated period, and peak limits to L_{Cpeak}, so a meter that shows only a momentary level with fast time weighting cannot prove compliance. Event limits and their logging are in chapter 5. Phone apps with a calibrated external microphone can come close to class 2, but phone apps on the internal microphone vary by several dB and stop working at high levels.

Analyzers

The real-time analyzer

A real-time analyzer (RTA) shows the level in each band, usually in 31 third-octave bands, of whatever the microphone picks up. Fed with pink noise, a flat system shows a flat line. It was long the tool for setting a graphic equalizer band by band.

An RTA shows only the magnitude and only of the sum of everything at the microphone. It cannot separate the direct sound of the loudspeaker from the reflections of the room, it shows nothing of phase or timing, and it needs pink noise rather than music. These limits led to the dual-channel analyzer.

The dual-channel FFT analyzer

A dual-channel analyzer compares the signal sent to the system, the reference, with the signal the microphone picks up, the measurement, and computes the transfer function between them, magnitude and phase over frequency. Because it compares, it works with any broadband signal, including music during a show. Smaart, Open Sound Meter and Room EQ Wizard (REW) are common programs, the first commercial and the other two free.

The analyzer shows four things.

- **Magnitude.** The response of the system in dB, like an RTA but independent of the signal's spectrum.
- **Phase.** The phase difference over frequency, which shows the alignment of drivers and boxes and their polarity.
- **Coherence.** A value between 0 and 1 per frequency that states how much of the measured signal is explained by the reference. Low coherence means noise, late reflections or a distorting system, and a data point with low coherence is not trusted.
- **Impulse response.** The response to a single click, computed from the same data. Its first peak gives the delay from the reference to the microphone, which sets the delay of fills and subwoofers, and its later peaks show reflections.

The reference must be taken from the signal as close as possible to the system input, usually from a spare output of the console or the processor, or by a loopback in the audio interface. The delay between reference and measurement is found with the delay finder, and 1 ms equals 0.34 m of sound travel.

Swept sine measurements, as in REW, measure the transfer function as well. They reach a higher signal-to-noise ratio in a given time and separate harmonic distortion from the response, but they need the system to play the sweep and cannot run during a show.

Resolution and windows

An FFT analyzer divides the spectrum into bins of equal width, the sample rate divided by the FFT length. A long FFT resolves fine detail at low frequencies, but it also takes in more of the room's reflections.

FFT length at 48 kHz	Bin width	Time window
1024	46.9 Hz	21 ms
8192	5.9 Hz	171 ms
65536	0.73 Hz	1.37 s

Analyzers for sound systems therefore combine several FFT lengths, a long one for the bass and short ones for the treble, which gives an approximately constant resolution per octave. A short time window, either by the FFT length or by gating the impulse response, shows the loudspeaker almost without the room, while a long window shows what the audience hears including the room. Both views are needed.

Averaging

Measurements in a room fluctuate, and averaging several reduces the random part. Averaging n uncorrelated measurements improves the signal-to-noise ratio by $10 \lg n$, so 4 averages gain 6 dB and 16 gain 12 dB. Averaging over several microphone positions, rather than over time at one position, is what removes the comb filtering and room modes that differ from seat to seat, and it gives a response that represents the area, not one seat.

Measurement microphones

A measurement microphone is a small omnidirectional condenser microphone with a flat response, usually supplied with an individual calibration file that the software applies. An omnidirectional capsule avoids the proximity effect and the off-axis colouration of a cardioid, so its reading does not depend on the exact angle.

- **Placement for a loudspeaker.** On axis, at a distance of several times the size of the box, so that all drivers have merged into one source. Close distances overstate the driver nearest to the microphone.
- **Ground plane.** A microphone lying on the floor, pointed at the box, avoids the floor reflection that would otherwise cause a comb filter. The pressure at the boundary is doubled, which reads 6 dB higher, and that offset is constant over frequency.
- **Placement in the audience.** At ear height, at several representative positions per zone, never on the centre line only, and away from walls and other reflecting surfaces close to the microphone.
- **Level.** High enough above the background noise for a coherent measurement, and not so high that the system compresses or the audience in the room is annoyed.

Two microphones of the same model can differ by a decibel or more. When several are used at once, they are compared at the same position first, and the difference is corrected or noted.

In the SdWa5 rig

The rig measures with an audio interface, two measurement microphones and the command `sdwa5-dsp:measure`, which records reference and microphone channels and computes the transfer function (automation). The bench results show the methods of this chapter on real equipment.

- S1, noise floor and hum separates the noise of the amplifiers' fans from the room by switching them on one at a time.
- C3, SPL offset ties the microphone's digital level to SPL with a simple sound level meter, and states how far that offset can be trusted.
- Transfer function reads the delay, polarity and coherent range of each cabinet from its impulse response and phase.

To verify in pass 2

- The ratio of DC resistance to nominal impedance of typical voice coils (20 % to 30 % lower).

- IEC 61672-1 class tolerances and the IEC 60942 calibrator classes and levels.
- Whether phone apps with external microphones reach class 2 performance, and the AES or NIOSH study on apps.
- The licensing of Smaart, Open Sound Meter and REW, and REW's dual-channel mode.
- Polarity-tester behaviour on multi-way boxes with inverted ways.
- Typical sensitivity spread between measurement microphones of one model.

Sources

- IEC 61672-1, IEC 60942.
- McCarthy, B.: *Sound Systems: Design and Optimization*, Focal Press, 3rd ed. 2016.
- SdWa5: measurements, automation.
- Davis, G., Jones, R.: *The Sound Reinforcement Handbook*, 2nd ed., Yamaha / Hal Leonard 1989, §16 (pp. 303–312), §8.10 (pp. 101–103).

17. Power, grounding and safety

> **Draft, pass 1.** This chapter was written from general knowledge and checked against reference literature for coverage. Standards, editions and figures are named but have not yet been verified against their sources. The list *To verify in pass 2* at the end names every claim that still needs one.

This chapter answers how a sound system is connected to mains power safely, how the power is distributed over the phases, why hum and buzz appear and how they are removed without touching the protective earth, how equipment is built into racks, and how a fault is found. It assumes the European system with 230 V and 400 V at 50 Hz. Electrical installation work itself belongs to a qualified electrician, and nothing in this chapter replaces the national rules.

Two kinds of ground

Grounding serves two purposes, and keeping them apart is the key to the whole chapter.

- **Protective earth (PE)** connects every metal housing to earth, so that a fault in a device trips the circuit breaker or the residual current device instead of putting the housing at mains voltage. It protects people, and it is never interrupted.
- **Signal reference** is the common zero against which the audio signals are measured. It connects through the shields of the audio cables and inside the devices to the chassis, and through the chassis to PE.

Hum and buzz are problems of the signal reference. Their solutions all act on the signal side, never on PE.

Mains supply

Systems and connectors

Public supplies in Austria and Germany are TN systems, in which the neutral is earthed at the transformer. Older installations combine neutral and earth in one conductor (PEN, TN-C) up to a point where they are separated (TN-C-S), and newer ones carry them separately from the start (TN-S). A sound system needs separate N and PE from its distribution onwards, which every CEE and Schuko outlet provides.

Supply	Connector	Power at 230 V per phase
Single phase, 16 A	Schuko or blue CEE	3.7 kW
Three phase, 16 A	Red CEE, 5 pins	11.0 kW
Three phase, 32 A	Red CEE, 5 pins	22.1 kW
Three phase, 63 A	Red CEE, 5 pins	43.5 kW

A mobile distribution box splits a CEE supply into the circuits of the system, each with its own circuit breaker and with residual current devices (RCD). An RCD with 30 mA rated residual current protects every final circuit that people can touch, and installation rules

for shows, exhibitions and temporary structures require it. Switch-mode power supplies can produce smooth DC fault currents that blind a common type A RCD, which is why distributions for event technology increasingly use type B or type F devices where the rules ask for them.

Phase balance and the neutral

A three-phase supply is used best when its load is spread evenly over the three phases. With linear loads, the currents of balanced phases cancel in the neutral. Switch-mode power supplies, in amplifiers, LED fixtures and computers, draw their current in short pulses, and the harmonics at three times the mains frequency add in the neutral instead of cancelling. The neutral current can then exceed the phase current, which is a reason for neutral conductors of full cross-section and for amplifiers with power factor correction (PFC), which draw a nearly sinusoidal current. Sound and lighting are best fed from separate circuits, or separate distributions, from a common supply point.

Voltage drop

Long mains cables lose voltage. For a single-phase circuit, both conductors carry the current, and the drop is $\Delta U = 2\rho L I / A$. For a balanced three-phase load, the neutral carries no current, and the drop per phase is half of that.

Current	Length	Cross-section	Drop single phase	Drop three phase, balanced
16 A	25 m	1.5 mm²	9.3 V (4.1 %)	4.7 V (2.0 %)
16 A	25 m	2.5 mm²	5.6 V (2.4 %)	2.8 V (1.2 %)
16 A	50 m	2.5 mm²	11.2 V (4.9 %)	5.6 V (2.4 %)
32 A	50 m	6 mm²	9.3 V (4.1 %)	4.7 V (2.0 %)
63 A	50 m	16 mm²	6.9 V (3.0 %)	3.4 V (1.5 %)

An amplifier at full power can draw short peaks well above its average current, and a drop of a few percent at the average becomes a larger dip at each peak, which lowers the amplifier's rail voltage and its headroom. Extension cables are therefore kept short and heavy, which means 2.5 mm² for 16 A runs of more than a few metres. A cable drum is fully unwound under load, because a coiled cable heats up.

Generators

Outdoor events often run on a generator. Its rating in kVA must cover the peak demand of the system with a reserve, because switch-mode loads and amplifier peaks stress a generator more than a resistive load does. Inverter generators supply a cleaner voltage than simple ones. Sound runs best on its own generator, or at least separated from the lighting and catering loads whose switching produces dips and spikes. The generator's earthing follows its manufacturer's instructions and the local rules, and it is set up by someone qualified to do so.

Checks before connecting

Every outlet is checked before use, and an outlet whose wiring is in doubt is not used. A socket tester shows a missing PE or swapped conductors, a meter confirms 230 V between phase and neutral and 400 V between phases, and a voltage tester finds a

voltage between the housings of two devices, such as a guitar amplifier and a microphone stand, before a performer bridges them with a hand and lips. Mobile distributions and cables are inspected at regular intervals by a qualified person, as the national rules require.

Switching on and off

Amplifiers are switched on last and off first. A console or a processor that starts up after the amplifiers sends switching transients into live loudspeakers, and many amplifiers starting at once draw an inrush current that can trip a breaker. The amplifiers are therefore switched on one at a time, or by a sequencer that delays each circuit by a second or two. The shutdown runs in reverse.

Clean and reliable power

Every console and processor is a computer, and an unstable supply can crash it. A small uninterruptible power supply (UPS) on the console, the processors and the network switches keeps them running through a dip or a short interruption, and gives time to shut them down cleanly in a real outage. It does not supply the amplifiers.

Hum, buzz and ground loops

How a ground loop forms

Two devices connected by an audio cable are also both connected to PE through their mains cables. Shield and PE together form a loop. A magnetic field through the loop, or a small voltage between the PE contacts of two outlets, drives a current around it, and part of that current flows through the shield of the audio cable. On an unbalanced connection the voltage drop along the shield adds directly to the signal, and on a balanced connection it can still enter through a device with the pin-1 problem, as chapter 15 explains. The result is hum at 50 Hz and its harmonics, or a buzz with many harmonics.

Grounding schemes

Studio installations often bring all signal grounds to one central point, but a portable system changes from event to event and cannot be built that way. Portable systems combine the following measures instead.

- **Balanced connections everywhere** with the shield connected at both ends, and devices that bond the shield to the chassis at the connector (AES48). A ground loop then carries its current on the shield and the chassis, outside the signal circuit, and causes no hum.
- **Everything on one power distribution.** The devices of one system, the console, the processors, the amplifiers and the backline, draw power from one distribution, so that their PE contacts are at the same potential. This is the central-point principle in a form a portable system can keep, and it also covers the backline on stage.
- **Lifting the shield on the signal side** where a loop remains. A ground-lift switch on a DI box or a device input disconnects the shield at one end, and it breaks the loop without touching PE.
- **An isolation transformer** in the signal path, either in a DI box or as a line isolator, breaks the loop completely, because the transformer passes the signal magnetically

and has no conductive path between its sides.

An adapter without a protective earth contact, or a cut earth conductor, would also break a loop on the mains side. That is dangerous and against the rules. A device without PE is protected only by its insulation, and in a fault its housing, and every shield connected to it, can carry mains voltage to the microphone in a performer's hand. Protective earth is never lifted, neither by an adapter, nor by a broken contact, nor by cutting the conductor.

Interference

Dimmers with SCRs, which chopped the mains waveform and produced a hard buzz in audio lines, have largely given way to LED fixtures. Their switch-mode drivers produce interference at higher frequencies, often heard as a whine or a buzz that changes with the brightness. The remedies have not changed. They are separate circuits for sound and light, distance between signal and power cables, balanced connections, and crossing at right angles. Wireless receivers and mobile phones are further sources, which chapter 10 covers.

Racks

Equipment is mounted in 19-inch racks with a height unit of 1.75 inches (44.45 mm). A few practical points keep a rack reliable.

- **Rear support** for heavy devices such as amplifiers, which otherwise bend their front panels in transport.
- **Ventilation** in the direction the devices blow, usually front to back, without blocking panels in the air path, and with space or a fan where the rack is closed.
- **Shock-mounted racks** for sensitive devices that travel.
- **Cable routing** with mains on one side and signals on the other, and with strain relief at the rear rails.
- **Labels** on every connector and every cable, matching the patch documentation (chapter 20).

Finding faults

Faults come in two kinds, the absence of the expected signal and the presence of an unwanted one, such as hum, noise, distortion or oscillation. The method is the same for both.

1. **Check the obvious first.** Every device is switched on, every fuse is intact, no channel is muted, no other channel is soloed, and the routing is what it should be. Experienced engineers are caught by this too.
2. **Change one thing at a time**, and undo the change if it did not help.
3. **Trace the signal.** A missing signal is followed from the source towards the output, an unwanted one from the output towards the source, by listening at each point with the console's solo, a headphone amplifier or a meter.
4. **Halve the chain.** In a long chain, checking the middle first tells which half holds the fault, and repeating the step on that half finds it in a few steps. With a digital console and a network, the meters at each stage make this fast.

5. **Substitute a known good part**, a cable, a microphone or a channel, and see whether the fault follows it.

Hum that disappears when one cable is unplugged at one end is a ground loop through that cable. Hum that stays when every input is unplugged comes from the device itself or from its power.

In the SdWa5 rig

The rig's mains section applies this chapter to one event on a 16 A three-phase CEE supply. It finds the circuit breakers to be the tightest limit of the rig, spreads the amplifiers over the phases to balance them, puts the processors on the lightest phase, and switches on the processors first and then the amplifiers one at a time, because four large amplifiers starting together would trip a 16 A breaker on inrush alone.

To verify in pass 2

- IEC 60364-7-711 (exhibitions, shows and stands) and IEC 60364-7-740 (temporary installations), and their RCD requirements.
- The Austrian rules for installations and event venues (ÖVE E 8101 and its part for temporary installations), and the inspection intervals for mobile distributions in Austria (and DGUV V3 for Germany).
- RCD types A, F and B and when event distributions need type B.
- Neutral current with switch-mode loads, the factor by which it can exceed the phase current.
- Generator sizing factors for amplifier and LED loads, and the earthing of mobile generators.
- Recommended maximum voltage drop for final circuits (3 % or 5 %).
- Voltage testers per EN 61243-3 and socket testers for Schuko outlets.

Sources

- IEC 60364-7-711, IEC 60364-7-740, IEC 60309, ÖVE E 8101, AES48.
- Whitlock, B.: Balanced lines in audio systems, fact, fiction and transformers, *JAES* 43, 1995.
- SdWa5: signal chain, mains.
- Davis, G., Jones, R.: *The Sound Reinforcement Handbook*, 2nd ed., Yamaha / Hal Leonard 1989, §17.4 to §17.8 (pp. 327–346).

18. Stage monitoring

> **Draft, pass 1.** This chapter was written from general knowledge and checked against reference literature for coverage. Standards, editions and figures are named but have not yet been verified against their sources. The list *To verify in pass 2* at the end names every claim that still needs one.

This chapter answers how performers hear themselves and each other on stage, who mixes what they hear, how a monitor system is set up for the most gain before feedback, and how in-ear monitoring changes the job. The console's side of monitoring, its sends and the placement of a monitor console, is in chapter 11, the physics of feedback in chapter 6, and the wireless side of in-ear monitors in chapter 10.

Why monitors

The main system points at the audience, and the stage is behind it. Without monitors, performers hear mostly the reflections of the room and the backline next to them, which makes it hard to sing in tune and to play in time. A monitor system gives each performer a mix of what they need to hear, at a level they can work with, and it is part of the performance as much as the main system.

A good monitor mix is not only for comfort. Performers who hear themselves play and sing better, and they do not raise their own level to compensate, which keeps the stage quieter and the house mix cleaner.

Ways to monitor

- **Wedges** are floor monitors angled up at the performer, usually a two-way box with a 12-inch or 15-inch driver and a horn, on a separate mix per position or per performer.
- **Side fills** are larger boxes at the sides of the stage that cover the whole stage area with a common mix, for performers who move around.
- **Drum fills** give the drummer a loud mix with low-frequency extension, often with a subwoofer, because the drum kit itself is loud.
- **In-ear monitors (IEM)** place the mix directly in the performer's ears, through wireless or wired bodypacks and earphones, and remove most of the monitor sound from the stage.

Most productions mix these. A band may use in-ear monitors for the singers and the keyboard, a wedge for a guest, and a drum fill for the drummer who wants to feel the kick drum.

Who mixes the monitors

On larger stages, a separate monitor console is the rule. The engineer at the front of house cannot hear the stage, has to communicate by hand signals, and has a house mix to

run. A monitor engineer at the side of the stage, with a direct view of the performers, can listen to each mix, react at once and keep the performers' trust.

Digital consoles have added two more options.

- **Front of house runs the monitors** on its own console, from sends that are independent of the house faders. On a small production, with a few mixes that change little during the show, this works well, and a tablet with the console's remote app lets the engineer adjust the mixes from the stage during the soundcheck.
- **Personal mixing.** Each performer adjusts their own mix, on a small mixer at their position or on a phone with the console's app, from a set of stems that the engineer prepares. This relieves the engineer, but it needs performers who know what they want and an engineer who keeps control of the overall level and the limiters.

On a shared stage with several acts, the monitor mixes for each act are stored as scenes and recalled at the changeover, as chapter 11 describes.

Gain before feedback on stage

A wedge points at the microphone it serves, at a short distance, which makes stage monitoring the hardest case of feedback. Chapter 6 explains the loop. The measures that help are, in order of their effect, the following.

Microphone and wedge placement

A directional microphone rejects sound from its null, and a wedge placed in that null contributes least to the loop. The null lies at a different angle for each pattern.

Pattern	Null from the front axis	Wedge position
Cardioid	180°	Directly behind the microphone, one wedge
Supercardioid	about 125°	About 55° to either side of the line behind the microphone, two wedges
Hypercardioid	about 110°	About 70° to either side of the line behind the microphone, two wedges

A performer who sings into the microphone with the wedge in front of them places the wedge behind the microphone, which suits a cardioid. Supercardioid and hypercardioid microphones are more directional overall, but their wedges belong to the sides, as chapter 9 shows. Real patterns differ from the ideal ones, especially at high frequencies, and the manufacturer's polar plots show where the null lies for a given model.

The other placement rules follow from the loop. The microphone is close to the source, a performer's hand does not cover the grille, which makes a cardioid behave more like an omnidirectional microphone, and the number of open microphones is kept as low as the music allows.

Polarity

A polarity reversal on a monitor output is worth a try, because each monitor mix is heard on its own and its polarity does not interact with the other mixes. A reversal changes

where the loop's phase adds, and it can move the first feedback frequency to a place where the loop gain is lower. The gain it wins is usually small and depends on the position, so it is a quick test during the soundcheck rather than a rule. Reversing the polarity at the microphone would affect the house mix and any recording, so it is done on the monitor output.

Ringing out

Ringing out finds the frequencies where the loop gain is highest and lowers them with an equalizer before the show, so that the first feedback comes later. It used to be done by ear with a graphic equalizer per monitor mix, and a current workflow combines measurement with listening.

1. **Equalize the wedge first.** A measurement of each wedge model, with the dual-channel analyzer of chapter 16, gives a corrective equalization that the processor stores per wedge. A wedge with a smooth response rings later than one with peaks.
2. **Raise the gain slowly** with the microphone at the performer's position, until the first ringing appears.
3. **Find its frequency** with the analyzer, or with the console's feedback display, and set a narrow parametric notch of a few dB at that frequency.
4. **Repeat a few times.** Three to five notches per mix are usually enough. Many more notches remove more of the voice than they win in gain, and the mix starts to sound thin.
5. **Leave headroom.** A performer moves, and the stage changes when the audience arrives, so the working level stays a few dB below the point where the last ringing appeared.

Automatic feedback suppressors, described in chapter 12, can set the notches during ringing out, and some engineers use them during the show as a safety net. They do not replace a well-placed wedge and microphone.

In-ear monitoring

In-ear monitors seal the ear, which blocks most of the stage noise and lets the performer listen at a lower level than on a wedge. They remove the wedges' contribution to feedback and to the noise on stage, which helps the house mix too. They also bring their own practice.

- **Fit.** Earphones only work when they seal. Universal tips work for a short time, and custom-moulded earphones are more comfortable and seal better.
- **Ambience.** A sealed ear hears no audience and no room, which isolates the performer. One or two ambient microphones aimed at the audience, mixed in at a low level, bring the room back.
- **Level and limiters.** The earphones deliver high levels directly into the ear canal. A limiter in each mix, set with the performer, and a clear agreement on the level protect the hearing, as chapter 5 explains. The bodypack's volume is not a limiter.
- **Mono, stereo and mix.** A stereo mix with the instruments panned to their positions on stage is easier to listen to at a lower level than a mono mix, because the ear separates the sources by direction.

- **Failure.** A bodypack that loses its signal leaves the performer in silence. Spare packs, a check of the frequencies before the show and a wedge or side fill as backup on the stage are common practice.

Latency

A performer hears their own voice through bone conduction at once, and the in-ear signal adds the latency of the whole chain. Each digital stage adds some.

Stage	Typical latency
Stage box with network to the console	about 0.5 ms to 1 ms
Digital console	about 1 ms to 2 ms
Network back to the stage	about 0.5 ms to 1 ms
Digital in-ear transmitter	up to a few milliseconds

A total of a few milliseconds equals the travel time from a wedge at 1 m to 2 m, which takes 2.9 ms to 5.8 ms. Musicians accept this, but a chain with plug-in processing or a slow wireless system can exceed it, and the performer then hears a comb filter or a slight echo on their own voice. Chapter 1 explains where latency arises in a digital system.

Talking to the stage

The monitor engineer needs to hear what the performers hear, and to talk with them.

- **Listening.** A cue wedge next to the monitor console, and a set of the same in-ear monitors the performers use, reproduce each mix through the console's PFL or AFL bus, so that the engineer hears exactly what was sent.
- **Talkback.** A talkback microphone on the console routes to any mix, so that the engineer can speak to one performer without the audience hearing it.
- **Shout microphones.** The performers in turn have a microphone, often a switched one, that goes only to the monitor engineer's cue, so they can ask for changes without hand signals.

The monitor console needs a position with a clear view of the stage, and with in-ear monitors the view matters even more, because the engineer sees a performer's request before hearing it.

To verify in pass 2

- The null angles of real supercardioid and hypercardioid microphones compared with the ideal patterns.
- The typical latency of stage boxes, digital consoles and digital in-ear systems.
- What latency performers accept on in-ear monitors (figures of about 5 ms to 10 ms are often quoted).
- Typical wedge driver sizes and the use of coaxial wedges.
- Advice on in-ear levels and limiters from hearing conservation guidance.

Sources

- Manufacturer documentation of digital consoles and in-ear systems, to be selected in pass 2.
- Davis, G., Jones, R.: *The Sound Reinforcement Handbook*, 2nd ed., Yamaha / Hal Leonard 1989, §11.9 (pp. 178–181), §11.11 (pp. 186–189).

19. Control and synchronization

> **Draft, pass 1.** This chapter was written from general knowledge and checked against reference literature for coverage. Standards, editions and figures are named but have not yet been verified against their sources. The list *To verify in pass 2* at the end names every claim that still needs one.

This chapter answers how devices on a show control each other, how MIDI, OSC and the remote interfaces of consoles are used, how timecode keeps playback, light and video in step, and how a playback rig is built so that it does not stop the show. It is about control data and time positions. Sample clocks, which keep digital audio itself in step, are covered in chapter 15.

MIDI

The Musical Instrument Digital Interface (MIDI) was defined in 1983 to connect synthesizers, and it is still the most widespread control protocol in music. On a show it switches presets of instruments and effects, recalls console scenes, fires cues and carries timecode.

Messages

MIDI 1.0 sends short messages of one to three bytes. A status byte names the type of message and, for channel messages, one of 16 channels, and one or two data bytes of 7 bits each follow, so most values range from 0 to 127.

Message	Meaning	Typical use on a show
Note On, Note Off	A key pressed or released, with velocity	Instruments, triggering samples or cues
Control Change	A controller moved, 128 controller numbers	Faders, effect parameters, footswitches
Program Change	Select one of 128 programmes	Recall a preset or a console scene
Pitch Bend, Aftertouch	Continuous expression	Instruments
System Exclusive	Manufacturer-specific data of any length	Device settings, backups, editors
Timing Clock	24 pulses per quarter note	Tempo sync of sequencers and delays
MIDI Timecode	Time position in hours to frames	Following timecode, see below

A device documents which messages it sends and understands in its MIDI implementation chart, which is the first thing to read when two devices refuse to talk.

Connections

The original connection is a 5-pin DIN cable carrying a current loop at 31 250 bit/s, isolated by an optocoupler at the receiver so that MIDI cables do not create ground loops. With 10 bits on the wire per byte, a three-byte message takes about 0.96 ms, and a chord

of ten notes takes several milliseconds, which is why dense MIDI streams on one cable can smear timing. A MIDI input, an output and a thru port pass the data on to the next device in a chain.

Today MIDI mostly travels in other ways.

- **USB-MIDI.** Class-compliant devices appear as MIDI ports on a computer without a driver, at a far higher speed than the DIN cable.
- **RTP-MIDI.** MIDI over Ethernet and Wi-Fi as a standardized network session, built into macOS and available on other systems, which replaces long DIN cables and merge boxes on a stage.
- **Bluetooth MIDI.** Convenient for a controller at a desk, but with a latency that varies too much for timing-critical use.

MIDI 2.0, published in 2020, adds higher resolution, two-way negotiation of capabilities and a new packet format. It is backwards compatible, and devices that support it are still the exception in live sound.

Common faults

- **No response.** The sender and the receiver are on different channels, the receiver ignores that message type, or the cable connects an input to an input.
- **Stuck notes.** A Note Off was lost, for example when a cable was unplugged during a note, and the note sounds until an All Notes Off message or a reset.
- **Loops.** A device that echoes its input to its output, connected back to the sender, creates a loop that floods both. The echo function (soft thru) is switched off on one of them.

OSC and remote interfaces

Open Sound Control (OSC) sends named messages over a network, such as `/ch/01/mix/fader 0.75`, as UDP or TCP packets. Its addresses are readable, its values can be floating point numbers or text, and it suits control from software, tablets and show-control systems. Many digital consoles, playback programs and lighting desks accept OSC, each with its own address scheme that its manufacturer documents.

Beyond OSC, most digital consoles and processors offer their own remote protocol, used by the manufacturer's app and often documented for third parties. A tablet app connected over Wi-Fi lets an engineer set monitor mixes from the stage, as chapter 18 describes, and a script can read meters or recall a scene. Such access goes over the control network, which is separate from the audio network or at least on its own VLAN, and is protected like any other network access, because a remote app can mute the whole system.

Show control

A show with several departments, sound, light, video and stage machinery, needs them to change state together. Two approaches are common.

- **A central cue list.** A playback or show-control program holds the cues, and each cue sends the commands for all departments, such as a scene recall by MIDI or OSC

for the console, a cue number for the lighting desk and a start command for video. An operator advances the cues.
- **Timecode.** For shows that run to a fixed time line, such as a musical set to a backing track or a video mapping, every department follows a common timecode and fires its cues at set times.

MIDI Show Control (MSC) is a MIDI extension for cue commands between show-control devices. In practice OSC and the programs' own network protocols have largely taken over its role.

Timecode

What timecode is

Timecode marks every frame of a time line with an address in hours, minutes, seconds and frames, as defined in SMPTE ST 12-1 (and in an equivalent EBU form in Europe). It was made for film and video, and it serves on a show to let every system know where in the time line it is.

Frame rate	Use	Remarks
24 fps	Film	
25 fps	Video in Europe (PAL regions)	The usual rate for shows in Austria
29.97 fps drop frame	Video in North America (NTSC regions)	Skips frame numbers to match clock time
29.97 fps non-drop	Video in North America	Drifts from clock time
30 fps	Audio-only work in North America	

North American colour video runs at 29.97 frames per second rather than 30. Counting 30 frame numbers per second at that rate drifts from the clock by 108 frames, or about 3.6 s, per hour. Drop-frame timecode skips the frame numbers 0 and 1 at the start of every minute except every tenth minute, which removes exactly those 108 numbers per hour. No frames are dropped, only numbers. All systems on a show must use the same rate and the same drop-frame setting, or their cues drift apart.

How it is carried

- **Linear timecode (LTC)** is an audio signal, a square-wave-like bit stream of 80 bits per frame, so 2000 bit/s at 25 fps. It travels on any audio cable or audio channel, can be recorded on a track of the playback, and is the most robust way to distribute timecode. It is loud and harsh, and it is kept out of every audio path that reaches a loudspeaker.
- **MIDI Timecode (MTC)** carries the same information as MIDI messages. It sends four quarter-frame messages per frame, so a complete time address takes two frames to arrive, and a full-frame message locates directly.
- **Network timecode.** Several lighting and media systems distribute timecode over the network in their own protocols.

A device that follows timecode chases it, which means it locates to the incoming position and keeps running with it. A short dropout in the timecode should not stop the show, so

receivers have a freewheel setting that keeps running for a set time without code.

Timecode and clocks are different things

Timecode says where in the time line a system is. It does not say how fast the samples run. Two audio devices that follow the same timecode but run on independent sample clocks still drift by a few samples per second relative to each other, and they are locked by a word clock or PTP, as chapter 15 explains. A playback computer that sends timecode and plays audio into the console's network uses the network clock for the audio and the timecode only to tell the other departments where it is.

Playback rigs

Backing tracks, click tracks and time-coded shows run from a computer, and a computer that stops in the middle of a song stops the show. Productions that depend on playback build it redundant.

- **Two computers** play the same session in sync, the main and the backup, each into its own interface or network outputs.
- **A switchover** selects the backup when the main stops, either by an operator or automatically when the main's signal or timecode disappears.
- **The click and the cues** go to the in-ear mixes of the musicians and the musical director, never to the main system, and the sound engineer can mute the playback at once.
- **A known state.** Operating system updates, notifications, sleep modes and network services are switched off on show computers, and the session is tested on the same machines before the tour.

Smaller productions accept a single computer, and then at least keep the songs as separate files that can be restarted quickly, rather than one long file.

In the SdWa5 rig

At MARK Salzburg, the video mapping machine received a full-range line feed from output 7 of the DSP. The rig's documentation names no timecode or control connection between audio and video for that event.

To verify in pass 2

- The MIDI 1.0 data rate (31 250 bit/s), the optocoupler requirement and the timing of running status.
- RTP-MIDI as RFC 6295 and its availability on Windows and Linux.
- The date and scope of MIDI 2.0 (MIDI Association, 2020) and the Universal MIDI Packet.
- SMPTE ST 12-1 edition, the EBU equivalent, and the 80-bit LTC frame.
- The drop-frame rule (frame numbers 0 and 1 at each minute except every tenth).
- MTC quarter-frame structure and the full-frame message.
- MIDI Show Control as a MIDI recommended practice, and its current use.
- OSC 1.0 and 1.1 specifications.

Sources

- MIDI Association: MIDI 1.0 Detailed Specification; MIDI 2.0 specifications, 2020.
- IETF RFC 6295: RTP Payload Format for MIDI.
- Wright, M.: The Open Sound Control 1.0 Specification, CNMAT, 2002.
- SMPTE ST 12-1, Time and Control Code.
- SdWa5: signal chain.
- Davis, G., Jones, R.: *The Sound Reinforcement Handbook*, 2nd ed., Yamaha / Hal Leonard 1989, §19 (pp. 371–393), §20 (pp. 395–408).

20. System design and tuning

> **Draft, pass 1.** This chapter was written from general knowledge and checked against reference literature for coverage. Standards, editions and figures are named but have not yet been verified against their sources. The list *To verify in pass 2* at the end names every claim that still needs one.

This chapter brings the book together. It answers how a sound system is planned from the requirements of an event, how its parts are chosen and documented, how its coverage and level are predicted, and how it is tuned on site until it does what the plan promised. Every section draws on earlier chapters, and links to them where the detail lives.

The brief

A system design starts with questions, not with loudspeakers. The answers decide the size, the cost and the compromises.

- **Programme.** Speech, background music, a band, electronic music or a mix of them. Speech needs intelligibility and little bass, and electronic music needs level and deep bass, as chapters 5 and 7 explain.
- **Level.** The target level at the mix position, usually as L_{Aeq} over the show, together with the peak level the programme reaches above it.
- **Audience area.** Its shape, the distance from the stage to the nearest and to the farthest listener, and whether it is seated, standing, indoors or outdoors.
- **Limits.** Noise limits at the neighbours, audience exposure limits, the venue's rules, the mains supply, the rigging points and their loads, and the time for setup.
- **Budget.** Money, crew and transport, which decide how much of the ideal design survives.

The answers are written down. A brief that exists only in a conversation is remembered differently by everyone on the day of the show.

Architecture and documentation

From the brief follows the architecture of the system, the chain of chapter 1 from the sources to the loudspeakers, with the number of inputs, the console and the stage boxes, the processing, the amplifiers, the loudspeaker groups and the monitors.

Three documents describe a system well enough that someone else can build it.

- **A block diagram** that shows the signal flow and the groups, such as left and right tops, subwoofers, front fills and delays.
- **A patch list** that names every input and output, from the microphone on the stage to the amplifier channel and the box it drives.
- **A settings table** for each processor and amplifier, with the crossover, delay, equalization, gain and limiter of each output.

These documents are best generated from one set of master data, such as a list of the boxes, amplifiers and their specifications, rather than copied from the last event and edited. Copies drift, and after a few events nobody knows which copy describes the system as it was set, which the rig documentation of SdWa5 describes from experience (see *In the SdWa5 rig* below).

Predicting coverage and level

Prediction software

Loudspeaker manufacturers supply software that predicts the level and the frequency response of their systems over a model of the audience area, such as EASE Focus, d&b ArrayCalc, L-Acoustics Soundvision and Meyer Sound MAPP 3D. For line arrays, this software is the only practical way to choose the number of boxes, the angles between them and the trim height. For independent comparisons, EASE and similar room acoustics programs model the room as well, using the directivity data of chapter 14.

A prediction is as good as its model. The software treats the audience as a plane and the air as uniform, and it knows nothing about the wind, a wrong angle set on site or a box that differs from its data. Its results are checked by measurement after the setup.

Coverage

The level across the audience should vary as little as possible. A common target is a variation of a few decibels from the front to the back. A single point source cannot meet it over a deep audience, because it loses 6 dB per doubling of distance.

Audience from	to	Level spread of a point source
10 m	40 m	12 dB
5 m	40 m	18 dB

The tools against this are those of chapter 6, which are a line array that is shaped to aim more energy at the back, a higher position of the main system, which equalizes the distances, front fills for the first rows that the main system misses, and delay systems for the back.

Level budget

The required capability of the system follows from the target level, the crest factor of the programme, the distance and a reserve. For a target of 100 dB L_{Aeq} at the mix position at 30 m, with a programme crest factor of 12 dB and a reserve of 6 dB, the system must reach the following.

Step	Value
Target level at 30 m	100 dB
Peaks of the programme, +12 dB	112 dB
Distance loss of a point source from 1 m to 30 m	29.5 dB
Reserve against compression and power loss	6 dB
Required peak level at 1 m, as a point source	147.5 dB

A single box that reaches 135 dB at 1 m falls 12.5 dB short, which at low frequencies corresponds to about four coherently coupled boxes, and at high frequencies to a line array whose behaviour the prediction software must show. Air absorption at high frequencies and over long distances adds further loss, as chapter 6 shows. The same budget is made for the subwoofers, with the programme's bass content, which for electronic music can be 10 dB above the rest of the spectrum.

Fills and delays

A fill system covers an area that the main system misses, and a delay system extends the main system to distant areas. Both are timed so that their sound arrives just after the sound from the main system, so that the audience localises the sound at the stage by the precedence effect of chapter 5.

The delay is the distance to the main system divided by the speed of sound, plus a small offset of some milliseconds. For a delay tower 40 m from the main system the delay is about 117 ms at 20 °C, and with an offset of 10 ms about 127 ms. The speed of sound changes with temperature, as chapter 6 explains, and the delay with it.

Temperature	Speed of sound	Delay for 40 m
0 °C	331 m/s	120.7 ms
10 °C	337 m/s	118.6 ms
20 °C	343 m/s	116.5 ms
30 °C	349 m/s	114.6 ms

A difference of 6 ms between a cold night and a hot afternoon matters for the alignment at high frequencies, so an outdoor delay is checked by measurement when the temperature has changed. The level of a delay system is set so that it supports the main system in its area, not so that it dominates it.

Subwoofers

The subwoofer design follows chapter 14. A central array or a flown array gives the most even bass across the audience. Left and right stacks are simpler but produce a power alley and seat-to-seat differences. A cardioid arrangement keeps the stage quieter and reduces the bass at the neighbours behind the stage.

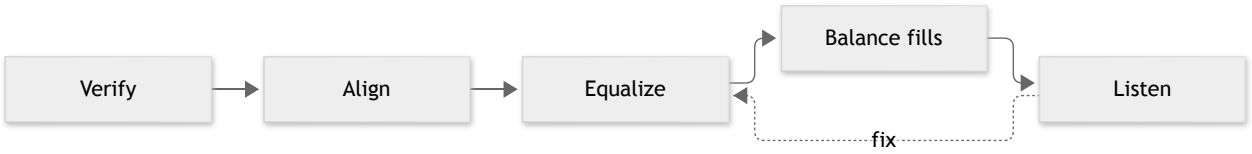
Setting up the processing

The processing of chapter 12 is set up in an order that follows the signal.

1. **Load the manufacturer's presets** for the boxes, or the measured settings for self-built boxes, with crossover, driver alignment, equalization and limiters.
2. **Set the gain structure** so that the console, the processor and the amplifiers reach their maximum together, as chapter 4 explains.
3. **Set the limiters** from the power handling of the drivers and the gain of the amplifiers, so that the limiter acts before the amplifier clips and before the driver is overloaded, as chapter 13 explains.
4. **Check every output** before the loudspeakers are connected, with a signal that verifies routing and levels.

Tuning

Tuning turns a system that is set up into one that works in the room. It uses the dual-channel analyzer of chapter 16 and the ears, in a fixed order, because each step depends on the ones before it.



1. **Verify.** Each box and each group is played alone, and its polarity, its frequency response and its level are compared with its siblings. A box that differs is wired wrongly, damaged or set wrongly, and it is fixed before anything else, because the following steps would otherwise correct a fault with equalization.
2. **Align.** The groups are aligned in time where their sounds overlap. Subwoofers and the main system are aligned at their crossover frequency at a representative position, by their phase in the analyzer. At 80 Hz, 1 ms equals 29° of phase, and a misalignment of 3.1 ms already shifts them by a quarter period, so the alignment is set to a fraction of a millisecond. Fills and delays are aligned to the main system in their own areas.
3. **Equalize.** The equalization of each group is set to the average of several measurement positions in its area, not to one position, because the response at a single seat is dominated by its local comb filters and room modes. The equalization corrects broad tendencies with a few filters, and it does not try to remove narrow notches, which move from seat to seat and cannot be filled with level.
4. **Balance fills.** Fills and delays are set to support the main system in their areas, with the main system on, until the overall response is even but before the fill becomes the dominant source.
5. **Listen.** Familiar recordings and the engineer's own voice through a microphone at the main positions tell whether the measured result sounds right. A system that measures well but sounds wrong has a fault that the measurement did not cover, such as distortion, a rattle or an excess of level at the mix position.

A target curve is a choice, not a law. Most engineers tune a system so that it falls gently from the bass to the treble, because a measured flat response in a room sounds bright, and they set the subwoofers to suit the programme.

During the show

Once the audience arrives, the room changes, with more absorption, warmer air and the noise of the audience. A measurement microphone at the mix position with the show's signal as the reference shows how the response moves, and the engineer corrects the overall equalization and the delays if needed. Noise monitoring against the event's limits, as chapter 5 describes, runs at the same time.

In the SdWa5 rig

The rig's signal chain applies the documentation rule of this chapter. The table is generated, not copied explains why the settings for an event are generated from master data rather than copied from a template, after the copies on the drive had drifted apart.

MARK Salzburg, 2026-09-19 is a worked case of the whole chapter. The number of outputs that the video mapping rig needed decided the topology, so that a second processor supplied the missing outputs for the tops. The amplifier gains and limiter thresholds follow from the gain structure in Why 32 dB and 26 dB, and the mains limits from the phase plan in Mains. Open, and checked on site lists what the plan could not settle, such as the alignment delay of the tops, which still has to account for the latency of the second processor. It shows the verification step of this chapter in practice, as a list of questions that a measurement on site answers.

To verify in pass 2

- Current names and scope of the prediction programs (EASE Focus 3, ArrayCalc, Soundvision, MAPP 3D).
- Typical targets for the level spread across the audience.
- The precedence offset for delay systems (often quoted as 5 ms to 15 ms) and the level limits for it.
- The bass content of electronic music relative to the rest of the spectrum (here assumed about 10 dB).
- Typical target curves for tuning and their slopes.
- Crest factors of typical programme material (12 dB assumed here).

Sources

- McCarthy, B.: *Sound Systems: Design and Optimization*, Focal Press, 3rd ed. 2016.
- Ahnert, W., Steffen, F.: *Sound Reinforcement Engineering*, Spon 1999.
- Davis, G., Jones, R.: *The Sound Reinforcement Handbook*, 2nd ed., Yamaha / Hal Leonard 1989, §17.1, §17.2 (pp. 313–324), §18 (pp. 347–370).
- SdWa5: signal chain.